



**BOARD OF CONTRACT AND SUPPLY
CITY OF PROVIDENCE, RHODE ISLAND**

REQUEST FOR PROPOSALS

Item Description: Hope High School- Boiler Demolition and Abatement

Procurement/MinuteTraq #: 53960

Date to be opened: 7/27/2026

Issuing Department: Department of Public Property

QUESTIONS

- Please direct questions relative to the Minority and Women's Business Enterprise Program and the corresponding forms (Pages 10-11) to the MBE/WBE Outreach Director for the City of Providence, Grace Diaz
 - Email: gdiaz@providenceri.gov
 - Please use subject line "MBE WBE Forms"
- Please direct questions relative to the specifications outlined (beginning on page 9) to vendor question submission portal link below. Questions will be answered by multiple subject matter experts via a publicly issued addendum.
 - Question submission link:
<https://forms.monday.com/forms/2623174bcb0d6268172c99a3aab9a409?r=use1>
Subject Matter Expert Paul DeMelo
 - Title: Project Manager
 - Email Address: pdemelo@downesco.com

Pre-submission Conference

There will be a Mandatory Pre-Bid Conference.

Date of Pre-Bid Conference: 6/30/2026 Time: 4:00 pm

Other details (e.g. location, links, question submission deadline): 324 Hope Street, Providence, RI 02906

Deadline for questions submissions:

Questions Due: July 9, 2026 @ 4:00 pm Final Addendum: July 14, 2026

INSTRUCTIONS FOR SUBMISSION

<u>Meeting Date: 07/27/2026</u>
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Bids may be submitted up to **2:15 P.M.** on the above meeting date at the **Department of the City Clerk, Room 311, City Hall, 25 Dorrance Street, Providence.** At 2:15 P.M. all bids will be publicly opened and read at the Board of Contract Meeting in Conference Room 305, on the 3rd floor of City Hall. **Do not wait until the last moment to submit your bid.**

- Bidders must submit **2 copies** of their bid in sealed envelopes or packages labeled with the captioned **Item Description** and the **City Department to which the solicitation and bid are related and must include the company name and address on the envelope as well.** (On page 1).
- If required by the Department, please keep the original bid bond and check in only one of the envelopes.
- Communications to the Board of Contract and Supply that are not competitive sealed bids (i.e. product information/samples) should have “**NOT A BID**” written on the envelope or wrapper.
- Only use form versions and templates included in this solicitation. If you have an old version of a form do not recycle it for use in this bid.
- The bid envelope and information relative to the bid must be addressed to:

**Board of Contract and Supply
Department of the City Clerk – City Hall, Room 311
25 Dorrance Street
Providence, RI 02903**

****PLEASE NOTE:** This bid may include details regarding information that you will need to provide (such as proof of licenses) to the issuing department before the formalization of an award.

This information is NOT requested to be provided in your initial bid by design.

All bids submitted to the City Clerk become public record. Failure to follow instructions could result in information considered private being posted to the city’s Open Meetings Portal and made available as a public record. The City has made a conscious effort to avoid the posting of sensitive information on the City’s Open Meetings Portal, by requesting that such sensitive information be submitted to the issuing department only at their request.

BID PACKAGE CHECKLIST

Digital forms are available in the City of Providence Purchasing Department Office or online at <http://www.providenceri.gov/purchasing/how-to-submit-a-bid/>

The bid package **MUST** include the following, in this order:

- Bid Form 1: Bidder's Blank as the cover page/ 1st page (*see page 6 of this document*)
- Bid Form 2: Certification of Bidder as 2nd page (*see page 7 of this document*)
- Bid Form 3: Certificate Regarding Public Records (*see page 8 of this document*)
- Bid Form 4: Affidavit of City Vendor (*see pages 9 and 10 of this document*)
- Forms from the Minority and Women Business Enterprise Program: Based on Bidder Category. See *forms and instructions enclosed (pages 10-11) or on:*
<https://www.providenceri.gov/purchasing/minority-women-owned-business-mbewbe-procurement-program/>

***Please note: MBE/WBE forms must be completed for EVERY bid submitted and must be inclusive of ALL required signatures. Forms without all required signatures will be considered incomplete.**

- Bidder's Proposal/Packet: Formal response to the specifications outlined in this RFP, including pricing information and details related to the good(s) or service(s) being provided. Please be mindful of formatting responses as requested to ensure clarity.
- Financial Assurance, *if requested* (as indicated on page 5 of this document under "Bid Terms")

All of the above listed documents are REQUIRED. (With the exception of financial assurances, which are only required if specified on page 5.)

*****Failure to meet specified deadlines, follow specific submission instructions, or enclose all required documents with all applicable signatures will result in disqualification, or in an inability to appropriately evaluate bids.**

NOTICE TO VENDORS

1. The Board of Contract and Supply will make the award to the lowest qualified and responsible bidder.
2. In determining the lowest responsible bidder, cash discounts based on preferable payment terms will not be considered.
3. Where prices are the same, the Board of Contract and Supply reserves the right to award to one bidder, or to split the award.
4. No proposal will be accepted if the bid is made in collusion with any other bidder.
5. Bids may be submitted on an "equal in quality" basis. The City reserves the right to decide equality. Bidders must indicate brand or the make being offered and submit detailed specifications if other than brand requested.
6. A bidder who is an out-of-state corporation shall qualify or register to transact business in this State, in accordance with the Rhode Island Business Corporation Act, RIGL Sec. 7-1.2-1401, et seq.
7. The Board of Contract and Supply reserves the right to reject any and all bids.
8. Competing bids may be viewed in person at the Department of the City Clerk, City Hall, Providence, immediately upon the conclusion of the formal Board of Contract and Supply meeting during which the bids were unsealed/opened. Bids may also be accessed electronically on the internet via the City's [Open Meetings Portal](#).
9. As the City of Providence is exempt from the payment of Federal Excise Taxes and Rhode Island Sales Tax, prices quoted are not to include these taxes.
10. In case of error in the extension of prices quoted, the unit price will govern.
11. The contractor will **NOT** be permitted to: a) assign or underlet the contract, or b) assign either legally or equitably any monies or any claim thereto without the previous written consent of the City Purchasing Director.
12. Delivery dates must be shown in the bid. If no delivery date is specified, it will be assumed that an immediate delivery from stock will be made.
13. A certificate of insurance will normally be required of a successful vendor.
14. For many contracts involving construction, alteration and/or repair work, State law provisions concerning payment of prevailing wage rates apply ([RIGL Sec. 37-13-1 et seq.](#))
15. No goods should be delivered, or work started without a Purchase Order.
16. **Submit 2 copies of the bid to the City Clerk, unless the specification section of this document indicates otherwise.**
17. Bidder must certify that it does not unlawfully discriminate on the basis of race, color, national origin, gender, gender identity or expression, sexual orientation and/or religion in its business and hiring practices and that all of its employees are lawfully employed under all applicable federal, state and local laws, rules and regulations. (See Bid Form 2.)

BID TERMS

1. Financial assurances may be required in order to be a successful bidder for Commodity or Construction and Service contracts. If either of the first two checkboxes below is checked, the specified assurance must accompany a bid, or the bid will not be considered by the Board of Contract and Supply. The third checkbox indicates the lowest responsible bidder will be contacted and required to post a bond to be awarded the contract.
 - a) ☐ A certified check for \$_____ must be deposited with the City Clerk as a guarantee that the Contract will be signed and delivered by the bidder.
 - b) ☐ A bid bond in the amount of _____ per centum (%) of the proposed total price, must be deposited with the City Clerk as a guarantee that the contract will be signed and delivered by the bidder; and the amount of such bid bond shall be retained for the use of the City as liquidated damages in case of default. Any person signing a bid bond as an attorney-in-fact shall include with the bid bond an original, or a photocopy or facsimile of an original, power of attorney.
 - c) ☐ A performance and payment bond with a satisfactory surety company will be posted by the bidder in a sum equal to one hundred per centum (100%) of the awarded contract.
 - d) ☒ No financial assurance is necessary for this item.
2. Awards will be made within **nighty (90) days of bid opening**. All bid prices will be considered firm, unless qualified otherwise. Requests for price increases will not be honored.
3. Failure to deliver within the time quoted or failure to meet specifications may result in default in accordance with the general specifications. It is agreed that deliveries and/or completion are subject to strikes, lockouts, accidents, and Acts of God.

The following entry applies only for COMMODITY BID TERMS:

4. Payment for partial delivery will not be allowed except when provided for in blanket or term contracts.

The following entries apply only for CONSTRUCTION AND SERVICE BID TERMS:

5. Only one shipping charge will be applied in the event of partial deliveries for blanket or term contracts.
6. Prior to commencing performance under the contract, the successful bidder shall attest to compliance with the provisions of the Rhode Island Worker's Compensation Act, [RIGL 28-29-1, et seq.](#) If exempt from compliance, the successful bidder shall submit a sworn Affidavit by a corporate officer to that effect, which shall accompany the signed contract.
7. Prior to commencing performance under the contract, the successful bidder shall, submit a certificate of insurance, in a form and in an amount satisfactory to the City.

BID FORM 1: Bidders Blank

1. Bids must meet the attached specifications. Any exceptions or modifications must be noted and fully explained.
2. Bidder's responses must be in ink or typewritten, and all blanks on the bid form should be completed.
3. The price or prices proposed should be stated both in **WRITING** and in **FIGURES**, and any proposal not so stated may be rejected. **Contracts exceeding twelve months must specify annual costs for each year.**
4. Bids **SHOULD BE TOTALED** so that the final cost is clearly stated (unless submitting a unit price bid), however **each item should be priced individually**. Do not group items. Awards may be made on the basis of *total* bid or by *individual items*.
5. All bids **MUST BE SIGNED IN INK.**

Name of Bidder (Firm or Individual): _____

Contact Name: _____

Business Address: _____

Business Phone #: _____

Contact Email Address: _____

Agrees to bid on (Write the "Item Description" here): _____

If the bidder's company is based in a state *other than Rhode Island*, list name and contact information for a local agent for service of process that *is located within Rhode Island* _____

Delivery Date (if applicable): _____

Name of Surety Company (if applicable): _____

Total Amount in Writing*: _____

Total Amount in Figures*: _____

****If you are submitting a unit price bid, please insert "Unit Price Bid"***

Use additional pages if necessary for additional bidding details.

Signature of Representation

Title

BID FORM 2: Certification of Bidder
(Non-Discrimination/Hiring)

Upon behalf of _____ (Firm or Individual Bidding),

I, _____ (Name of Person Making Certification),

being its _____ (Title or "Self"), hereby certify that:

1. Bidder does not unlawfully discriminate on the basis of race, color, national origin, gender, sexual orientation and/or religion in its business and hiring practices.
2. All of Bidder's employees have been hired in compliance with all applicable federal, state and local laws, rules and regulations.

I affirm by signing below that I am duly authorized on behalf of Bidder, on

this _____ day of _____ 20____.

Signature of Representation

Printed Name

BID FORM 3: Certificate Regarding Public Records

Upon behalf of _____ (Firm or Individual Bidding),

I, _____ (Name of Person Making Certification),

being its _____ (Title or "Self"), hereby certify an

understanding that:

1. All bids submitted in response to Requests for Proposals (RFP's) and Requests for Qualification (RFQ's), documents contained within, and the details outlined on those documents become public record upon receipt by the City Clerk's office and opening at the corresponding Board of Contract and Supply (BOCS) meeting.
2. The Purchasing Department and the issuing department for this RFP/RFQ have made a conscious effort to request that sensitive/personal information be submitted directly to the issuing department and only at request if verification of specific details is critical the evaluation of a vendor's bid.
3. The requested supplemental information may be crucial to evaluating bids. Failure to provide such details may result in disqualification, or an inability to appropriately evaluate bids.
4. If sensitive information that has not been requested is enclosed or if a bidder opts to enclose the defined supplemental information prior to the issuing department's request in the bidding packet submitted to the City Clerk, the City of Providence has no obligation to redact those details and bears no liability associated with the information becoming public record.
5. The City of Providence observes a public and transparent bidding process. Information required in the bidding packet may not be submitted directly to the issuing department at the discretion of the bidder in order to protect other information, such as pricing terms, from becoming public. Bidders who make such an attempt will be disqualified.

I affirm by signing below that I am duly authorized on behalf of Bidder, on

this _____ day of _____ 20_____.

Signature of Representation

Printed Name

BID FORM 4: Affidavit of City Vendor

Per our Code of Ordinances [Sec. 21.-28.1 \(e\)](#), this form applies to a) the business, b) any political action committee whose name includes the name of the business, c) all persons holding ten (10) percent or greater equity interest or five thousand dollars (\$5,000.00) or greater cash value interest in the business at any time during the reporting period, d) all executive officers of the business entity, e) any spouse or dependent child of any individual identified in a) through d) above.

Executive officers who are not residents of the state of Rhode Island are exempted from this requirement.

Per [R.I.G.L. § 36-14-2](#), “Business” means a sole proprietorship, partnership, firm, corporation, holding company, joint stock company, receivership, trust, or any other entity recognized in law through which business for profit or not for profit is conducted.

Name of the person making this affidavit: _____

Position in the “Business” _____

Name of Entity _____

Address: _____

Phone number: _____

The number of persons or entities in your entity that are required to report under [Sec. 21.-28.1 \(e\)](#): _____

Read the following paragraph and answer one of the options:

Within the 12 month period preceding the date of this bid submission with the City of Providence, or with respect to the contracts that are not in writing within the 12 month period preceding the date of notification that the contract has reached the \$100,000 threshold, have you made campaign contributions within a calendar year to (please list all persons or entities required under [Sec. 21.-28.1 \(e\)](#)).

a. Members of the Providence City Council? ☐ Yes ☐ No

- If Yes, please complete the following:

Recipient(s) of the Contribution:

Contribution Date(s):

Contribution Amount(s):

b. Candidates for election or reelection to the Providence City Council? ☐ Yes ☐ No

- If Yes, please complete the following:

Recipient(s) of the Contribution:

Contribution Date(s):

Contribution Amount(s):

c. The Mayor of Providence? ☐ Yes ☐ No

- If Yes, please complete the following:

Recipient(s) of the Contribution:

Contribution Date(s):

Contribution Amount(s):

d. Candidates for election or reelection to the office of Mayor of Providence? ☐ Yes ☐ No

- If Yes, please complete the following:

Recipient(s) of the Contribution:

Contribution Date(s):

Contribution Amount(s):

Signed under the pains and penalties of perjury.

Position

MBE/WBE Participation Plan

Please complete separate forms for each MBE/WBE subcontractor/supplier to be utilized on the solicitation.

Bidder's Name:					
Bidder's Address:					
Point of Contact:					
Telephone:					
Email:					
Procurement #:					
Project Name:					
Which one of the following describes your business' status in terms of Minority and/or Woman Owned Business Enterprise certification with the State of Rhode Island? (Check all that apply).	☐ MBE ☐ WBE ☐ Neither MBE nor WBE				
This form is intended to capture commitments between the prime contractor/vendor and MBE/WBE subcontractors and suppliers, including a description of the work to be performed and the percentage of the work as submitted to the prime contractor/vendor. Please note that all MBE/WBE subcontractors/suppliers must be certified by the Office of Diversity, Equity and Opportunity at the time of bid. The MBE/WBE Directory can be found here. Please visit, the City's MBE/WBE page for details of the program (e.g. instructions and requirements). • Nonprofit organizations are not required to complete the rest of this form. • Construction projects unable to identify subcontractors prior to bid submission (e.g. Design Build) are required to provide updates to the MBE/WBE Outreach Office					
Name of Subcontractor/Supplier:					
Type of RI Certification:	☐ MBE ☐ WBE ☐ Neither				
Address:					
Point of Contact:					
Telephone:					
Email:					
Detailed Description of Work to Be Performed by Subcontractor or Materials to be Supplied by Supplier Per the Scope of Work provided in the RFP					
Total Contract Value (\$):		Subcontract Value (\$):		Participation Rate (%):	
Anticipated Date of Performance:					
I certify under penalty of perjury that the forgoing statements are true and correct.					
Prime Contractor/Vendor Signature		Title		Date	
Subcontractor/Supplier Signature		Title		Date	

***If you did not meet the 20% MBE/WBE combined participation goal, submit a Waiver Request Form.**

MBE/WBE Waiver Request Form

Fill out this form only if you did not meet the 20% MBE/WBE participation goal.

State-certified MBE or WBE Prime Bidders are NOT REQUIRED to fill out this form.

Submit this form to the City of Providence MBE/WBE Outreach Director, Grace Diaz, at gdiaz@providenceri.gov, for review **prior to bid submission**. This waiver applies only to the current bid which you are submitting to the City of Providence and does not apply to other bids your company may submit in the future. **In case a waiver is needed, City Department Directors should not recommend a bidder for an award if this form is not included, absent or is not signed by the city of Providence MBE/WBE director.**

Prime Bidder: _____ Contact Email and Phone _____
Company Name, Address: _____ Trade _____
Project /Item Description (as seen on RFP): _____

To receive a waiver, you must list the certified MBE and/or WBE companies you contacted, the name of the primary individual with whom you interacted, and the reason the MBE/WBE company could not participate on this project.

MBE/WBE Company Name	Individual's Name	Company Name	Why did you choose not to work with this company?

I acknowledge the City of Providence's goal of a combined MBE/WBE participation is 20% of the total bid value. I am requesting a waiver of _____ % MBE/WBE (20% minus the value of **Box F** on the Subcontractor Disclosure Form). If an opportunity is identified to subcontract any task associated with the fulfillment of this contract, a good faith effort will be made to select MBE/WBE certified businesses as partners.

Signature of Prime Contractor /
or Duly Authorized Representative

Printed Name

Date Signed

Signature of City of Providence
MBE/WBE Outreach Director /
or Duly Authorized Representative

Printed Name of City of Providence
MBE/WBE Outreach Director

Date Signed

FOR CONSTRUCTION PROJECTS

APPRENTICE REQUIREMENTS (Construction Projects Valued at \$100,000 or More).

Attention of prospective bidders is called to the fact that this project is to be bid upon and executed under the City of Providence Code of Ordinances Chapter 21 Art. II [Section 21-28.1](#) c(1) and (2) related to utilizing apprentices in the contract. This ordinance outlines requirements for utilizing not less than 15% of total hours worked by apprentices. The City may lower this percentage only if it determines in writing that compliance is not feasible or that it would be unduly cost prohibitive to the project. The attention of prospective bidders is also called to the fact that reporting the efforts undertaken and progress towards achieving the requirements in this ordinance is a condition for payment. Compliance reporting shall be submitted with any contract payment requisition, in a format to be specified by the City. This demonstration of compliance through such reports shall be a condition of the requisition for payment to be processed. Upon the contract being awarded to the successful bidder, a mandatory meeting will be scheduled to review the project requirements relative to apprenticeship requirements and the process and protocols by which these goals will be achieved. At this meeting, specific forms and procedures for the documentation and achievement of these requirements by the successful bidder will be provided, discussed and agreed upon for the execution of the contract.

“FIRST SOURCE” REQUIREMENTS.

Attention of prospective bidders is called to the fact that this project is to be bid upon and executed under the City of Providence Code of Ordinances [Chapter 21 Art. III 1/2 First Source Agreements](#) Sec. 21-91 through 21-96. This ordinance outlines requirements for hiring Providence residents to work on this project. The City may waive this requirement only upon a determination in writing that qualified residents of Providence are not available for the project, pursuant to Sec. 21-94(e). The attention of prospective bidders is called to the fact that reporting the efforts undertaken and progress towards achieving the requirements in this ordinance is a condition for payment. Compliance reporting shall be submitted with any contract payment requisition, in a format to be specified by the City. This demonstration of compliance through such reports shall be a condition of the requisition for payment to be processed. Upon the contract being awarded to the successful bidder, a mandatory meeting will be scheduled to review the project requirements relative to the First Source Agreements and the process and protocols by which these goals will be achieved. At this meeting, specific forms and procedures for the documentation and achievement of these requirements by the successful bidder will be provided, discussed, and agreed upon for the execution of the contract.

BID PACKAGE SPECIFICATIONS

Overview:

The City of Providence is seeking comprehensive, competitive proposals from licensed and qualified environmental and demolition contractors. The objective is to safely and legally perform the hazardous materials abatement and selective demolition/removal of the existing boiler systems at Hope High School located at 324 Hope Street, Providence, RI 02906.

This project will be issued by the city in collaboration with the Providence Public School District and managed by Downes Construction Company is the leading Owner's Project Manager (OPM).

Other Key Information:

Architect's Name: StudioJAED

Drawings

G-000 COVER SHEET

M-000 MECHANICAL COVER

MD101

E-000

E-101

MECHANICAL DEMOLITION PLANS ELECTRICAL COVER ELECTRICAL PLANS

Specifications

See attached Table of Contents prepared by SStudioJAED

Proposed Schedule:

The timeline for the selection process is presented below. A mandatory pre-bid conference is scheduled for this item. Complete responses must be received by Monday, July 27, 2026, at 2:15 pm and sent to the City Clerk (25 Dorrance Street Providence, RI 02903 Room 311). No faxed or electronic submissions will be accepted. Upon awarding the bid, the City would plan to commence work starting upon the execution of an official contract. Any changes will be posted to the City of Providence Purchasing Department official website located at <https://www.providenceri.gov/purchasing/openrfpsummary/>. Respondents are responsible for checking the website for any scheduled changes.

<u>Events</u>	<u>Date</u>
Request for Proposals Issued	Monday, June 29, 2026
Pre-Bid Conference	Tuesday June 30, 2026, at 4PM
Deadline to submit bid questions (via email to subject matter experts)	Thursday, July 9, 2026, at 4 PM
Final response to questions (via public addendum)	No later than Thursday July 23, 2026, by 2PM
Bids Due	Monday, July 27, 2026

SUPPLEMENTAL INFORMATION

If the issuing department for this RFP determines that your firm's bid is best suited to accommodate their need, you will be asked to provide proof of the following prior to formalizing an award.

An inability to provide the outlined items at the request of the department may lead to the disqualification of your bid.

*This information is **NOT** requested to be provided in your initial bid that you will submit to the City Clerk's office by the "date to be opened" noted on page 1. This list only serves as a list of items that your firm should be ready to provide on request.*

All bids submitted to the City Clerk become public record. Failure to follow instructions could result in information considered private being posted to the city's Open Meetings Portal and made available as a public record.

You must be able to provide:

- Business Tax ID will be requested after an award is approved by the Board of Contract and Supply.
- Proof of Insurance.
- Certificate of Good Standing with the Rhode Island Secretary of State.



**BOARD OF CONTRACT AND SUPPLY
CITY OF PROVIDENCE, RHODE ISLAND**

CITY OF PROVIDENCE STANDARD TERMS & CONDITIONS

1. The terms “you” and “your” contained herein refer to the person or entity that is a party to the agreement with the City of Providence (“the City”) and to such person’s or entity’s employees, officers, and agents.
2. The Request For Proposals (“RFP”) and these Standard Terms and Conditions together constitute the entire agreement of the parties (“the Agreement”) with regard to any and all matters. By your submission of a bid proposal or response to the City’s RFP, you accept these Standard Terms & Conditions and agree that they supersede any conflicting provisions provided by bid or in any terms and conditions contained or linked within a bid and/or response. Changes in the terms and conditions of the Agreement, or the scope of work thereunder, may only be made by a writing signed by the parties.
3. You are an independent contractor and in no way does this Agreement render you an employee or agent of the City or entitle you to fringe benefits, workers’ compensation, pension obligations, retirement or any other employment benefits. The City shall not deduct federal or state income taxes, social security or Medicare withholdings, or any other taxes required to be deducted by an employer, and this is your responsibility to yourself and your employees and agents.
4. You shall not assign your rights and obligations under this Agreement without the prior written consent of the City. Any assignment without prior written consent of the City shall be voidable at the election of the City. The City retains the right to refuse any and all assignments in the City’s sole and absolute discretion.
5. Invoices submitted to the City shall be payable sixty (60) days from the time of receipt by the City. Invoices shall include support documentation necessary to evidence completion of the work being invoiced. The City may request any other reasonable documentation in support of an invoice. The time for payment shall not commence, and invoices shall not be processed for payment, until you provide reasonably sufficient support documentation. In no circumstances shall the City be obligated to pay or shall you be entitled to receive interest on any overdue invoice or payment. In no circumstances shall the City be obligated to pay any costs associated with your collection of an outstanding invoice.
6. For contracts involving construction, alteration, and/or repair work, the provisions of applicable state labor law concerning payment of prevailing wage rates (R.I. Gen. Laws §§ 37-13-1 et seq., as amended) and the City’s First Source Ordinance (Providence Code of Ordinances §§ 21-91 et seq., as amended) apply.
7. With regard to any issues, claims, or controversies that may arise under this Agreement, the City shall not be required to submit to dispute resolution or mandatory/binding arbitration. Nothing prevents the parties from mutually agreeing to settle any disputes using mediation or non-binding arbitration.
8. To the fullest extent permitted by law, you shall indemnify, defend, and hold harmless the City, its employees, officers, agents, and assigns from and against any and all claims, damages, losses, allegations, demands, actions, causes of action, suits, obligations, fines, penalties, judgments, liabilities, costs and expenses, including but not limited to attorneys’ fees, of any nature whatsoever arising out of, in connection with, or resulting from the performance of the work provided in the Agreement.
9. You shall maintain throughout the term of this Agreement the insurance coverage that is required by the RFP or, if none is required in the RFP, insurance coverage that is considered in your industry to be commercially reasonable, and you agree to name the City as an additional insured on your general liability policy and on any umbrella policy you carry.
10. The City shall not subject itself to any contractual limitations on liability. The City shall have the time permitted within the applicable statute of limitations, and no less, to bring or assert any and all causes of action, suits, claims or demands the City may have arising out of, in connection with, or resulting from the performance of the work provided in the Agreement, and in no event does the City agree to limit your liability to the price of the Agreement or any other monetary limit.
11. The City may terminate this Agreement upon five (5) days’ written notice to you if you fail to observe any of the terms and conditions of this Agreement, or if the City believes your ability to perform the



BOARD OF CONTRACT AND SUPPLY
CITY OF PROVIDENCE, RHODE ISLAND

terms and conditions of this Agreement has been materially impaired in any way, including but in no way limited to loss of insurance coverage, lapsing of a surety bond, if required, declaration of bankruptcy, or appointment of a receiver. In the event of termination by the City, you shall be entitled to just and equitable compensation for any satisfactory work completed and expenses incurred up to the date of termination.

12. Written notice hereunder shall be deemed to have been duly served if delivered in person to the individual or member of the firm or entity or to an officer of the entity for whom it was intended, or if delivered at or sent by registered or certified mail to the last business address known by the party providing notice.
13. In no event shall the Agreement automatically renew or be extended without a writing signed by the parties.
14. You agree that products produced or resulting from the performance of the Agreement are the sole property of the City and may not be used by you without the express written permission of the City.
15. For any Agreement involving the sharing or exchange of data involving potentially confidential and/or personal information, you shall comply with any and all state and/or federal laws or regulations applicable to confidential and/or personal information you receive from the City, including but not limited to the Rhode Island Identity Theft Protection Act, R.I. Gen. Laws § 11-49.3-1, during the term of the Agreement. You shall implement and maintain appropriate physical, technical, and administrative security measures for the protection of, and to prevent access to, use, or disclosure of, confidential and/or personal information. In the event of a breach of such information, you shall notify the City of such breach immediately, but in no event later than twenty-four (24) hours after discovery of such breach.
16. The Agreement is governed by the laws of the State of Rhode Island. You expressly submit yourself to and agree that any and all actions arising out of, in connection with, or resulting from the performance of the Agreement or relationship between the parties shall occur solely in the venue and jurisdiction of the State of Rhode Island or the federal court located in Rhode Island.
17. The failure of the City to require performance of any provision shall not affect the City's right to

require performance at any time thereafter, nor shall a waiver of any breach or default of this Agreement constitute a waiver of any subsequent breach or default or a waiver of the provision itself.

18. If any term or provision of this Agreement, or the application thereof to any person or circumstance shall, in any extent, be invalid or unenforceable, the remainder of this Agreement shall not be affected thereby, and each term and provision shall be valid and enforceable to the fullest extent permitted by law.

PROVIDENCE PUBLIC SCHOOL DEPARTMENT

Hope HS Boiler Demolition

324 Hope St,
Providence, RI 02906

May 14, 2026

Prepared By
StudioJAED
Phone # (401) 648-0884
www.studiojaed.com

**SECTION 00 01 10
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PROCUREMENT AND CONTRACTING REQUIREMENTS

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- A. 01 10 00 - Summary
- B. 01 20 00 - Price and Payment Procedures
- C. 01 21 00 - Allowances
- D. 01 25 00 - Substitution Procedures
- E. 01 30 00 - Administrative Requirements
- F. 01 31 00 - Project Management And Coordination
- G. 01 32 00 - Construction Progress
- H. 01 33 00 - Submittal Procedures
- I. 01 35 53 - Security Procedures
- J. 01 40 00 - Quality Requirements
- K. 01 42 16 - Definitions
- L. 01 50 00 - Temporary Facilities and Controls
- M. 01 60 00 - Product Requirements
- N. 01 61 16 - Volatile Organic Compound (VOC) Content Restrictions
- O. 01 70 00 - Execution and Closeout Requirements
- P. 01 74 19 - Construction Waste Management and Disposal
- Q. 01 76 10 - Temporary Protective Coverings
- R. 01 78 00 - Closeout Submittals
- S. 01 79 00 - Demonstration and Training
- T. 01 81 13 - SUSTAINABLE DESIGN REQUIREMENTS

2.02 DIVISION 02 -- EXISTING CONDITIONS

- A. 02 41 00 - Demolition

2.03 DIVISION 26 -- ELECTRICAL

- A. 26 05 05 - Selective Demolition for Electrical

END OF SECTION

SECTION 00 31 00
AVAILABLE PROJECT INFORMATION

PART 1 GENERAL

1.01 EXISTING CONDITIONS

- A. Hazardous Material Survey: Entitled ASBESTOS REMOVAL SPECIFICATION, dated 5.19.2026.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION



*PO Box – E * Pawtucket, RI 02861 * Office (401)-727-4941 *
Email: emeryenvironmental@verizon.net*

ASBESTOS REMOVAL SPECIFICATION

PROJECT LOCATION:

BOILER REPLACEMENT PROJECT
PROVIDENCE SCHOOL
HOPE HIGH SCHOOL
324 HOPE STREET
PROVIDENCE, RI 02906

CLIENT:

MR. PHIL CONTE
STUDIO JAED ARCHITECTS
42 WEYBOSSET STREET; SUITE #403
PROVIDENCE, RI 02903

REPORT PREPARED BY:

MR. PATRICK A. EMERY
EMERY ENVIRONMENTAL ASSOCIATES
P.O. BOX - E
PAWTUCKET, RI 02861

A handwritten signature in black ink, appearing to read "Pat A. Emery", is written over a horizontal line.

DATE: 5.19.2026

PATRICK A. EMERY
RIDOH ASBESTOS PROJECT DESIGNER; APD00505

EEA PROJECT #: 260405-A

Environmental Consulting and Mitigation Contractors
Lead Paint * Asbestos * Mold * Indoor Air Quality

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***APPENDIX – A: ASBESTOS ABATEMENT SPECIFICATION /
RIDOH ASB-16, ASB-16A & ASB-16B ASBESTOS
ABATEMENT APPLICATION FORMS***

APPENDIX – B: ANALYTICAL SAMPLE RESULTS

APPENDIX – C: ASBESTOS CONSULTANT LICENSURE

APPENDIX – D: ASBESTOS TESTING LABORATORY LICENSURE

APPENDIX – E: ASBESTOS PROJECT DRAWING(S)

APPENDIX – A:

***ASBESTOS ABATEMENT SPECIFICATION /
RIDOH ASB-16, ASB-16A & ASB-16B ASBESTOS
ABATEMENT APPLICATION FORMS***

**RHODE ISLAND DEPARTMENT OF HEALTH
NOTARIZED CERTIFICATION OF ASBESTOS ABATEMENT PLAN**

Facility: Hope High School

Address: 324 Hope Street

City/Town: Providence ZIP: 02906 Amendment/Phase No.: _____

Abatement Plan Written By: Patrick A. Emery Certification No.: APD0505

Summary of specific waivers/variances being requested: _____

Type of Asbestos Abatement ☒ Removal ☐ Enclosure ☐ Encapsulation ☐ Demolition
 ☐ Glovebag ☐ Asphalt Roofing / Exterior Abatement
 ☐ Other (Specify) _____

Is this plan being submitted in response to a Notice of Violation and/or a Notice of Requirement to Submit an Asbestos Abatement Plan?

☐ YES ☒ NO

If Yes, Indicate Notice/Building Evaluation No(s): _____

Contractor: To Be Selected License No: _____

Estimated Start Date: 6.1.2026

Pre-Abatement Sampling Information:

Bulk Samples Collected By: Patrick A. Emery Certification No.: AI0505

Bulk Samples Analyzed By: EMSL Analytical Inc. Certification No.: PLM0139

Air Samples Collected By: Patrick A. Emery Certification No.: AI0505

Air Samples Analyzed By: Emery Environmental Assoc. Certification No.: PCM0146

In-Process / Clearance Air Sampling Information:

Air Samples to be Collected By: Patrick A. Emery Certification No.: AI00505

Air Samples Analyzed By: EMSL Analytical Inc. Certification No.: PCM00139

Air Samples Analyzed By: Emery Environmental Associates Certification No.: PCM00146

Air Samples Analyzed By: EMSL Analytical Inc. Certification No.: TEM00139

CERTIFICATION

I certify that this asbestos abatement plan is prepared and submitted under the provisions of Section 23-24.5-6 of the RI Asbestos Control Act and Parts A and C of the Rhode Island Rules and Regulations for Asbestos Control; all abatement/management activities performed in conjunction with this plan must be in compliance with the specifications prescribed in this plan (when approved) and the most current revision of all applicable federal and state regulations; and the asbestos abatement/ management activities described in this plan must be performed by a RI licensed asbestos abatement contractor.

Certified by: _____ Title: _____
(Signature of Building Owner or Agent)

Certified by: _____ Date: _____
(Typed/Printed Name of Certified)

Subscribed and sworn before me this _____ day of _____, 20_____

(Notary Public) My Commission Expires: _____

AFFIX NOTARY SEAL HERE

STATE OF RHODE ISLAND AND PROVIDENCE PLANTATIONS

Department of Health

Division of Occupational & Radiological Health

APPLICATION FOR APPROVAL OF AN ASBESTOS ABATEMENT PLAN

1. Building Owner's Name:

City Of Providence –

Dept. Of Public Property

3. Building Owner's Mailing Address and Telephone Number:

Street: City Hall Providence

25 Dorrance Street – Room #407

2. Application Prepared By:

Mr. Patrick A. Emery

City/Town: Providence

Zip: 02903

R.I. Certification No. Aac-505 PD

Telephone No: 401-421-7740

(Area Code, No., Ext.)

Telephone No.: (401)-727-4941

(Area Code, No., Ext.)

4. Person To Be Contacted Regarding This Application:

Name: Mr. Patrick Emery

Telephone No.: 401-641-0074

(Area Code, No., Ext.)

5. Location where abatement work will be performed:

Name: (if applicable) Hope High School

Street: 324 Hope Street

City/Town: Providence

ZIP: 02906

6. Is this application being submitted in response to a "Notice of Requirement to Submit an Asbestos Abatement Plan"?

() YES (X) NO

If yes, what is the due date for submittal of Abatement Plan?

(Mo.) (Day) (Yr.)

Evaluation Number on the Notice:

7. Contractor who will be performing abatement work (if selected):

Name: To Be Selected

R.I. License No.: LAC-

8. Estimated Starting Date of Abatement Work: 6 1 2026
(Month) (Day) (Year)

9. Estimated Completion Date of Abatement Work: 7 1 2026
(Month) (Day) (Year)

10. Type of Asbestos Abatement: (Check all that apply)

- ☒ **Removal** ☐ Enclosure
☐ Encapsulation ☐ Demolition
☐ Operations and Maintenance Only
☐ Other: (Specify) _____
-

11. Type of Building: ☒ **School**
☐ Privately Owned Building
☐ Publicly Owned Building
☐ Residence
☐ Other: (Specify) _____

12. Building Access: ☒ **Public Access ($\geq 25\%$ of Building Area)**
☐ Limited Public Access (< 25 % of Building Area)
☐ No Public Access

13. Bulk Sample Collection and Analysis:

A. Person collecting bulk samples:

Name: Patrick A. Emery R.I. Certification No.: AI00505

B. Sampling Methodology:

- ☒ EPA AHERA Sampling Requirements (40 CFR 763.86).
☐ EPA's Asbestos Containing Material in School Buildings: A Guidance Document (EPA-450/2-78-014) or
Guidance for Controlling Asbestos Containing Materials - 1985 Edition (EPA-560/5-85-024).
☐ Other (Specify) _____

C. Laboratory performing the analysis of the bulk samples:

Name: EMSL Analytical Inc. R.I. Certification No.: PLM00139

D. Analytical Methodology:

- ☒ EPA Interim Method for the Determination of Asbestos in Bulk Insulation Samples (PLM method only).
☐ Other (Specify) _____
-

14. Pre-Abatement Air Sample Collection and Analysis:

A. Person collecting pre-abatement air samples:

Name: Patrick A. Emery Affiliation: Emery Environmental Associates

B. Laboratory performing analysis of pre-abatement air samples:

Air Samples Analyzed By: Emery Environmental Associates RI Certification No.: PCM0146

C. Methodology used in the collection and analysis of pre-abatement samples:

(X) NIOSH Method 7400 (Most Current Revision)

() OSHA 29 CFR 1926.58 - Appendix A & B

() Other (Specify) _____

15. A. Indicate how the regulated asbestos containing material (RACM) will be removed from the abatement site. If a hauler or broker will be used to transport the RACM to a disposal site, they must also be identified.

To Be Provided By Contractor

B. Provide the name and location of the authorized asbestos waste facility to which the removed material will be transferred for disposal (if known).

To Be Provided By Contractor

16. Person designated as compliance monitor for abatement work. **(NOT REQUIRED)**

Name: _____

Affiliation: Emery Environmental Associates

17. In-Process & Clearance Air Sampling: **(SEE ATTACHMENT #1)**

A. Describe on an attachment the type, number and location of air samples that will be collected outside the work area during the abatement project.

B. Describe on an attachment the plan of action to be followed if the Indoor Non-Occupational Air Exposure Standard for Asbestos (0.01 fibers per cubic centimeter) is exceeded outside the work area during the abatement project.

C. Describe on an attachment the type, number and location of air samples that will be collected as part of the final clearance testing.

D. Describe on an attachment the plan of action to be followed if the Indoor Non-Occupational Air Exposure Standard for Asbestos (0.01 fibers per cubic centimeter) is exceeded during final clearance testing.

18. A separate and fully completed Form ASB-16A must be submitted for each area to be abated. List below the entry in Item 1 from each attached ASB-16A.

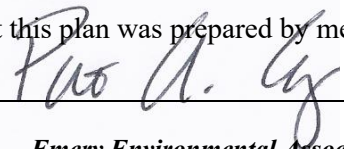
Asbestos Control Area #1 / Basement – Boiler Room

Control Area #1 / Basement – Boiler Room

NOTE: SEE PROJECT DRAWINGS – ATTACHED / APPENDIX - E

CONSULTANT CERTIFICATION

19. I certify that this plan was prepared by me and I am responsible for its content.

Signature:  Date: 5 19 2026

(Month) (Day) (Year)

Affiliation: ***Emery Environmental Associates***

20. ASBESTOS ABATEMENT PLAN APPLICATION FEE:

- | | |
|--|--------|
| () Operations & Maintenance Only | \$ 75 |
| () Up to One (1) NESHAP Unit | \$ 75 |
| () Between One (1) & Ten (10) NESHAP Units | \$ 300 |
| () Between Ten (10) & Fifty (50) NESHAP Units | \$ 600 |
| () Over Fifty (50) NESHAP Units | \$ 900 |
| () AMENDMENT FEE | \$ 150 |

STATE OF RHODE ISLAND AND PROVIDENCE PLANTATIONS

Department of Health

Division of Occupational & Radiological Health

APPLICATION FOR APPROVAL OF AN ASBESTOS ABATEMENT PLAN

SUPPLEMENTAL INFORMATION: AREA DESCRIPTION AND PROPOSED REMEDY

BUILDING LOCATION: *Hope High School*

324 Hope Street; Providence, RI 02906

INSTRUCTIONS: All items on this form must be addressed. All references to attachments must be clearly identified. All attachments must be marked with the specific item numbers on this form to which they pertain.

(1) Area Location/Identification (Room Name / No., Evaluation Number, etc.):

Control Area #1A + Asbestos Control Area #1 / Basement – Boiler Room

See Attachment #2 & Attached Drawing EEA1/MD101 (Appendix - E) For Exact Locations

(2) Attach a description of each type (e.g. pipe, ceiling, etc.) of regulated asbestos containing material (RACM) in this area, including condition, location, quantity and asbestos content. Attach a copy of the laboratory report(s) for all samples. (**NOTE:** All laboratory reports must include the name of the building(s) and the location(s) of the sample(s)).

See Attachment #2

(3) Attach a current scale drawing of this area, showing direction of North and East, which has been clearly annotated to show the type, location and quantity of all RACM in this area. This drawing must include a legend which acts as a guide to the scale, symbols and nomenclature used in the drawing. If a master plan or multiple drawings are provided, indicate the specific location(s) and drawing number(s) which depict this area. If the location of the decontamination chamber is known, it should be so indicated on the appropriate drawing(s).

Refer to Attached Drawings EEA1/MD101 + EEA2/MD101 / (Appendix - E)

(4) **PROPOSED REMEDIES:**

A) Attach a description of the interim Operations and Maintenance Plan that will be implemented in accordance with C.1.2(b).

See Attachment #3

(4) **PROPOSED REMEDIES: (Cont.)**

B) Will any portion of this area be abated by use of B.8 work procedures?
(☒) YES () NO

If Yes, indicate below which RACM in this area will be abated by use of the following B.8 work procedures:

B.8.2 & B.8.3	(REMOVAL): <u>THE ASBESTOS CONTRACTOR SHALL BE REQUIRED TO ABATE ALL ASBESTOS CONTAINING MATERIALS AS DEFINED WITHIN "APPENDIX - 2" OF THE ATTACHED SPECIFICATION. ALL ASBESTOS REMOVAL OPERATIONS MUST BE CONDUCTED IN ACCORDANCE WITH THE WORK REQUIREMENTS OF SECTION 1.14.2 / 1.14.3 OF THE CURRENT RIDOH ASBESTOS CONTROL REGULATIONS, IN ADDITION TO ALL WORK REQUIREMENTS AS DEFINED WITHIN ATTACHMENTS #1 - #4 (ATTACHED)</u>
B.8.2 & B.8.4	(ENCAPSULATION): _____
B.8.2 & B.8.5	(ENCLOSURE): _____
B.8.6	(DEMOLITION): _____
B.8.7	(GLOVEBAG): _____
B.8.8	(ASPHALT ROOFING): _____

C) Are you requesting any waivers to the above selected B.8 procedures for any of the abatement activities in this area?

() YES (☒) NO

If Yes, attach a detailed description of the waivers requested you are proposing to utilize. All items must be keyed to the specific section(s) of the regulations for which waivers are requested.

D) Are you proposing alternative procedures under B.11 for any of the abatement activities in this area?

() YES (☒) NO

If Yes, attach a detailed description of the alternative procedures requested you are proposing to utilize. Alternate procedures must include a justification for not following specific section(s) of the regulations and be as protective of public health.

E) Will any **accessible** RACM remain in this area after abatement?

(☒) YES () NO

If Yes, attach a description of the RACM that will remain and the details of the on-going Operations and Maintenance Plan that will be implemented in accordance with C.1.2(b).

All remaining ACM within this area and the envelope of the structure at the completion of the project shall be maintained by the building owner in accordance with the in-place AHERA Management Plan as developed and managed by the building owner / LEA.

AGENCY USE ONLY

ATTACHMENT #1

IN-PROCESS AIR MONITORING / INTERIOR: ASBESTOS CONTROL AREA #1 INTERIOR: CONTROL AREA #1A

During the scheduled abatement operations, the building owner's asbestos consultant will collect air sample(s) outside each asbestos control area, periodically for the duration of the asbestos removal project. The compliance air samples shall be collected at random locations adjacent to the control areas (i.e., control area barriers or where the project representative determines, during the performance of the project.

If there is a deviation in proper procedures on the part of the asbestos abatement contractor during the set-up, the removal of the final clean up, the selected project monitor shall be notified immediately. These deviations include inadequate on-site paperwork, deviations in outlined work procedures, or if compliance air sampling outside the work area indicate fiber concentrations in excess of 0.01 f/cc, the on-site project monitor shall determine the extent of the contamination, and the asbestos abatement contractor shall be responsible for extending containment / control area to include the containment and initiating the cleanup.

The owner shall also conduct dust monitoring to document the demolition contractor has not exceeded the OSHA established permissible exposure limit (PEL) for respirable dust at 5.0 milligrams per cubic meter (mg/m³) at any time during the removal of the existing Boiler Breech Insulation.

CLEARANCE AIR MONITORING REQUIREMENTS / INTERIOR: CONTROL AREA #1:

A visual clearance inspection within the asbestos abatement work area shall be performed prior to the performance of clearance air sampling. The visual inspection(s) will include an inspection of all vertical and horizontal surfaces within the control area, (as applicable). The on-site project monitor shall determine the cleanliness of the work area(s). After the containment area has passed the on-site project monitors visual clearance inspection, the asbestos monitor shall collect the following air samples within the control area:

Upon the successful completion of the visual inspection, the on-site project monitor shall collect clearance air samples in accordance with the sampling requirements of 40 CFR Part 763, EEA shall use the clearance air sampling guidelines as outlined in EPA guidance document EPA 560/5-89-001. ["Guidelines for Conducting the AHERA TEM Clearance Test to Determined Completion of an Asbestos Abatement Project"].

NOTE: If clearance air monitoring results determine that the area is above an arithmetic mean of greater than 70s/mm² under the initial screening and / or the Z-test screening process]. The asbestos abatement contractor shall be responsible for the re-cleaning (i.e., wet-wiping, HEPA vacuuming...etc.) and re-encapsulating of the work area. A period of no less 24 hours shall elapse prior to the collection of the next set of clearance samples. This process shall be repeated until the air sample fiber concentrations are less than a 70s/mm² as an arithmetic mean. **The fees for any required re-testing due to clearance testing failure(s) within any asbestos work area shall be the financial responsibility of the asbestos contractor and must be paid to the building owner at the completion of the project.**

ATTACHMENT #2

CHART OF ACM:

****NOTE: THE SELECTED ASBESTOS CONTRACTOR SHALL BE RESPONSIBLE FOR THE FIELD VERIFICATION OF ALL ASBESTOS QUANTITIES, AS REPORTED FOR EACH ASBESTOS CONTROL AREA AS DEFINED BY THIS SPECIFICATION - PRIOR TO SIGNING THE PROJECT CONTRACT AND START OF THE PROJECT. NO CHANGE ORDERS WILL BE APPROVED FOR DISCREPANCIES ASSOCIATED WITH THE REPORTED QUANTITIES.**

****CONTROL AREA #1 / ASBESTOS NPE**

LOCATION	ACM QUANTITY	TYPE / DESCRIPTION OF ACM MATERIAL TO BE REMOVED.	ASBESTOS CONTENT
**Basement – Boiler Room @ Boiler #2 + Boiler #3	+/- 100 Square Feet (Estimated)	Plumbing / Boiler Gaskets @ Pipe Flange(s) + Gasketing @ Boiler Sections	65% Chrysotile / PACM

****Refer to Appendix E of this plan for the approximate limits of the asbestos control area defined as Control Area #1. All work associated with each defined asbestos control area must be conducted in strict compliance with the designed work practices as defined by Attachments #1 - #4 of this specification and clearance air sampled in accordance with the requirements of Attachment #1 of this specification (prior to the removal of any areas of demarcation / engineering controls as erected by the RIDOH licensed asbestos contractor).**

*****CONTROL AREA #1A / NPE**

LOCATION	ACM QUANTITY	TYPE / DESCRIPTION OF ACM MATERIAL TO BE REMOVED.	ASBESTOS CONTENT
***Basement – Boiler Room – Boiler Breeching @ Boiler #3	+/- 150 Square Feet	Insulation @ Boiler Breech Exhaust (Boilers to Chimney)	NO ASBESTOS
***Basement – Boiler Room – Boiler Breeching @ Boiler #2	+/- 220 Square Feet	Insulation @ Boiler Breech Exhaust (Boilers to Chimney)	NO ASBESTOS
***Basement – Boiler Room – Boiler Breeching Box @ Chimney	+/- 192 Square Feet	Insulation @ Boiler Breech Exhaust (Boilers to Chimney)	NO ASBESTOS

***** Note: Waste generated for the removal of existing boiler breech insulation can be disposed as non-asbestos waste material if the material is removed prior to the removal of the ACM material as defined by Asbestos Control Area #1. However, Boiler Breech insulation must be removed using wet methods and must be containerized in a leak tight container of six-mil polyethylene bags or fiber drum. At no time shall this material be removed dry or transported through the school or in a waste dumpster unless it properly containerized.**

ATTACHMENT #3

INTERIM AND ON-GOING OPERATIONS AND MAINTENANCE PROGRAM:

The building owner representatives, maintenance personnel, building staff, outside contracting personnel, and other building occupants have been or will be made aware of the presence, amount, location, and condition, of the asbestos-containing building materials within this area of the building. These people have been or will be educated and advised not to disturb the asbestos-containing materials due to the potential health effects if asbestos fibers become airborne.

EMPLOYEE NOTIFICATIONS:

The building owner as well as outside contractors have been or will be notified as to the presence of asbestos containing building materials within the **interior Boiler Room area of the Hope High School; 324 Hope Street; Providence, RI 02906**. All outside contractor(s) will sign a document stating that he has been made aware of the presence and location of the asbestos-containing materials within this areas. Also, the building owner representative(s) are responsible for presenting information to the building occupants of any asbestos abatement activities being conducted. This will be accomplished by posting memo's and/or posting of caution / warning signs at all the entrances to the building during such activities.

ACCIDENTAL DISTURBANCE OF ASBESTOS-CONTAINING MATERIALS:

The building owner was, [at the time of the inspection], has been made aware of the presence of asbestos materials within the **interior Boiler Room area of the Hope High School; 324 Hope Street; Providence, RI 02906**. Due to the presence of the asbestos-containing materials. The building owner will institute the following procedures in an event of an accidental asbestos fiber release within the building prior to the start of the asbestos abatement project.

If an asbestos-containing material becomes disturbed within the criteria of a minor fiber release (less than 3 linear feet of 3 square feet of ACBM), a trained "R.I. Competent Person" may perform the clean-up, removal, encapsulation, or enclosure abatement activities utilizing spot removal/repair techniques. During these spot abatement techniques, access to the area shall be restricted to only those trained individuals, signs shall be posted, and HVAC (if applicable) shall be shut down and locked out. If a major fiber release occurs (greater than 3 linear feet or 3 square feet of ACM), the clean-up, removal, encapsulation, or enclosure abatement activities must be completed by a RI Department of Health (RIDOH) certified asbestos abatement contractor. Regardless of the amount of asbestos to be abated, the affected area must be isolated and entry to the area restricted to only those trained/ certified personnel.

EMPLOYEE TRAINING:

Any employee of the company, firm, agency, or other organization as well as any outside contractor employed by the building owner who, as a consequence of their work activities, disturbs asbestos-containing building materials must be properly trained and certified by the R.I. Department of Health as a "R.I. Competent Person" in accordance with the R.I. Rules and Regulations for Asbestos Control.

ATTACHMENT #4

ADDITIONAL WORK REQUIREMENTS:

During the asbestos mitigation operations the selected asbestos contractor shall be responsible for the work to be conducted in accordance with all applicable asbestos control regulations: USEPA – AHERA 40 CFR 763, USEPA – NESHAP 40 CFR Part 61, US OSHA 29 CFR 1926.1101 & US OSHA 29 CFR 1910.1001 and the RIDOH Asbestos Control Regulation [216-RICR-50-15-1]. The building owner (City of Providence / Providence School Department), project architect (StudioJAED Architects), and/or the selected asbestos consultant (Emery Environmental Associates) will stop the work at any time violations of any asbestos control regulations are observed. In addition to the above referenced asbestos control regulations, the selected asbestos contractor shall be required to comply with the following additional work requirements.

ASBESTOS CONTROL AREA #1:

GENERAL SCOPE OF WORK: THE SCOPE OF THE PROJECT IS THE PLANNED DEMOLITION OF BOILER #2 + BOILER #3 AND THE DEMOLITION / REMOVAL ANY/ALL SECTION'S OF BOILER BREECHING / ALL ASSOCIATED BOILER BREECHING INSULATION) CONNECTED TO BOILER #2 AND BOILER #3 . LIMITS OF DEMOLITION WORK IS DEFINED ON STUDIOJAED DEMOLITION DRAWING MD-101 AND SHOULD INCLUDE ANY ADDITIONAL WORK AS DEFINED ON STUDIOJAED DRAWING M-101

REFER TO ATTACHED STUDIOJAED PROJECT DRAWING AD101 / EEA1, FOR APPROXIMATE LIMITS OF ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A.

EXACT LIMITS OF BOILER DEMOLITION AND EXISTING BREECHING REMOVAL SHALL BE AT THE DIRECTION OF THE PROJECT ARCHITECT AND/ OR THE CONTRACTED MECHANICAL CONTRACTOR RESPONSIBLE FOR THE INSTALLATION OF NEW BOILER(S) AND ASSOCIATED MECHANICAL SYSTEMS WITHIN THE EXISTING BOILER ROOM.

EXISTING ENCLOSED GASKETING AND PACKINGS WITHIN THE INTERNAL AREAS OF THE BOILER(S) ARE ASSUMED TO CONTAIN ASBESTOS AND PER THE DESIGN OF THIS SPECIFICATION, DEMOLITION AND REMOVAL OF THE EXISTING BOILERS ARE A PLANNED TO ONLY BE REMOVED AS AN ASBESTOS RESPONSE ACTION USING WORK PRACTICES AS DEFINED BY SECTION 1.14.2 / 1.14.3 OF THE CURRENT RIDOH ASBESTOS CONTROL REGULATION [216-RICR-50-15-1].

CONTROL AREA #1A:

INSPECTION AND TESTING OF EXISTING BOILER BREECHING INSULATION ASSOCIATED WITH BOILER #1, BOILER #2 AND BOILER #3 HAS DETERMINED THE EXISTING TSI MATERIAL IS NON-ASBESTOS, HOWEVER DUE TO THE FRIABILITY OF MATERIAL AND THE OWNERS RESPONSIBILITY TO COMPLY WITH THE OSHA RESPIRABLE DUST STANDARDS IN ASSOCIATION WITH ADDITIONAL REQUIREMENT'S FOR COMPLIANCE WITH NESHAP STANDARD OF NO VISIBLE EMISSIONS. THE PLANNED REMOVAL OF EXISTING BOILER BREECHING INSULATION IS DESIGNATED TO ONLY BE REMOVED USING WET METHODS, CREATE NO VISIBLE EMISSIONS AND ONLY WITHIN A **NEGATIVE PRESSURE CONTAINMENT ENCLOSURE**. THE AREA DEFINED AS ASBESTOS CONTROL AREA #1A IS DEFINED ON ATTACHED PROJECT DRAWING EEA2. THE ASBESTOS CONTRACTOR MAY CONSTRUCT ONE NPE CONTAINMENT FOR BOTH BREECHING INSULATION REMOVAL AND THE DEMOLITION OF BOILERS #2 + #3 (ASBESTOS CONTROL AREA #1). **A FINAL INSPECTION OF THE NPE FOR WORK ASSOCIATED WITH CONTROL AREA #1A IS REQUIRED PRIOR TO REMOVAL.**

PROJECT SUBMITTALS: Prior to the start of any asbestos work operations, the selected asbestos contractor shall be required to submit copies of the following items for review and approval. **NOTE:** *Additional submittals not requested at this time may be required for submittal and approval in the future.* No asbestos work operations may begin on site until the following items are submitted and approved in writing by the owner (City of Providence / Providence School Department) and/or the owners project representative (StudioJAED Architects)

1. *Copy of the Revised RIDOH 10 day written start work notification for this project,*
2. *Copies of the Providence police and fire department written notification(s),*
3. *Copy of contractors current insurance certificate,*
4. *Copy of contractor's current written health and safety program,*
5. *Copy of contractor's current written asbestos respirator program,*
6. *Copy of contractors current Health & Safety program,*
7. *Copies of MSDS sheet for the polyethylene sheeting, proposed for use,*
8. *Copies of MSDS sheet for the duct tape, proposed for use,*
9. *Copies of MSDS sheet for the final spray encapsulant, proposed for use,*
10. *Copies of MSDS sheet for duct tape, proposed for use,*
11. *Copies of MSDS sheet for protective suits, proposed for use,*
12. *Copy of manufactures efficiency report for each "HEPA vacuum system" proposed for use*
13. *Copies of licensure for the waste landfill, proposed for use,*
14. *Copies of RIDOH Asbestos licensure, training certification, medical information, fit test, First Aid / CPR card(s), for each worker proposed for use on the project site,*
15. *Copies of RIDOH certification for laboratory used to analyze the contractors daily OSHA personal air samples.*

ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A /

NEGATIVE PRESSURE CONTAINMENT REQUIREMENTS: The Asbestos Abatement contractor shall set-up and maintain negative pressure ventilation units with HEPA filtration, in sufficient number, dependent upon the abatement areas, to provide at least one (1) work-place air change every fifteen minutes. These units shall operate continuously (24 hours per day) from the time of barrier construction through the time of successful final visual inspection by the building owners selected consultant.

Local exhaust filtered air will be evacuated to the outside of the building at all times away from public access areas and building ventilation intake vents. HEPA filters will be leak proof and the ventilation unit will maintain a minimum pressure differential of -0.02 inches of water column relative to the adjacent areas of the building. Documentation of negative pressure shall be monitored by a manometer with printed reading. If on-site readings indicate adequate negative pressure has not / is not being maintained. The asbestos contractor shall increase the number of units until proper negative pressure can be established.

COVERING OF NON-ASBESTOS BUILDING MATERIALS: The asbestos contractor shall be responsible for the covering of all floor and wall surface areas (**as applicable**), materials and objects within each work area that is not scheduled for asbestos abatement with two layers of polyethylene sheeting. All items scheduled to be covered within each work area must be first pre-cleaned using wet-methods and HEPA vacuumed to a dust free condition prior to covering. This shall include but not limited to: fiberglass pipe, non-asbestos doors, floors, walls, light fixtures, HVAC duct insulation, etc. All items not covered and not decontaminated at the completion of the work shall be considered asbestos contaminated and must be cleaned or disposed of as an asbestos containing building material.

ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A /

CONTAINMENT BARRIERS: The asbestos contractor shall be responsible for isolation of the asbestos control / work area and prevent the movement of any/all dust or debris beyond the limits of the asbestos control / work area(s), as defined by Attachment #2 of this specification and the attached print(s). Containment barriers shall be installed in compliance with this specification and sections: 1.14.2 A & B. of the RIDOH asbestos control regulation [216-RICR-50-15-1].

CONTROL AREA #1A / REMOVAL OF THERMAL INSULATION AT BOILER BREECH

EXHAUST: The selected asbestos contractor shall be responsible for the complete removal of any/all thermal insulation material as present on existing boiler breeching at Boiler #2 and Boiler #3, within the Basement – Boiler Room area, as determined applicable by the project architect (StudioJAED Architects) drawing EEA2 / MD101. Thermal insulation at each of the three boilers appears to be homogenous and testing of the material has determined it is a non-asbestos material. However, Due to the high friability and silica type composition of the insulation material - the owner will require the contractor removing the insulation material to be done only using wet methods and only within a negative pressurized containment as defined by section 1.14.2 A, B, C, D, E, F, G, H, I, J & K of the current RIDOH asbestos control regulation 216-RICR-50-15-1.

ASBESTOS CONTROL AREA #1 /

BOILER(S) DISMANTELING (REQUIRED DEMOLITION) : The selected asbestos contractor shall be responsible for the complete removal of any/all accessible and inaccessible asbestos and presumed asbestos materials as defined by Attachment #2 of this project specification. In addition, the asbestos contractor shall be responsible for the COMPLETE demolition (dismantling) of all boiler mechanical equipment as defined for removal by Attachment #2 of this specification. **Work shall include but not limited to:** Dismantle and removal of Boiler #2 & Boiler #3 and the removal of ANY/ALL accessible and inaccessible presumed asbestos containing materials as defined by Attachment #2 of this specification. During and/or at the completion asbestos abatement response action, the asbestos shall be responsible for the removal / dismantling of the Existing boiler(s), and the movement of all clean mechanical “equipment / parts” from the work area (Boiler Room) to an area outside the defined NPE (negative pressure enclosure). This work must be completed prior to a final visual inspection of the Boiler Room area and clearance air testing by Emery Environmental Associates in association with requirements of Attachment #1 & Attachment #4 of this specification.

NOTE - 1: Any/all plumbing and/or utility service disconnects as required for the planned work shall be the responsibility of the asbestos contractor prior to the start of the asbestos abatement operations.

NOTE - 2: Any/all equipment or rigging, ladders or staging as required for the safe access, removal, cleaning or installation of engineering controls, shall be the responsibility of the asbestos contractor prior to the start to and / or during the asbestos abatement response action

LEAD PAINT / OSHA: All workers coming into contact with the painted walls, ceilings, window frame(s) and door frames(s), etc., shall be required to comply with all worker protection requirements of the current OSHA lead regulation 29 CFR 1926.62.

MOVABLE OBJECTS: It is the responsibility of the asbestos contractor to wet clean, HEPA vacuum and remove all moveable non-asbestos objects from the work area prior to the start of any asbestos mitigation operations.

LOCKOUT / TAG OUT: The asbestos contractor shall be responsible for the lock-out / tag-out of any and all gas or electric equipment operating within the asbestos control area(s). (This shall include generators, boiler system equipment, hot water heaters, etc.) In addition, the asbestos contractor shall be responsible for the isolation and lock-out / tag-out of all electrical service(s) within the asbestos control area(s).

WATER SUPPLY FOR SITE / WORK OPERATIONS: For the project - the building owner shall provide the asbestos contractor with water as required for the performance of the asbestos abatement operations. The water will be provided from the structure being renovated or from an adjacent building at the site. It shall be the selected asbestos contractor’s responsibility to retrieve the supplied utility (i.e. water) as required for the completion and for the duration of the project.

ELECTRICITY SUPPLY FOR SITE / WORK OPERATIONS During the performance of the asbestos abatement operations, the building owner shall be responsible for providing any/all electricity as required for the performance of the contracted asbestos abatement operations.

ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A /

FINAL CLEANING: The selected contractor shall be responsible for the complete cleaning of all surface areas within the asbestos control area using wet methods and HEPA vacuuming prior to the completion of the project.

ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A /

POST-ABATEMENT FINAL CLEANING: Upon completion of the project the selected asbestos contractor shall be responsible for the removal of all asbestos contaminated polyethylene sheeting and any associated waterproof tape used as part of the containment system. In addition, the contractor shall be responsible for the following:

- *Cleaning of all tape residue from light fixtures, wall / ceiling and floor surface areas,*
- *Removal of any/all tack staples used during the project,*
- *Cleaning of any/all residues left from mastic removal products on floor or wall surface area(s),*
- *Cleaning of any/all paint chips that has been dislodged during the containment removal operations,*

ASBESTOS CONTROL AREA #1 /

POWER SUPPLY TO WORK AREA FOR AIR TESTING: The selected asbestos contractor is required to supply each asbestos work area with at least six (6) independent electric cords for the purpose of air testing / project oversight. The cords must be operational (“live”) and directly attached to a GFI / GFCI.

ASBESTOS CONTROL AREA #1 + CONTROL AREA #1A /

WATER SUPPLY TO WORK AREA FOR ASBESTOS RESPONSE ACTION WORK: The selected asbestos contractor is required to supply each asbestos work area with water from a operating hose as required to complete the asbestos response action. A 1-5 gallon handheld pump sprayer is NOT considered adequate water.

END OF SPECIFICATION

APPENDIX – B:

ANALYTICAL SAMPLE RESULTS



EMSL Analytical, Inc.

5 Constitution Way, Unit A Woburn, MA 01801

Tel/Fax: (781) 933-8411 / (781) 933-8412

<http://www.EMSL.com/bostonlab@emsl.com>

EMSL Order: 132602537

Customer ID: EMER53

Customer PO:

Project ID:

Attention: Patrick Emery
Emery Environmental Associates
PO BOX E
Pawtucket, RI 02861

Phone: (401) 727-4941
Fax: (401) 724-0926
Received Date: 04/20/2026 8:30 AM
Analysis Date: 04/21/2026
Collected Date: 04/17/2026

Project: 260405-A - Hope HS; 524 Hope Street; Providence, RI

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
260405-1.1 132602537-0001	Boiler Breeching at Boiler #3 - Thermal Insulation at Boiler Breeching Box	White Fibrous Homogeneous	10% Synthetic	90% Non-fibrous (Other)	None Detected
260405-1.2 132602537-0002	Boiler Breeching at Boiler #3 - Thermal Insulation at Boiler Breeching Box	Gray Fibrous Homogeneous	5% Synthetic	95% Non-fibrous (Other)	None Detected
260405-1.3 132602537-0003	Boiler Breeching at Boiler #2 - Thermal Insulation at Boiler Breeching Box	Gray/White Fibrous Homogeneous	10% Synthetic	90% Non-fibrous (Other)	None Detected
260405-2.1 132602537-0004	Pipe Breeching at Boiler #1 - Thermal Insulation at Round Breeching	White Fibrous Homogeneous	5% Synthetic	95% Non-fibrous (Other)	None Detected
260405-2.2 132602537-0005	Debris at Top Boiler #1 - Thermal Insulation at Round Breeching	White Fibrous Homogeneous	5% Synthetic	95% Non-fibrous (Other)	None Detected

Analyst(s)

Ramon Buenaventura (5)

Steve Grise, Laboratory Manager
or Other Approved Signatory

EMSL maintains liability limited to cost of analysis. Interpretation and use of test results are the responsibility of the client. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. The report reflects the samples as received. Results are generated from the field sampling data (sampling volumes and areas, locations, etc.) provided by the client on the Chain of Custody. Samples are within quality control criteria and method specifications unless otherwise noted. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method") but augmented with procedures outlined in the 1993 ("final") version of the method. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Non-friable organically bound materials present a problem matrix and therefore EMSL recommends gravimetric reduction prior to analysis. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Woburn, MA NVLAP Lab Code 101147-0, CT PH-0315, MA AA000188, RI PLM00139, VT AL998919, ME LB-0039

Initial report from: 04/21/2026 06:00:29

EMSL ANALYTICAL, INC.
LABORATORY PRODUCTS TRAINING

Asbestos Bulk Building Material Chain of Custody

EMSL Order Number (Lab Use Only):

132602537

Company: Emery Environmental Associates		EMSL-Bill to: ☐ Same ☐ Different If Bill to is Different note instructions in Comments**	
Street: PO Box - E		Third Party Billing requires written authorization from third party	
City: Pawtucket	State/Province: RI	Zip/Postal Code: 02861	Country: USA
Report To (Name): Patrick Emery		Telephone #: 401-641-0074	
Email Address: EMERYENVIRONMENTAL@ICLOUD.COM		Fax #:	Purchase Order:
Project Name/Number: SEE BELOW		Please Provide Results: ☐ Fax ☒ Email	
U.S. State Samples Taken:		CT Samples: ☐ Commercial/Taxable ☐ Residential/Tax Exempt	

Turnaround Time (TAT) Options* - Please Check

☐ 3 Hour ☐ 6 Hour ☒ 24 Hour ☐ 48 Hour ☐ 72 Hour ☐ 96 Hour ☐ 1 Week ☐ 2 Week

*For TEM Air 3 hr through 6 hr, please call ahead to schedule. *There is a premium charge for 3 Hour TEM AHERA or EPA Level II TAT. You will be asked to sign an authorization form for this service. Analysis completed in accordance with EMSL's Terms and Conditions located in the Analytical Price Guide.

PLM - Bulk (reporting limit)

- ☒ PLM EPA 600/R-93/116 (<1%)
☐ PLM EPA NOB (<1%) CALL FIRST
 Point Count ☐ 400 (<0.25%) ☐ 1000 (<0.1%)
 Point Count w/Gravimetric ☐ 400 (<0.25%) ☐ 1000 (<0.1%)
☐ NIOSH 9002 (<1%)
☐ NY ELAP Method 198.1 (friable in NY)
☐ NY ELAP Method 198.6 NOB (non-friable-NY)
☐ OSHA ID-191 Modified
☐ Standard Addition Method

TEM - Bulk

- ☐ TEM EPA NOB - EPA 600/R-93/116 Section 2.5.5.1
☐ NY ELAP Method 198.4 (TEM)
☐ Chatfield Protocol (semi-quantitative)
☐ TEM % by Mass - EPA 600/R-93/116 Section 2.5.5.2
☐ TEM Qualitative via Filtration Prep Technique
☐ TEM Qualitative via Drop Mount Prep Technique

Other

☐☒ Check For Positive Stop - Clearly Identify Homogenous Group

Date Sampled: 4/17/2026

Samplers Name: PATRICK EMERY

Samplers Signature: Pat A. Emery

Sample #	HA #	Sample Location	Material Description
260405	1-1	BOILER BREECHING @ BOILER #3	THERMAL INSULATION @
260405	1-2	BOILER BREECHING @ BOILER #3	BOILER BREECHING BOX
260405	1-3	BOILER BREECHING @ BOILER #2	↓
260405	2-1	PIPE BREECHING @ BOILER #1	THERMAL INSULATION @
260405	2-2	DEBRIS @ TOP BOILER #1	ROUND BREECHING ↓

Client Sample # (s): 260405HA1-1 - 260405HA2-2	Total # of Samples: 5
Relinquished (Client): Pat A. Emery	Date: 4/19/26 Time: 01:00
Received (Lab):	Date: Time:
Comments/Special Instructions: P# 260405-A PROJECT: HOPE #5 524 HOPE STREET: PROVIDENCE, RI	



EMSL Analytical, Inc.

5 Constitution Way, Unit A Woburn, MA 01801

Tel/Fax: (781) 933-8411 / (781) 933-8412

<http://www.EMSL.com> / bostonlab@emsl.com

EMSL Order: 132602805

Customer ID: EMER53

Customer PO:

Project ID:

Attention: Patrick Emery
Emery Environmental Associates
PO BOX E
Pawtucket, RI 02861

Phone: (401) 727-4941
Fax: (401) 724-0926
Received Date: 04/29/2026 3:45 PM
Analysis Date: 04/30/2026
Collected Date: 04/29/2026

Project: 260405-A - Hope HS; 324 Hope Street; Providence, RI

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
260405-1.4-Outer Layer 132602805-0001	TSI at Boiler Breech / Boiler #3 / Middle - South Side - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.4-Inner Layer 132602805-0001A	TSI at Boiler Breech / Boiler #3 / Middle - South Side - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	10% Cellulose	90% Non-fibrous (Other)	None Detected
260405-1.5-Outer Layer 132602805-0002	TSI at Boiler Breech / Breech Box at Chimney (Bottom) - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.5-Inner Layer 132602805-0002A	TSI at Boiler Breech / Breech Box at Chimney (Bottom) - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	10% Cellulose	90% Non-fibrous (Other)	None Detected
260405-1.6-Outer Layer 132602805-0003	TSI at Boiler Breech / Boiler #2 / North Side - Middle - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.6-Inner Layer 132602805-0003A	TSI at Boiler Breech / Boiler #2 / North Side - Middle - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	10% Cellulose	90% Non-fibrous (Other)	None Detected
260405-1.7-Outer Layer 132602805-0004	TSI at Boiler Breech Box at Chimney - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.7-Inner Layer 132602805-0004A	TSI at Boiler Breech Box at Chimney - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	15% Cellulose	85% Non-fibrous (Other)	None Detected
260405-1.8-Outer Layer 132602805-0005	TSI at Boiler Breech / Boiler #2 / South Side - Low - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.8-Inner Layer 132602805-0005A	TSI at Boiler Breech / Boiler #2 / South Side - Low - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	15% Cellulose	85% Non-fibrous (Other)	None Detected

Initial report from: 04/30/2026 10:22:26



EMSL Analytical, Inc.

5 Constitution Way, Unit A Woburn, MA 01801

Tel/Fax: (781) 933-8411 / (781) 933-8412

<http://www.EMSL.com / bostonlab@emsl.com>

EMSL Order: 132602805

Customer ID: EMER53

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
260405-1.9-Outer Layer 132602805-0006	TSI at Boiler Breech / Boiler 3 North Side - High Area - TSI Boiler Breech Insul. Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-1.9-Inner Layer 132602805-0006A	TSI at Boiler Breech / Boiler 3 North Side - High Area - TSI Boiler Breech Insul. Gray & White	White Fibrous Homogeneous	15% Cellulose	85% Non-fibrous (Other)	None Detected
260405-2.3-Outer Layer 132602805-0007	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-2.3-Inner Layer 132602805-0007A	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	White Fibrous Homogeneous	10% Synthetic	90% Non-fibrous (Other)	None Detected
260405-2.4-Outer Layer 132602805-0008	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-2.4-Inner Layer 132602805-0008A	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	White Fibrous Homogeneous	10% Synthetic	90% Non-fibrous (Other)	None Detected
260405-2.5-Outer Layer 132602805-0009	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	Gray Fibrous Homogeneous	15% Min. Wool	85% Non-fibrous (Other)	None Detected
260405-2.5-Inner Layer 132602805-0009A	TSI at Round Boiler Breech / Boiler #1 - TSI at Boiler Breech Exhaust Gray & White	White Fibrous Homogeneous	10% Synthetic	90% Non-fibrous (Other)	None Detected

Analyst(s)

Kevin McKenzie (18)

Steve Grise, Laboratory Manager
or Other Approved Signatory

EMSL maintains liability limited to cost of analysis. Interpretation and use of test results are the responsibility of the client. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. The report reflects the samples as received. Results are generated from the field sampling data (sampling volumes and areas, locations, etc.) provided by the client on the Chain of Custody. Samples are within quality control criteria and met method specifications unless otherwise noted. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method") but augmented with procedures outlined in the 1993 ("final") version of the method. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Non-friable organically bound materials present a problem matrix and therefore EMSL recommends gravimetric reduction prior to analysis. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Woburn, MA NVLAP Lab Code 101147-0, CT PH-0315, MA AA000188, RI PLM00139, VT AL998919, ME LB-0039

Initial report from: 04/30/2026 10:22:26



EMSL ANALYTICAL, INC.
TESTING LABS • PRODUCTS • TRAINING

Asbestos Bulk Building Materials - Chain of Custody

EMSL Order Number / Lab Use Only

EMSL Analytical, Inc.
5 Constitution Way, Unit A
Woburn, MA 01801

PHONE: (781) 933-8411

EMAIL: bostonlab@emsl.com

132602805

Customer Information		Billing Information	
Customer ID:	Company Name:	Billing ID:	Company Name:
Company Name: EMERY ENVIRONMENTAL ASSOC.	Billing Contact:	Billing Contact:	Billing Contact:
Contact Name: PATRICK EMERY	Street Address:	Street Address:	Street Address:
Street Address: PO BOX E PANTUCKET, RI	City, State, Zip:	City, State, Zip:	City, State, Zip:
City, State, Zip:	Country: USA	Country: SAME	Country:
Phone: 401-641-0074	Phone:	Phone:	Phone:
Email(s) for Report:	Email(s) for Invoice:	Email(s) for Invoice:	Email(s) for Invoice:

Project Information			
Project Name/No: SEE BELOW	Purchase Order:	State of Connecticut (CT) must select project location:	
EMSL LIMS Project ID: (If applicable, EMSL will provide)	US State where samples collected: RI	☐ Commercial (Taxable)	☐ Residential (Non-Taxable)
Sampled By Name: Pat 66 PATRICK EMERY	Sampled By Signature: <i>[Signature]</i>	Date Sampled: 4/29/26	No. of Samples in Shipment: 9
Turnaround-Time (TAT)			
☐ 3 Hour	☐ 6 Hour	☒ 24 Hour	☐ 32 Hour
☐ 48 Hour	☐ 72 Hour	☐ 96 Hour	☐ 1 Week
☐ 2 Week	Please call ahead for large projects and/or turnaround times 6 Hours or Less. *32 Hour TAT available for select tests only; samples must be submitted by 11:30am.		

Test Selection	
PLM - Bulk (reporting limit) ☒ PLM EPA 600/R-93/116 (<1%) ☐ PLM EPA NOB (<1%) ☐ POINT COUNT ☐ 400 (<0.25%) ☐ 1,000 (<0.1%) ☐ POINT COUNT w/ GRAVIMETRIC ☐ 400 (<0.25%) ☐ 1,000 (<0.1%) ☐ NIOSH 9002 (<1%) ☐ NYS 198.1 (Friable - NY) ☐ NYS 198.6 NOB (Non-Friable - NY) ☐ NYS 198.8 (Vermiculite SM-V)	TEM - Bulk ☐ TEM EPA NOB ☐ NYS NOB 198.4 (Non-Friable - NY) ☐ TEM EPA 600/R-93/116 w Milling Prep (0.1%) Other Tests (please specify) INSULATION HAS 2 LAYERS - HARD "GRAY" LAYER @ OUTSIDE - (SOFT) WHITE LAYER @ INSIDE LAYER REPORT RESULTS FOR BOTH LAYERS ☒ Positive Stop - Clearly Identified Homogeneous Areas (HA)

Sample Number	HA Number	Sample Location	Material Description
260405	* 1.4	TSI @ BOILER BREACH / BOILER #3 / MIDDLE - SOUTH SIDE	TSI BOILER BREACH INSUL.
260405	* 1.5	TSI @ BOILER BREACH / BREACH BOX @ CHIMNEY (BOTTOM)	GRAY + WHITE
260405	* 1.6	TSI @ BOILER BREACH / BOILER #2 / NORTH SIDE - MIDDLE	
260405	* 1.7	TSI @ BOILER BREACH BOX @ CHIMNEY / SOUTH SIDE	
260405	* 1.8	TSI @ BOILER BREACH / BOILER #2 / SOUTH SIDE - LOW	

Special Instructions and/or Regulatory Requirements (Sample Specifications, Processing Methods, Limits of Detection, etc.)			
PA 260405-A HOPE HS - 324 HOPE ST. ; PROVIDENCE, RI			
Method of Shipment: DELIVERY	Sample Condition Upon Receipt:	Date/Time: APR 29 2026	
Relinquished by: Pat 66	Date/Time: 4/29/26	Received by: REC'D	Date/Time: APR 29 2026
Relinquished by:	Date/Time:	Received by: EMSL-BOSTON	Date/Time: APR 29 2026

Controlled Document - Asbestos Bulk R7 9/14/2021

☒ AGREE TO ELECTRONIC SIGNATURE (By checking, I consent to signing this Chain of Custody document by electronic signature.)

EMSL Analytical, Inc.'s Laboratory Terms and Conditions are incorporated into this Chain of Custody by reference in their entirety. Submission of samples to EMSL Analytical, Inc. constitutes acceptance and acknowledgment of all terms and conditions by Customer.

* INSULATION HAS A HARD OUTER LAYER "GRAY" & A INNER INSULATION-WHITE (SOFT). REPORT RESULTS FOR EACH SEPERATE LAYER

Method of Shipment:		Sample Condition Upon Receipt:	
Relinquished by:	Date/Time:	Received by: <u>MM</u>	Date/Time
Relinquished by:	Date/Time:	Received by: <u>EMSL BOSTON</u>	Date/Time

Page 2 Of 2

APPENDIX – C:

ASBESTOS PROJECT CONSULTANT LICENSURE

Rhode Island Department of Health

Asbestos Program

Asbestos Inspector

PATRICK EMERY

Exp. Date: **4/30/2027**

License #: **AI00505**

Member of C.O.N.E.S.



Rhode Island Department of Health

Asbestos Program

Asbestos Management Planner

PATRICK EMERY

Exp. Date: **4/30/2027**

License #: **AMP00505**

Member of C.O.N.E.S.



Rhode Island Department of Health

Asbestos Program

Asbestos Project Designer

PATRICK EMERY

Exp. Date: **4/30/2027**

License #: **APD00505**

Member of C.O.N.E.S.



Rhode Island Department of Health

Asbestos Program

Asbestos Inspector

PATRICK EMERY

Exp. Date: **4/30/2027**

License #: **AI00505**

Member of C.O.N.E.S.



APPENDIX – D:

ASBESTOS TESTING LABORATORY LICENSURE



***State of Rhode Island
DEPARTMENT OF HEALTH
CENTER FOR HEALTHY HOMES & ENVIRONMENT – ASBESTOS PROGRAM***

ASBESTOS ANALYTICAL SERVICES CERTIFICATION

Pursuant to the Asbestos Abatement Act, Chapter 24.5 of Title 23 of the Rhode Island General Laws, and Regulation 216-RICR-50-15-1 – Asbestos Control, this license is hereby issued as designated below. This license is subject to all applicable rules, regulations, orders and notices of the Department of Health now or hereafter in effect and to any conditions delineated below.

**Certificate Holder: EMSL ANALYTICAL INC
Address: 5 CONSTITUTION WAY UNIT A
WOBURN MA 01801**

**Certification Number: PLM00139
Expiration Date: 04/30/2026
Type of Certification: Analytical Service - PLM**

Except as specifically provided otherwise in this Certificate, Certificate Holders shall conduct their program in accordance with statements, procedures and representations contained in their application, including any attachments. Regulation 216-RICR-50-15-1 - Asbestos Control shall govern unless the statements representations and procedures in the Certificate Holder's application and documentation are more restrictive than the Regulation.

STEVE GRISE
EMSL ANALYTICAL INC
5 CONSTITUTION WAY UNIT A
WOBURN MA 01801

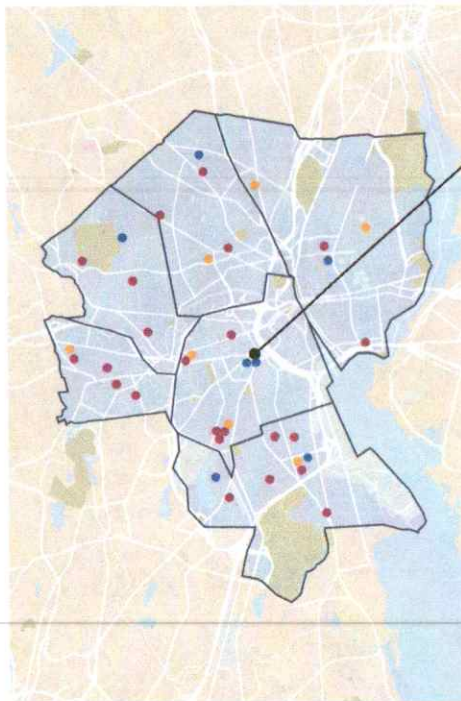
APPENDIX – E:

ASBESTOS PROJECT DRAWING(S)

CITY OF PROVIDENCE PROVIDENCE PUBLIC SCHOOL DEPARTMENT BOILER REPLACEMENT AT

HOPE HIGH SCHOOL
324 HOPE ST,
PROVIDENCE, RI 02906

PROJECT LOCATION MAPS



LOCATION MAP

PROJECT LOCATION
HOPE HIGH SCHOOL
324 HOPE ST,
PROVIDENCE, RI 02906

HOPE HIGH SCHOOL

SITE



AREA MAP



SITE MAP

SHEET INDEX	
SHEET NUMBER	SHEET TITLE
GENERAL	
G-000	COVER SHEET
MECHANICAL	
M-000	MECHANICAL COVER
M-101	MECHANICAL DEMOLITION FLOOR PLANS
M-101	MECHANICAL FLOOR PLANS



PROJECT
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DATE: 04/20/2024

CITY OF PROVIDENCE
PROVIDENCE PUBLIC SCHOOL DEPARTMENT
BOILER REPLACEMENT AT
HOPE HIGH SCHOOL
324 HOPE ST.
PROVIDENCE, RI 02906

REVISIONS		
NO.	DESCRIPTION	DATE

COVER SHEET

CODE INFORMATION

1. APPLICABLE CODES AND STANDARDS

2021 IBC BUILDING CODE (IBC 2021)
2021 IBC MECHANICAL CODE (IMC 2021)
2021 IBC PLUMBING CODE (IPC 2021)
2021 IBC ELECTRICAL CODE (IEC 2021)
2021 IBC ENERGY CONSERVATION CODE (IECC 2021)
2021 IBC SAFETY CODE (ISC 2021)

INTERNATIONAL BUILDING CODE (IBC) 2018
INTERNATIONAL ELECTRICAL CODE (IEC) 2018
INTERNATIONAL ENERGY CONSERVATION CODE (IECC) 2018
NATIONAL FIRE ALARM AND SIGNALING CODE (NFPA 72) 2018

NFPA 72 NATIONAL FIRE ALARM AND SIGNALING CODE, 2018
NFPA 13 STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS, 2016
NFPA 70 NATIONAL ELECTRICAL CODE (NEC), 2017

2011 ADA STANDARDS FOR ACCESSIBLE DESIGN (ADAAG)
2010 ADA STANDARDS FOR ACCESSIBLE DESIGN (ADAAG)

2. PROJECT DATA

OWNER: PROVIDENCE PUBLIC SCHOOL DEPARTMENT
LOCATION: HOPE HIGH SCHOOL
ADDRESS: 324 HOPE ST., PROVIDENCE, RI 02906
OCCUPANCY CLASSIFICATION: EDUCATION GROUP 1

CONSTRUCTION DOCUMENTS
APRIL 20, 2024

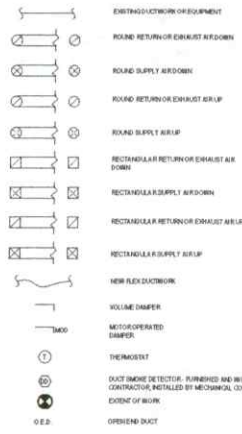
DRAWN: BMS
CHECKED: FRC
DATE: 2024

SHEET NO.
G-000

GENERAL HVAC NOTES

- ALL NEW SUPPLY AND RETURN AIR REGISTERS SHALL BE EQUIPPED WITH SUPPLY VOLUME DAMPERS.
- FIELD DAMPERS SHALL BE PLACED IN ALL DUCT OUTPENS AND INLET PENS OF ALL NEW RISE AND RETURN. SEE ARCHITECTURAL DRAWINGS FOR FIELD INSTRUCTIONS. PROVIDE ACCESS DOORS AS NOTED. FIELD DAMPERS AND INSTALLATION ARE TO BE IN ACCORDANCE WITH CURRENT EDITION OF THE INTERNATIONAL MECHANICAL CODE.
- DUCT BRACKET/STAYS ARE TO BE SUPPLIED AND INSTALLED WHEN SHOWN IN ACCORDANCE WITH NFPA 72. ACCESS SHALL BE PROVIDED TO MAKE REPAIRS TO THE DUCTWORK. ELECTRICAL CONTRACTORS SHALL MAKE APPROPRIATE AND REQUIRED ELECTRICAL CONNECTIONS TO THE DUCT BRACKET DETECTORS AND REQUIRED FOR PROPER OPERATION. DUCT BRACKET DETECTORS SHALL BE ENERGIZED BY THE AIR HANDLING UNIT OR EXHAUST FAN. THEY ARE ASSOCIATED WITH AND ACTIVATE A WIRELESS AND WIRELESS SENSING SYSTEM IN THE BUILDING. THE PROTECTIVE SYSTEM SET A COORDINATE ATTENDED LOCATION AND SYSTEM THAT IS ENERGIZED BY THE ALARM SHALL NOT RESTART EXCEPT BY MANUAL RESET ON THE UNIT. THROUGH THE BAG SYSTEM (BAG) DETECTOR SHALL COMPLY WITH THE CURRENT EDITION MECHANICAL CODE IN ENERGY.
- INSTALL ALL NEW DUCTWORK AND EQUIPMENT AS SPECIFIED UNLESS OTHERWISE NOTED. ALL NON-ALLOWABLE DUCTWORK SHALL BE METAL. DUCTWORK IS PROHIBITED.
- PROVIDE AND INSTALL ALL GRILLES AND METERS AS SHOWN AND SPECIFIED.
- METERS OF ALL COILS SHALL ENSURE THAT THE COILS ARE CLEAN AND FREE OF ANY RESIDUE FROM MANUFACTURING AND SHIPPING PROCESS. IF COILS ARE FOUND TO BE DIRTY OR HAVE IMPROPER MATERIALS PROVIDED TO THEM, THE CONTRACTOR WILL BE RESPONSIBLE FOR CLEANING OF THE COILS, AS WELL AS, CLEANING THE BUILDING FROM MOISTURE, COIL EMANATIONS, OR VAPORS.
- THE RETURN AIR UNITS SHOWN ON THE PLANS CORRESPOND TO THE OCCUPANCY MODE WITH MINIMUM OUTSIDE AIR FLOW. THE BALANCE REPORT IS TO BE SUBMITTED TO THE OWNER IN THE MONTH OF OPERATION.
- VAV EQUIPMENT SHALL BE PROVIDED WITH A MOTOR STATUE DUTY FOR THE OFF ON CONTROL OF THE MOTOR BY A REMOTE TERMINAL FOR AS CONTACT. THESE STATUES SHALL COMPLY WITH NATIONAL ELECTRIC CODE.
- ALL RETURN AIR UNITS ARE TO BE INSTALLED TO PROVIDE ADEQUATE, UNOBSTRUCTED, ACCESS AND CLEARANCE FOR AIR FILTER REPLACEMENT.
- ALL VAVS TO BE INSTALLED BY THE MECHANICAL CONTRACTOR (OR A MECHANICAL SUBCONTRACTOR) SHALL BE PROTECTED AGAINST OVERLOAD IN ACCORDANCE WITH NFPA 70, ARTICLE 410, SECTION 4.
- VENTS AND FLUING VENTS SHALL BE LOCATED A MINIMUM OF 10' FROM ANY AIR USE AIR CLOSING.
- MECHANICAL EQUIPMENT AND DEVICES SHALL OPERATE WITHOUT OBSTRUCTIBLE NOISE AND VIBRATION BEING TRANSMITTED TO OCCUPANT PORTIONS OF THE BUILDING OR PART OF THE BUILDING STRUCTURE BY APPROPRIATE SILENCERS, DUCTWORK, CONNECTIONS, OR COMPONENTS OF THE MECHANICAL SYSTEM. SECURE AND PROTECT ALL PIPING AND DUCTWORK. PROVIDE SILENCERS, CONNECTIONS, VIBRATION ISOLATORS, OR OTHER DEVICES WHERE INDICATED OR REQUIRED TO PREVENT THE TRANSMISSION OF NOISE AND VIBRATION TO THE BUILDING. SEE NFPA 70.
- THE CONTRACTOR SHALL PROVIDE ALL WALL AND ROOF PENETRATIONS REQUIRED FOR DUCTWORK. THE CONTRACTOR SHALL PATCH AND WIRE ALL PENETRATIONS LEFT BY DEMOLITION WORK AND BY CLEARANCE REQUIRED THAT REMAIN AFTER DUCT INSTALLATION.
- NOTIFY THE OWNER IN THE EVENT ANY EXISTING OR NEW MATERIALS, SUCH AS ROOFING, PORTS, LEAD, ETC., ARE ENCOUNTERED ON THE PROJECT. THE OWNER WILL ARRANGE WITH A QUALIFIED SPECIALIST FOR THE IDENTIFICATION, TESTING, REMOVAL, HANDLING AND PROTECTION AGAINST EXPOSURE TO ENVIRONMENTAL POLLUTION TO COMPLY WITH APPLICABLE REGULATIONS, LAWS AND ORDINANCES.
- ALL THERMISTATS SHOWN ON DISCONNECTS ARE TO BE LOCATED 4' IF MINIMUM AFF (TOP OF UNIT) UNLESS NOTED OTHERWISE. CONTRACTOR IS RESPONSIBLE FOR LOCATING A MINIMUM ALL THERMISTATS. PLUMBING RATES CHARGES MUST BE MAINTAINED.
- THE CONTRACTOR SHALL PROVIDE VIBRATION ISOLATORS AND TRUCKS TO BE USED WITH VAVS FOR ALL DUCTWORK CHANGED IN DIRECTION. IF TUBING VAVS DUCT CONSTRUCTION FINDS ARCHITECTURAL AND PLUMBING ISOLATION TRUCKS IN EXISTING VAVS AND MISSED (BURNS) ARE NOT PERMITTED.
- ALL QUANTITIES TO BE REMOVED BY CONTRACTOR.
- PROVIDE TO QUANTITIES OF THE CONTRACTOR SHALL VISIT THE SITE AND BE THOROUGHLY FAMILIAR WITH THE EXISTING CONDITIONS AND PROVIDE CONTRACTOR. CONTRACTOR SHALL INCLUDE ALL MATERIALS, LABOR AND ALL INCIDENTALS FOR A COMPLETE INSTALLATION. WHERE SPECIFICALLY INDICATED OR NOT, ALL EXISTING, DISCONNECTED AND MISSED ITEMS SHALL BE REMOVED TO THE ATTENTION OF THE ENGINEER DURING THE BEGINNING PROCESS BY THE CONTRACTOR. THESE ITEMS SHALL BE INCLUDED IN THE BIDDING PRICE. NO EXTRA COST WILL BE ALLOWED FOR ANY DISCONNECT WHICH CAN HAVE BEEN REMOVED AT THE SITE BY THE CONTRACTOR.
- INSTALL ALL EXISTING AND MATERIALS REMOVED BY CONTRACTOR WITH THE DEMOLITION.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL ELECTRICAL COSTS INCURRED AS A RESULT OF SUBSTITUTIONS OR DEVIATIONS FROM THE BASIS OF DESIGN SHOWN ON THESE DRAWINGS.
- CONTRACTOR IS RESPONSIBLE FOR PROVIDING TRANSITIONS AND FLEXIBLE CONNECTIONS BETWEEN ALL UNITS AND SUPPLY, RETURN, OR OUTSIDE AIR DUCTS.
- CONTRACTOR SHALL VERIFY THAT ALL EXISTING EQUIPMENT TO BE REMOVED IS NON-HAZARDOUS.
- INSTALL REMOTE EMERGENCY SHUT-OFF SWITCH FOR BOILERS ADJACENT TO EACH BOILER ROOM DOOR.

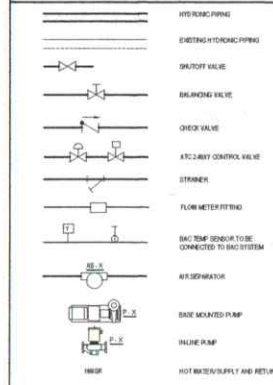
HVAC LEGEND



GENERAL PIPING NOTES

- INSTALL ALL GRADES AND METERS AS SHOWN ON DISCONNECTS AND AS SPECIFIED.
- INSTALL ALL PIPING AS SPECIFIED.
- UNLESS OTHERWISE SHOWN OR SPECIFIED, ALL CONDENSATE PIPING FROM ALL HVAC UNITS SHALL BE TRAPPED AND PIPED THROUGH THE ROOF AND DRAINAGE THROUGH THE EXTERIOR WALL OF THE BUILDING. EXISTING NOT TO EXCEED 1/4" ABOVE GRADE. THE ROOF SHALL MATCH THE SIZE OF DRAIN CONNECTION AT A MINIMUM 1/4" DRAIN SHALL NOT BE CONNECTED TO THE EXTERIOR UNLESS APPROVED BY ARCHITECT/ENGINEER. UNLESS OTHERWISE SHOWN OR SPECIFIED, CONDENSATE PIPING SHALL NOT BE EXPOSED.
- ALL FLOOR PENETRATIONS SHALL BE SLEAVED AND SEALED AS SPECIFIED.
- THE CONTRACTOR IS RESPONSIBLE FOR ALL NEW PIPING PENETRATIONS IN WALLS AND FLOORS. THE PIPING AND NOT BE SPECIFICALLY NOTED WHERE PIPING PENETRATES WALLS AND FLOORS. IT IS THE CONTRACTOR'S RESPONSIBILITY PRIOR TO THE TO START THE PIPING AND IF NECESSARY EVALUATE THE SITE, TO ACCOUNT FOR ALL PIPING PENETRATIONS OF EXISTING STRUCTURE.
- THE CONTRACTOR SHALL PROVIDE NEW SUPPORTS OF COMPARABLE MATERIAL FOR ALL NEW PIPING.
- PROVIDE HOT WATER SUPPLY AND RETURN SYSTEM TO THE EXTENT SHOWN ON PLANS AND NOTED AND NOTED.
- ALL CONTROL VALVES TO BE INSTALLED OVER OR UNDER PIPING UNITS.
- ALL PIPING SHALL BE SIZED AS CALLED OUT ON DISCONNECTS, AND MAY NOT BE DISCONNECTED BETWEEN CONTROL VALVE AND COIL.
- ALL GAS PIPING IS TO BE INSTALLED PER NFPA 54, AND ALL LOCAL AND NATIONAL CODES.
- BEFORE ANY SYSTEM OF GAS PIPING IS FINALLY PUT IN SERVICE, IT SHALL BE TESTED TO ENSURE THAT IT IS LEAK TIGHT. TO TEST FOR TIGHTNESS, THE PIPING SHALL BE FILLED WITH AIR OR INERT GAS, BUT NOT WITH ANY OTHER GAS OR LIQUID. TESTING, INSPECTION AND PURGING OF GAS PIPING SYSTEM SHALL COMPLY WITH NFPA 54.
- GAS PIPING LOCATED AT GRADE LEVEL IS TO BE IN PIPING SADDLES, AND NOT TO BE IN CONTACT WITH CONCRETE OR GRADE.
- PROVIDE REDUCERS, FITTINGS, AND CONNECTIONS TO EQUIPMENT FOR A COMPLETE GAS PIPING INSTALLATION. FITTINGS, JOINTS AND CONNECTIONS SHALL BE OF AN APPROVED TYPE FOR GAS PIPING SYSTEM.
- ALL NEW EXPOSED PIPING LOCATED AREAS SHALL BE PROVIDED IN ACCORDANCE WITH NFPA 54. EXISTING PIPING OF 1/4" OR SMALLER, EXPOSED SHALL BE BY PLACED IN A RIGID, RIGID, OR TUBING COMPANY. LINTS SHALL BE REMOVED. COILS SHALL BE REMOVED BY ARCHITECT.
- PROVIDE AIR VENTS AT ALL HIGH POINTS IN SYSTEM. PROVIDE DRAIN VALVES TO ALL LOW POINTS IN SYSTEM.
- COORDINATE INSTALLATION OF ALL NEW PIPING WITH EXISTING FIRE PROTECTION, CONDUIT, DUCT WORK, AND STRUCTURE.

PIPING LEGEND

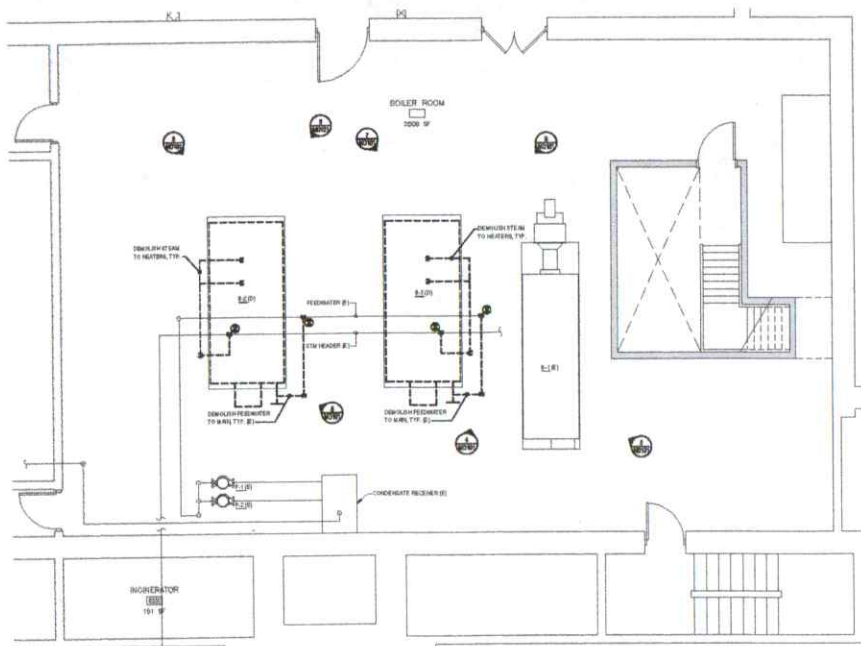


MECHANICAL DEMO NOTES & LEGEND

- ALL DEMOLISHED MECHANICAL UNITS ARE TO BE REMOVED AND TAKEN OFF SITE AND BECOME PROPERTY OF THE CONTRACTOR UNLESS NOTED OTHERWISE.
- ALL DEMOLISHED DUCTWORK, OUTPENS, REGISTERS, GRILLES, SILENCERS, AND TUBING ARE TO BE REMOVED AND TAKEN OFF SITE AND BECOME PROPERTY OF THE CONTRACTOR UNLESS NOTED OTHERWISE.
- ALL DEMOLISHED HEATING SUPPLY AND RETURN PIPING, VALVES, FITTINGS, ISOLATION AND HANGERS ARE TO BE REMOVED AND TAKEN OFF SITE AND BECOME PROPERTY OF THE CONTRACTOR UNLESS NOTED OTHERWISE. PIPING THAT IS TO BE REMOVED IN PLACE SHALL BE CUT AND CAPPED BELOW FLOOR OR BOTTOM TRENCH, AND THE OPENING SHALL BE NON-CONCRETE.
- EXPOSED TEMPORARY COVER AND CAPS FOR TEMPORARILY EXPOSED EXISTING PIPING, EQUIPMENT AND WALL OPENINGS. TEMPORARY COVER AND CAPS SHALL BE SUFFICIENT TO PREVENT WEATHER AND CONTAMINATION. DEBRIS WILL NOT ENTER ANY SYSTEM OR WALL, ROOF OR FLOOR OPENING.
- REMOVAL AND DISPOSAL OF GAS PIPING AND GAS FIRED APPLIANCES SHALL BE IN ACCORDANCE WITH NFPA 54.
- DEMOLISH ALL REGISTERS AND AIR VENTS AT DEMOLISHED GAS FIRED EQUIPMENT.
- EXISTING GAS PIPING SHALL FIRST BE DISCONNECTED FROM ALL SOURCES OF GAS AND THEN THOROUGHLY PURGED WITH AIR, WATER, OR INERT GAS BEFORE CUTTING OR REMOVAL IS DONE.
- WHEN GAS PIPING IS OPENED, THE SECTION TO BE REMOVED OR SHALL BE TURNED OFF FROM THE GAS SUPPLY AT THE NEAREST CONVENIENT POINT, AND THE LINE PRESSURE VENTED TO THE OUTDOORS, OR TO ANOTHER AREA OF SUFFICIENT SIZE TO PREVENT ACCUMULATION OF A HAZARDOUS MIXTURE. IF THIS SECTION EXCEEDS IN FEET OF 10 FEET, THE REMAINING GAS PIPING SHALL BE DISPLACED WITH INERT GAS.
- SEE ELECTRICAL PLANS FOR INFORMATION CONCERNING REMOVAL OF DEMO EQUIPMENT.
- REFER TO ARCHITECTURAL DRAWINGS FOR DEMOLITION PLAN.
- THE CONTRACTOR SHALL PROVIDE PROTECTION FOR ALL ROOF TOP AND WALL OPENINGS CREATED BY DEMOLITION DURING THE DEMOLITION WEATHER CONDITIONS.
- ALL HAZARDOUS CONTAMINATING MATERIALS WILL BE REMOVED BY OTHERS PRIOR TO DEMOLITION.
- THE INFORMATION SHOWN ON THESE DRAWINGS WAS COMPILED FROM EXISTING DRAWINGS AND SITE VISITS. HOWEVER, THE INFORMATION IS NOT A GUARANTEE OF THE ACTUAL INSTALLATION. CONTRACTORS RESPONSIBLE FOR FIELD VERIFICATION FOR DEMOLITION WORK.



NO.	DESCRIPTION	DATE



1 BOILER ROOM STEAM + FEEDWATER DEMOLITION PLAN
SCALE: 1/8" = 1'-0"



3 REFERENCE PHOTO
MD101



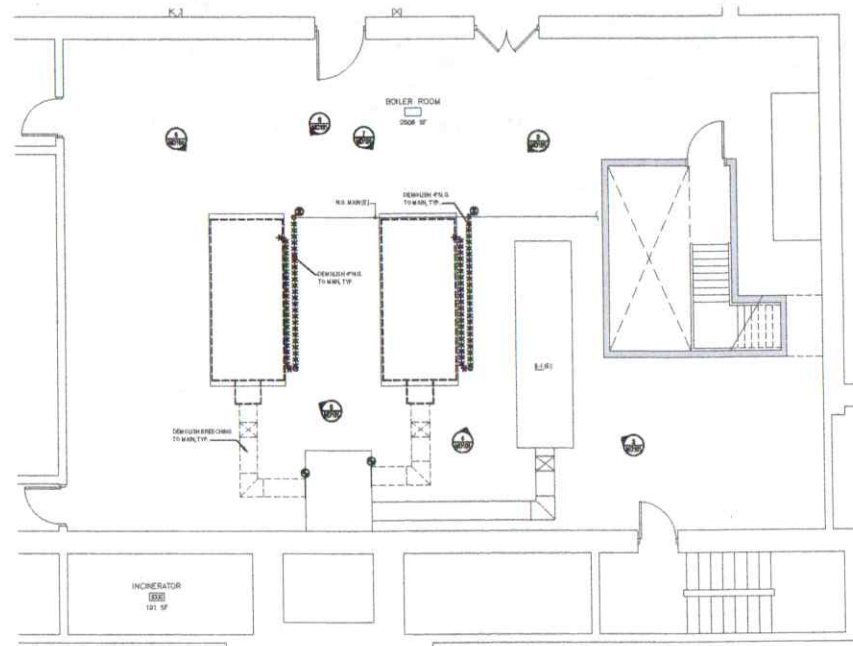
7 REFERENCE PHOTO
MD101



4 REFERENCE PHOTO
MD101



6 REFERENCE PHOTO
MD101



2 BOILER ROOM N.G. AND BREECHING DEMOLITION PLAN
SCALE: 1/8" = 1'-0"



5 REFERENCE PHOTO
MD101



9 REFERENCE PHOTO
MD101

REVISIONS

NO.	DESCRIPTION	DATE

SHEET TITLE
MECHANICAL
DEMOLITION
PLANS

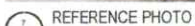
CONSTRUCTION
DOCUMENTS

APRIL 30, 2024
BMS QWY
PROJ. NO. 26026

MD101
BMS AND ENGINEERING



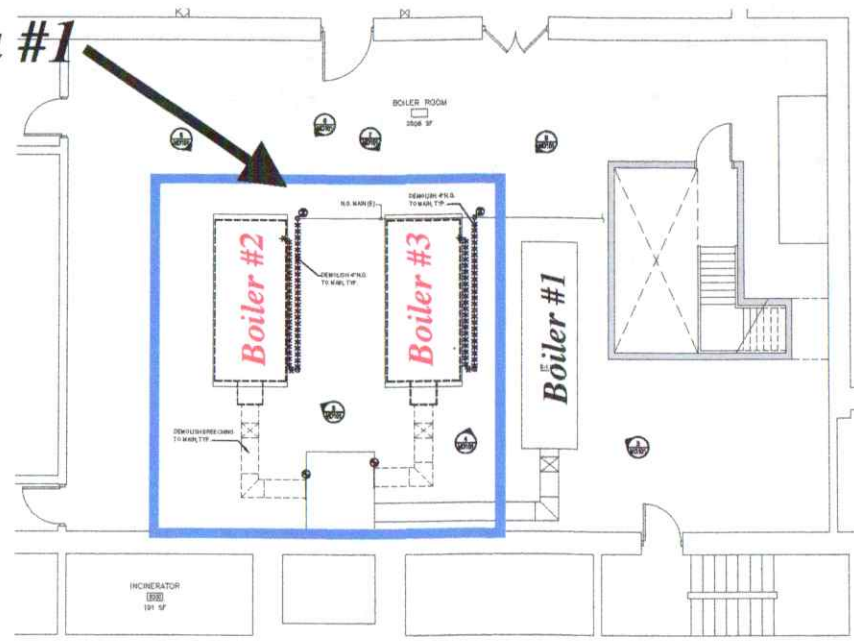
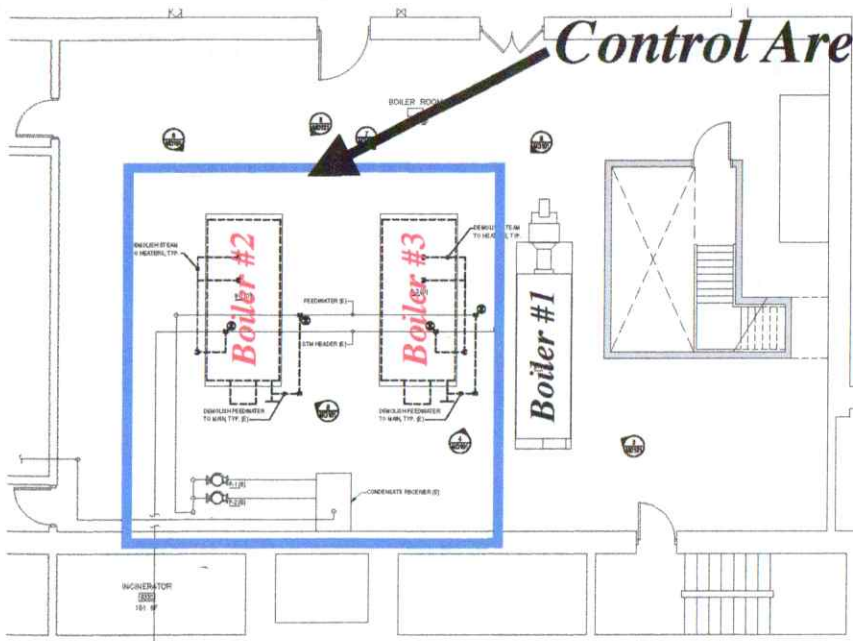
NO. 101 SCALE: 1/8" = 1'-0"



MD101 SCALE: 1/4" = 1'-0"



EEA



1 BOILER ROOM STEAM + FEEDWATER DEMOLITION PLAN
SCALE: 1/8" = 1'-0"

2 BOILER ROOM N.G. AND BREECHING DEMOLITION PLAN
SCALE: 1/8" = 1'-0"



3 REFERENCE PHOTO



4 REFERENCE PHOTO



5 REFERENCE PHOTO



6 REFERENCE PHOTO



7 REFERENCE PHOTO



8 REFERENCE PHOTO



9 REFERENCE PHOTO

EEA2

STUDIO JAED

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DATE: 04/20/2024

CITY OF PROVIDENCE
PROVIDENCE PUBLIC SCHOOL DEPARTMENT
BOILER REPLACEMENT AT
HOPE HIGH SCHOOL
324 HOPE ST.
PROVIDENCE, RHODE ISLAND

REVISIONS		
NO.	DESCRIPTION	DATE

SHEET TITLE
MECHANICAL
DEMOLITION
PLANS

CONSTRUCTION
DOCUMENTS
APRIL 20, 2024

DESIGN: BMS
CHECK: FRC
PROJECT NO: 26026

SHEET NO:
MD101
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SECTION 00 73 00

MINORITY BUSINESS ENTERPRISE REQUIREMENTS AND WOMEN BUSINESS ENTERPRISE PARTICIPATION REQUIREMENTS / PREVAILING WAGES

Under Rhode Island General Laws §37-14.1, Minority business enterprises shall be included in all procurements and construction projects under this chapter and shall be awarded a minimum of fifteen percent (15%) of the dollar value of the entire procurement or project. Of that fifteen percent (15%), minority business enterprises owned and controlled by a minority owner, shall be awarded a minimum of seven and one-half percent (7.5%), and minority business enterprises owned and controlled by a woman shall be awarded a minimum of seven and one-half percent (7.5%).

The successful bidder shall report on applicable participation within ten (10) days after receipt of Notice of Award.

Prevailing Wages: Any construction over \$ 1,000.00 is required by state law to pay state prevailing wage. A current copy of the most recent Prevailing Wage scale is obtained from the State of Rhode Island, Department of Labor and Training, Division of Professional Regulation or at the Davis-Bacon Wage Determination link. It is also by request for each construction bid document. All vendors supplying construction services must submit certified payrolls from any contractor and sub-contractor who work on the project. No payments are made until all payrolls are up to date, and correct. Click here for Prevailing Wage Tables. <http://www.dlt.ri.gov/pw>.

END OF SECTION

SECTION 01 10 00 SUMMARY

PART 1 GENERAL

1.01 PROJECT

- A. Project Name: Hope HS Boiler Demolition.
- B. Project Locations: Hope High School
- C. Owner's Name: Providence Public School Department.
- D. Architect / Engineer's Name: Studio JAED (SJ)
- E. The Project consists of:
 - 1. Demolition of existing HVAC equipment as described in the documents
 - 2. Abatement as follows:
 - a. In general - the plan is to have at the demolition / asbestos contractor in charge of the boiler demolition construct a negative pressurized containment around the boilers (Control Area #1) + boiler breechings (Control Area #1A).
 - b. Conduct the removal of the boiler breeching as a general work operation (non asbestos) while keeping the breeching insulation wet and contained within a airtight NPE to prevent dust from traveling throughout the boiler room, adjacent interior areas of the school or the exterior ambient air. Waste would be required to be wetted and containerized into bags so visible emission from the debris is not escaping to the ambient air of the boiler room, adjacent interior areas of the school or the exterior. Waste should be continued to be containerized while on site or when transported off site in enclosed vehicles. **PPE would need to be compliant with the OSHA respirable dust standard (Adequate respiratory protection and compliance with OSHA 29CFR134, suits and body washing facilities).** During work the area outside the containment / NPE will be monitored for visible emissions and respirable dust levels. If levels are reported to be above the OSHA PEL - the contractor in charge of the work will be required to institute additional engineering controls to reduce elevated ambient dust levels.
 - c. After the existing boiler breeching insulation is removed (as required for the limits of work for this project)- the contractor could use the same NPE for the demolition of the boilers. The boiler demolition is planned as an asbestos work operation (Asbestos Control Area #1) and subject to work requirements of sections 1.14.2 / 1.14.3 of the current RIDOH Asbestos Control Regulation and all requirements of AHERA. (Removal as a asbestos response action, NPE, licensed asbestos workers, start work notifications and clearance air testing)
 - d. If the work is done in reverse the asbestos contractor would need to conduct the demo of the boilers as a asbestos response action, then the NPE would be clearance air sampled and the contractor would be required to erect a NPE for the removal of the boiler breeching. Breeching removal is required to be done wet, with no visible emissions and the handling of the waste would be required to be handled and disposed with no emissions to the Boiler Room, other interior areas of the school or the exterior ambient air as previously stated.

1.02 CONTRACT DESCRIPTION

- A. Contract Type: A single prime contract based on a Stipulated Price as described in Division 00.

1.03 DESCRIPTION OF ALTERATIONS WORK

- A. Scope of demolition and removal work is shown on drawings.

1.04 OWNER OCCUPANCY

- A. Owner intends to continue to occupy adjacent portions of the existing building during the entire construction period.
- B. Cooperate with Owner to minimize conflict and to facilitate Owner's operations.
- C. Schedule the Work to accommodate Owner's occupancy.

1.05 CONTRACTOR USE OF SITE AND PREMISES

- A. Construction Operations: Limited to the building premises.
- B. Provide access to and from site as required by law and by Owner:
 - 1. Emergency Building Exits During Construction: Keep all exits required by code open during construction period; provide temporary exit signs if exit routes are temporarily altered.
 - 2. Do not obstruct roadways, sidewalks, or other public ways without permit.
 - 3. Adhere to owner's guidelines regarding entrance and egress to the site as identified during the pre-bid meeting.
- C. Utility Outages and Shutdown:
 - 1. Prevent accidental disruption of utility services to other facilities.
 - 2. Any shutdown of a critical building system (including but not limited to HVAC, sprinkler, fire alarm, water, electric) is to occur during a coordinated, off-hours shutdown (either daytime or overnight hours and either weekdays or weekends at owner's option). Shutdown work is to proceed continuously until the critical building system is restored.
 - 3. Coordinate any interruption and/or shutdown of utilities with the owner at least 14 days in advance of the anticipated interruption and/or shutdown. Limit any interruptions/shutdowns to the absolute minimum amount of time.
 - 4. The owner reserves the right to reschedule construction shutdowns with minimal warning to the contractor as required to respond to emergencies.

1.06 GENERAL STANDARDS

- A. General Project Requirements
 - 1. Manufacturers of all coils shall ensure that the coils are clean and free of any residue from the manufacturing and shipping process. If coils are found to be dirty or smoke when hot water is provided to them, the contractor will be responsible for cleaning of the coils, as well as, cleaning the building from smoke, coil emanations, or vapors.
 - 2. The return airflows indicated on the plans correspond to the occupied mode with minimum outside air being supplied. The balance report is to indicate these airflows in this mode of operation.
 - 3. HVAC equipment shall be furnished with a motor starter suitable for the off-on control of the motor by a remote thermostat or bas contact. These starters shall comply with national electric code.
 - 4. All air handling units are to be installed to provide adequate, unobstructed, access and clearance for air filter replacement.
 - 5. Notify the owner in the event any existing hazardous materials, such as asbestos, pcb's, lead, etc., are encountered on the project. The owner will arrange with a qualified specialist for the identification, testing, removal, handling and protection against exposure or environmental pollution, to comply with applicable regulations, laws and ordinances.
 - 6. All thermostats shown on hvac plans are to be mounted 4'-0" maximum aff (top of unit), unless noted otherwise. Contractor is responsible for mounting & wiring all thermostats. Plenum rated cables must be used.
 - 7. Prior to submitting bid, the contractor shall visit the site and be thoroughly familiar with the existing conditions and proposed construction. Contractor shall include in their bid all

materials, labor and all incidentals for a complete installation whether specifically indicated or not. All errors, discrepancies and missed items shall be brought to the attention of the engineer during the bidding process by the contractor. These items shall be included in the bid price. No extra cost will be allowed for any discrepancy which could have been noticed at the site by the contractor.

8. The contractor shall be responsible for all additional costs incurred as a result of substitutions or deviations from the basis of design shown on these drawings.
9. Contractor is responsible for providing transitions and flexible connections between all units and supply, return, or outside air ducts.
10. All roof and wall penetrations shall be sleeved and sealed for assembly rating.
11. The contractor is responsible for all new piping penetrations in walls and floors. The piping plans do not specifically note where piping penetrates walls and floors. It is the contractor's responsibility prior to bid to study the plans and if necessary evaluate the site, to account for all piping penetrations of the existing structure.
12. All gas piping is to be installed per NFPA 54, and all local and national codes.
13. Before any system of gas piping is finally put in service, it shall be tested to ensure that it is gas tight. To test for tightness, the piping shall be filled with air or an inert gas, but not with any other gas or liquid. Testing, inspection and purging of gas piping systems shall comply with NFPA 54.
14. Provide splash blocks for all condensate drains.
15. Notify the owner in the event any existing hazardous materials, such as asbestos, pcb's, lead, etc., are encountered on the project. The owner will arrange with a qualified specialist for the identification, testing, removal, handling and protection against exposure or environmental pollution, to comply with applicable regulations, laws and ordinances.
16. Prior to submitting bid, the contractor shall visit the site and be thoroughly familiar with the existing conditions and proposed construction. Contractor shall include in their bid all material, labor, and all incidentals for a complete installation whether specifically indicated or not. All errors, discrepancies and missed items shall be brought to the attention of the engineer during the bidding process by the contractor. These items shall be included in the bid price. No extra cost will be allowed for any discrepancy which could have been noticed at the site visit by the contractor.
17. Perform work as required by applicable codes, regulations, and laws of local, state, and federal governments and other authorities with lawful jurisdiction. All work shall be in accordance with the latest edition of the national electric code.
18. Material and equipment shall be UL, NEMA, ANSI, IEEE, ADA & CMB approved for intended purpose. Material and installation shall meet requirements of national and local electrical code.
19. Provide all labor, materials, tools, equipment, coordination, additional design and all incidentals necessary to provide a complete and operable system as detailed on plans to the satisfaction of the engineer and the owner. Coordinate all work with the engineer before the start of work.
20. Prior to submitting bid, the contractor shall visit the site and be thoroughly familiar with the existing conditions and proposed construction. Contractor shall include in their bid all material, labor, and all incidentals for a complete installation whether specifically indicated or not. All errors, discrepancies and missed items shall be brought to the attention of the engineer during the bidding process by the contractor. These items shall be included in the bid price. No extra cost will be allowed for any discrepancy which could have been noticed at the site visit by the contractor.
21. Perform work as required by applicable codes, regulations, and laws of local, state, and federal governments and other authorities with lawful jurisdiction. All work shall be in accordance with the latest edition of the national electric code.

22. Material and equipment shall be ul, nema, ansi, ieee, ada & cmb approved for intended purpose. Material and installation shall meet requirements of national and local electrical code.
23. The contractor shall be responsible for all additional costs incurred as a result of substitutions or deviations from the basis of design shown on these drawings.
24. Give notices, file plans, obtain permits, and licenses, pay fees and back charges, and obtain necessary approvals from authorities that have jurisdiction.
25. Maintain record drawings on site. Record set must be complete and current and available for inspection when requisitions for payment are submitted.
26. Guarantee work in writing per specifications, repair or replace defective materials or installation at no cost to owner during the guarantee period. Correct damage caused in making necessary repairs and replacements under guarantee at no cost to owner. Submit guarantee to owner before final payment.
27. Coordinate all electrical items with existing field conditions. Locations shown are approximate and may require minor adjustment in the field to satisfy the design intent.
28. Damage to existing facilities and equipment shall be repaired or replaced immediately by the contractor at no additional expense to the owner.
29. The locations on these plans are approximate and require coordination with all other trades and verification of existing conditions. The contractor is responsible for field verification of all existing associated conditions. Contractor is responsible for obtaining all other trade's drawings and specifications and coordinating with all other trades during bidding and construction.
30. Contractor shall be responsible for maintaining continuity of all power, control, fire alarm, security systems, and communications functions to all areas affected by demolition and/or new construction.
31. Repair and patch any disturbed areas to match adjacent construction.
32. Disconnect and make safe any equipment to be removed by others. Coordinate removal of equipment with other trades prior to demolition.
33. In any area requiring the performance of any trade's work, this contractor shall carefully remove and store any or all electrical items in path of work, reinstalling, and reconnecting same as required, in accordance with the plans and/or as directed after completion of other trade's work in that area.
34. Prior to the start of demolition, contractor shall field verify all branch circuits and maintain those circuits that extend outside the scope of work.
35. After renovating existing electrical work, the contractor shall ensure that all remaining and new equipment will operate properly, including but not limited to backfeeding of existing power and lighting circuits. Refer to single line diagram.
36. All electrical work indicated to remain shall be suitably protected to prevent any damage.
37. Where electrical systems pass through renovated areas to serve other portions of the premises, systems shall be suitably protected to prevent damage or relocated and the systems restored to normal operation. Any outages in systems shall be coordinated with owner. Restore power to existing to remain equipment if interrupted by demolished circuits in the area.
38. Contractor shall submit for review, shop drawings for all equipment and materials used on the project. Submittals shall be reviewed by the architect before purchase of materials.
39. All wiring shall be copper, 600v, 75°/90° rated, flame-retardant, heat and moisture resistant.
40. Permanently label all new electrical equipment, including but not limited to, device designation and supply circuit designation. Update or replace panel directories to include new circuit information resulting from this project.

41. Provide temporary power and lighting for all trades as required to complete the project. All temporary and interim equipment shall be installed in accordance with all applicable codes and standards including, but not limited to NFPA 110 and NFPA 70.
42. Refer to specifications for additional information that is not shown on the drawings.
43. Openings in existing concrete walls and floors required for conduit installation shall be core drilled. Maximum core drill size shall be 5" in diameter. Core drill locations shall be spaced a minimum of 6" from each other measured from the outside edge of the core drill. All core drill openings shall be properly sealed according to their location and application.
44. All outages shall be kept to a minimum. All work that requires a sustained equipment outage shall be performed continuously around the clock until work is completed unless noted otherwise. Coordinate outages with owner representative.
45. Provide for each branch circuit and feeder circuit a dedicated equipment ground wire. For single phase branch circuits of 120 v/1ph or 277v/1 phase, provide dedicated hot, dedicated neutral and dedicated equipment ground wires. Sharing of neutral or equipment ground wires is not permitted. Wiring to all HVAC equipment or other trade equipment shall be in conduit. All equipment and feeder wiring in boiler room/electrical room shall be in rigid conduit. Use of mc cable is limited to branch circuit wiring above recessed ceiling or concealed in wall. Wiring to outlets on table shall be provide in either EMT conduit or flexible metal conduit.
46. Provide identification labels for all branch circuits and feeders circuits at junction boxes, panelboards, troughs, and splice boxes.
47. Provide unspliced feeders from panelboard or switchboard to all equipment. Splicing is permitted for single phase circuits for lighting and outlets only.
48. All wiring devices located in the basement are to be surface mounted with circuit wiring routed in surface mounted conduit per specifications. All other wiring devices shall be recessed unless noted otherwise.

B. Electrical Project Requirements

1. Notify the owner in the event any existing hazardous materials, such as asbestos, pcb's, lead, etc., are encountered on the project. The owner will arrange with a qualified specialist for the identification, testing, removal, handling and protection against exposure or environmental pollution, to comply with applicable regulations, laws and ordinances.
2. Prior to submitting bid, the contractor shall visit the site and be thoroughly familiar with the existing conditions and proposed construction. Contractor shall include in their bid all material, labor, and all incidentals for a complete installation whether specifically indicated or not. All errors, discrepancies and missed items shall be brought to the attention of the engineer during the bidding process by the contractor. These items shall be included in the bid price. No extra cost will be allowed for any discrepancy which could have been noticed at the site visit by the contractor.
3. Perform work as required by applicable codes, regulations, and laws of local, state, and federal governments and other authorities with lawful jurisdiction. All work shall be in accordance with the latest edition of the national electric code.
4. Material and equipment shall be UL, NEMA, ANSI, IEEE, ADA & CMB approved for intended purpose. Material and installation shall meet requirements of national and local electrical code.
5. Provide all labor, materials, tools, equipment, coordination, additional design and all incidentals necessary to provide a complete and operable system as detailed on plans to the satisfaction of the engineer and the owner. Coordinate all work with the engineer before the start of work.
6. Prior to submitting bid, the contractor shall visit the site and be thoroughly familiar with the existing conditions and proposed construction. Contractor shall include in their bid all material, labor, and all incidentals for a complete installation whether specifically indicated

or not. All errors, discrepancies and missed items shall be brought to the attention of the engineer during the bidding process by the contractor. These items shall be included in the bid price. No extra cost will be allowed for any discrepancy which could have been noticed at the site visit by the contractor.

7. Perform work as required by applicable codes, regulations, and laws of local, state, and federal governments and other authorities with lawful jurisdiction. All work shall be in accordance with the latest edition of the national electric code.
8. Material and equipment shall be ul, nema, ansi, ieee, ada & cmb approved for intended purpose. Material and installation shall meet requirements of national and local electrical code.
9. The contractor shall be responsible for all additional costs incurred as a result of substitutions or deviations from the basis of design shown on these drawings.
10. Give notices, file plans, obtain permits, and licenses, pay fees and back charges, and obtain necessary approvals from authorities that have jurisdiction.
11. Maintain record drawings on site. Record set must be complete and current and available for inspection when requisitions for payment are submitted.
12. Guarantee work in writing per specifications, repair or replace defective materials or installation at no cost to owner during the guarantee period. Correct damage caused in making necessary repairs and replacements under guarantee at no cost to owner. Submit guarantee to owner before final payment.
13. Coordinate all electrical items with existing field conditions. Locations shown are approximate and may require minor adjustment in the field to satisfy the design intent.
14. Damage to existing facilities and equipment shall be repaired or replaced immediately by the contractor at no additional expense to the owner.
15. The locations on these plans are approximate and require coordination with all other trades and verification of existing conditions. The contractor is responsible for field verification of all existing associated conditions. Contractor is responsible for obtaining all other trade's drawings and specifications and coordinating with all other trades during bidding and construction.
16. Contractor shall be responsible for maintaining continuity of all power, control, fire alarm, security systems, and communications functions to all areas affected by demolition and/or new construction.
17. Repair and patch any disturbed areas to match adjacent construction.
18. Disconnect and make safe any equipment to be removed by others. Coordinate removal of equipment with other trades prior to demolition.
19. In any area requiring the performance of any trade's work, this contractor shall carefully remove and store any or all electrical items in path of work, reinstalling, and reconnecting same as required, in accordance with the plans and/or as directed after completion of other trade's work in that area.
20. Prior to the start of demolition, contractor shall field verify all branch circuits and maintain those circuits that extend outside the scope of work.
21. After renovating existing electrical work, the contractor shall ensure that all remaining and new equipment will operate properly, including but not limited to backfeeding of existing power and lighting circuits. Refer to single line diagram.
22. All electrical work indicated to remain shall be suitably protected to prevent any damage.
23. Where electrical systems pass through renovated areas to serve other portions of the premises, systems shall be suitably protected to prevent damage or relocated and the systems restored to normal operation. Any outages in systems shall be coordinated with owner. Restore power to existing to remain equipment if interrupted by demolished circuits in the area.

24. Contractor shall submit for review, shop drawings for all equipment and materials used on the project. Submittals shall be reviewed by the architect before purchase of materials.
25. All wiring shall be copper, 600v, 75°/90° rated, flame-retardent, heat and moisture resistant.
26. Permanently label all new electrical equipment, including but not limited to, device designation and supply circuit designation. Update or replace panel directories to include new circuit information resulting from this project.
27. Provide temporary power and lighting for all trades as required to complete the project. All temporary and interim equipment shall be installed in accordance with all applicable codes and standards including, but not limited to NFPA 110 and NFPA 70.
28. Refer to specifications for additional information that is not shown on the drawings.
29. Openings in existing concrete walls and floors required for conduit installation shall be core drilled. Maximum core drill size shall be 5" in diameter. Core drill locations shall be spaced a minimum of 6" from each other measured from the outside edge of the core drill. All core drill openings shall be properly sealed according to their location and application.
30. All outages shall be kept to a minimum. All work that requires a sustained equipment outage shall be performed continuously around the clock until work is completed unless noted otherwise. Coordinate outages with owner representative.
31. Provide for each branch circuit and feeder circuit a dedicated equipment ground wire. For single phase branch circuits of 120 v/1ph or 277v/1 phase, provide dedicated hot, dedicated neutral and dedicated equipment ground wires. Sharing of neutral or equipment ground wires is not permitted. Wiring to all HVAC equipment or other trade equipment shall be in conduit. All equipment and feeder wiring in boiler room/electrical room shall be in rigid conduit. Use of mc cable is limited to branch circuit wiring above recessed ceiling or concealed in wall. Wiring to outlets on table shall be provide in either EMT conduit or flexible metal conduit.
32. Provide identification labels for all branch circuits and feeders circuits at junction boxes, panelboards, troughs, and splice boxes.
33. Provide unspliced feeders from panelboard or switchboard to all equipment. Splicing is permitted for single phase circuits for lighting and outlets only.
34. All wiring devices located in the basement are to be surface mounted with circuit wiring routed in surface mounted conduit per specifications. All other wiring devices shall be recessed unless noted otherwise.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 20 00
PRICE AND PAYMENT PROCEDURES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Procedures for preparation and submittal of applications for progress payments.
- B. Documentation of changes in Contract Price and Contract Time.
- C. Change procedures.
- D. Correlation of Contractor submittals based on changes.
- E. Procedures for preparation and submittal of application for final payment.

1.02 SCHEDULE OF VALUES

- A. Use Schedule of Values Form: AIA G703, edition stipulated in the Agreement.
- B. Forms filled out by hand will not be accepted.
- C. Submit Schedule of Values in duplicate within 15 days after date of Owner-Contractor Agreement.
- D. Format: Utilize the Table of Contents of this Project Manual. Identify each line item with number and title of the specification section. Identify site mobilization.
- E. Revise schedule to list approved Change Orders, with each Application For Payment.

1.03 APPLICATIONS FOR PROGRESS PAYMENTS

- A. Payment Period: Submit at intervals stipulated in the Agreement.
- B. Use Form AIA G702 and Form AIA G703, edition stipulated in the Agreement.
- C. Electronic media printout including equivalent information will be considered in lieu of standard form specified; submit sample to Architect for approval.
- D. Forms filled out by hand will not be accepted.
- E. For each item, provide a column for listing each of the following:
 - 1. Item Number.
 - 2. Description of work.
 - 3. Scheduled Values.
 - 4. Previous Applications.
 - 5. Work in Place and Stored Materials under this Application.
 - 6. Authorized Change Orders.
 - 7. Total Completed and Stored to Date of Application.
 - 8. Percentage of Completion.
 - 9. Balance to Finish.
 - 10. Retainage.
- F. Execute certification by signature of authorized officer.
- G. Use data from approved Schedule of Values. Provide dollar value in each column for each line item for portion of work performed and for stored products.
- H. List each authorized Change Order as a separate line item, listing Change Order number and dollar amount as for an original item of work.
- I. Submit one copy of each Application for Payment electronically.
- J. Include the following with the application:
 - 1. Transmittal letter as specified for submittals in Section 01 30 00.

2. Construction progress schedule, revised and current as specified in Section 01 30 00.
 3. Current construction photographs specified in Section 01 30 00.
 4. Partial release of liens from major subcontractors and vendors.
 5. Affidavits attesting to off-site stored products.
- K. When Architect requires substantiating information, submit data justifying dollar amounts in question. Provide one copy of data with cover letter for each copy of submittal. Show application number and date, and line item by number and description.

1.04 MODIFICATION PROCEDURES

- A. Submit name of the individual authorized to receive change documents and who will be responsible for informing others in Contractor's employ or subcontractors of changes to the Contract Documents.
- B. For minor changes not involving an adjustment to the Contract Price or Contract Time, Architect will issue instructions directly to Contractor.
- C. For other required changes, Architect will issue a document signed by Owner instructing Contractor to proceed with the change, for subsequent inclusion in a Change Order.
1. The document will describe the required changes and will designate method of determining any change in Contract Price or Contract Time.
 2. Promptly execute the change.
- D. For changes for which advance pricing is desired, Architect will issue a document that includes a detailed description of a proposed change with supplementary or revised drawings and specifications, a change in Contract Time for executing the change with a stipulation of any overtime work required and the period of time during which the requested price will be considered valid. Contractor shall prepare and submit a fixed price quotation within 5 days.
- E. Contractor may propose a change by submitting a request for change to Architect, describing the proposed change and its full effect on the work, with a statement describing the reason for the change, and the effect on the Contract Sum and Contract Time with full documentation. Document any requested substitutions in accordance with Section 01 60 00.
- F. Computation of Change in Contract Amount: As specified in the Agreement and Conditions of the Contract.
1. For change requested by Architect for work falling under a fixed price contract, the amount will be based on Contractor's price quotation.
 2. For change requested by Contractor, the amount will be based on the Contractor's request for a Change Order as approved by Architect.
 3. For pre-determined unit prices and quantities, the amount will be based on the fixed unit prices.
 4. For change ordered by Architect without a quotation from Contractor, the amount will be determined by Architect based on the Contractor's substantiation of costs as specified for Time and Material work.
- G. Substantiation of Costs: Provide full information required for evaluation.
1. On request, provide the following data:
 - a. Quantities of products, labor, and equipment.
 - b. Taxes, insurance, and bonds.
 - c. Overhead and profit.
 - d. Justification for any change in Contract Time.
 - e. Credit for deletions from Contract, similarly documented.
 2. Support each claim for additional costs with additional information:
 - a. Origin and date of claim.
 - b. Dates and times work was performed, and by whom.

- c. Time records and wage rates paid.
 - d. Invoices and receipts for products, equipment, and subcontracts, similarly documented.
- 3. For Time and Material work, submit itemized account and supporting data after completion of change, within time limits indicated in the Conditions of the Contract.
- H. Execution of Change Orders: Architect will issue Change Orders for signatures of parties as provided in the Conditions of the Contract.
- I. After execution of Change Order, promptly revise Schedule of Values and Application for Payment forms to record each authorized Change Order as a separate line item and adjust the Contract Price.
- J. Promptly revise progress schedules to reflect any change in Contract Time, revise sub-schedules to adjust times for other items of work affected by the change, and resubmit.
- K. Promptly enter changes in Project Record Documents.

1.05 APPLICATION FOR FINAL PAYMENT

- A. Prepare Application for Final Payment as specified for progress payments, identifying total adjusted Contract Price, previous payments, and sum remaining due.
- B. Application for Final Payment will not be considered until the following have been accomplished:
 - 1. All closeout procedures specified in Section 01 70 00.

1.06 INITIAL PROGRESS PAYMENT PROCEDURES

- A. The following items must be complete and submitted to the owner prior to acceptance and/or payment of the initial application for payment:
 - 1. Approved Schedule of Values
 - 2. Complete listing of Subcontractors
 - 3. Contractors Safety Plan
 - 4. Emergency Action Plan and Emergency Contacts
 - 5. Contractors Certificate of Insurance
 - 6. Contractors Submittal Schedule
 - 7. Approved Contractor's Construction Schedule (Preliminary if not final)

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 21 00
ALLOWANCES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Glycol fluid allowance.
- B. Payment and modification procedures relating to allowances.

1.02 RELATED REQUIREMENTS

- A. Section 01 20 00 - Price and Payment Procedures: Additional payment and modification procedures.

1.03 ALLOWANCES SCHEDULE

- A. Allowance No. 1 Glycol fluid: Include the stipulated sum/price of \$10,000 for use to fill Glycol system.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 25 00
SUBSTITUTION PROCEDURES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Procedural requirements for proposed substitutions.

1.02 RELATED REQUIREMENTS

- A. Requests for substitution must be made ten days prior to bid. Requests for substitution made after the bid may be considered by the Owner/Architect at their discretion and only under extraordinary conditions that could not be determined during the bidding period.
- B. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.03 SUMMARY

- A. This Section includes administrative and procedural requirements for handling requests for substitutions.
- B. Related Sections: The following Divisions contain requirements that relate to this Section:
 - 1. Division 01 specifies that applicability of industry standards to products specified.
 - 2. Division 01 specifies requirements for submitting the Contractor's Construction Schedule and the Submittal Schedule.
 - 3. Division 01 specifies requirements governing the Contractor's selection of products and product options.

1.04 DEFINITIONS

- A. Definitions in this Article do not change or modify the meaning of other terms used in the Contract Documents
- B. Substitutions: Changes in products, materials, equipment, and methods of construction required by the Contract Documents proposed by the Contractor. The following are not considered to be requests for substitutions:
 - 1. Revisions to the Contract Documents requested by the Owner or Architect.
 - 2. Specified options of products and construction methods included in the Contract Documents.
 - 3. The Contractor's determination of and compliance with governing regulations and orders issued by governing authorities.

1.05 SUBMITTALS

- A. Substitution Request Submittal: Substitution requests will only be considered during the bidding period. Substitutions will not be considered after the bids are accepted.
 - 1. Submit three copies of each request for substitution for consideration. Submit requests in the form and according to procedures required for change order proposals and utilizing the CSI Substitution Request Form 13.1A. The contractor is solely responsible for obtaining the required forms to submit before the stated time period expires.
 - 2. Identify the product or the fabrication or installation method to be replaced in each request. Include related Specification Section and Drawing numbers.
 - 3. Provide complete documentation showing compliance with the requirements for substitutions, and the following information, as appropriate:
 - a. Coordination information, including a list of changes or modifications needed to other parts of the Work and to construction performed by the Owner and separate contractors that will be necessary to accommodate the proposed substitution.

- b. A detailed comparison of significant qualities of the proposed substitution with those of the Work specified. Significant qualities may include elements, such as performance, weight, size, durability, and visual effect.
 - c. Product Data, including Drawings and descriptions of products and fabrication and installation procedures.
 - d. Samples, where applicable or requested.
 - e. A statement indicating the substitution's effect on the Contractor's Construction Schedule compared to the schedule without approval of the substitution. Indicate the effect of the proposed substitution on overall Contract Time.
 - f. Cost information, including a proposal of the net change, if any in the Contract Sum.
 - g. The Contractor's certification that the proposed substitution conforms to requirements in the Contract Documents in every respect and is appropriate for the applications indicated.
 - h. The Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of the failure of the substitution to perform adequately.
4. Architect's Action: If necessary, the Architect will request additional information or documentation for evaluation within one week of receipt of a request for substitution. The Architect will notify the Contractor of acceptance or rejection of the substitution within two weeks of receipt of the request, or one week of receipt of additional information or documentation, whichever is later.
- a. Use the product specified if the Architect cannot make a decision on the use of a proposed substitute within the time allocated.

PART 2 PRODUCTS

2.01 SUBSTITUTIONS

- A. Conditions: The Architect will receive and consider the Contractor's request for substitution when the following conditions are satisfied, as determined by the Architect. If the following conditions are not satisfied, the Architect will return the requests without action except to record noncompliance with these requirements.
- 1. Revisions to the Contract Documents are not required.
 - 2. Proposed changes are in keeping with the general intent of the Contract Documents.
 - 3. The request is timely, fully documented, and properly submitted.
 - 4. The specified product or method of construction cannot be provided within the Contract Time. The Architect will not consider the request if the product or method cannot be provided as a result of failure to pursue the Work promptly or coordinate activities properly.
 - 5. The requested substitution offers the Owner a substantial advantage, in cost, time, energy conservation, or other considerations, after deducting additional responsibilities the Owner must assume. The Owner's additional responsibilities may include compensation to the Architect for redesign and evaluation services, increased cost of other construction by the Owner, and similar considerations.
 - 6. The specified product or method of construction cannot receive necessary approval by a governing authority, and the requested substitution can be approved.
 - 7. The specified product or method of construction cannot be provided in a manner that is compatible with other materials and where the Contractor certifies that the substitution will overcome the incompatibility.
 - 8. The specified product or method of construction cannot be coordinated with other materials and where the Contractor certifies that the proposed substitution can be coordinated.

- B. The Contractor's submittal and the Architect's acceptance of Shop Drawings, Product Data, or Samples for construction activities not complying with the Contract Documents do not constitute an acceptable or valid request for substitution, nor do they constitute approval.

PART 3 EXECUTION

3.01 GENERAL REQUIREMENTS

- A. A Substitution Request for products, assemblies, materials, and equipment constitutes a representation that the submitter:
 - 1. Has investigated proposed product and determined that it meets or exceeds the quality level of the specified product, equipment, assembly, or system.
 - 2. Agrees to provide the same warranty for the substitution as for the specified product.
 - 3. Agrees to provide same or equivalent maintenance service and source of replacement parts, as applicable.
 - 4. Agrees to coordinate installation and make changes to other work that may be required for the work to be complete, with no additional cost to Owner.
 - 5. Waives claims for additional costs or time extension that may subsequently become apparent.
- B. Document each request with complete data substantiating compliance of proposed substitution with Contract Documents. Burden of proof is on proposer.
- C. Content: Include information necessary for tracking the status of each Substitution Request, and information necessary to provide an actionable response.
 - 1. Contractor's Substitution Request documentation must include the following:
 - a. Project Information:
 - 1) Official project name and number, and any additional required identifiers established in Contract Documents.
 - 2) Owner's, Architect's, and Contractor's names.
 - b. Substitution Request Information:
 - 1) Discrete and consecutive Substitution Request number, and descriptive subject/title.
 - 2) Indication of whether the substitution is for cause or convenience.
 - 3) Issue date.
 - 4) Reference to particular Contract Document(s) specification section number, title, and article/paragraph(s).
 - 5) Description of Substitution.
 - 6) Reason why the specified item cannot be provided.
 - 7) Differences between proposed substitution and specified item.
 - 8) Description of how proposed substitution affects other parts of work.
 - c. Attached Comparative Data: Provide point-by-point, side-by-side comparison addressing essential attributes specified, as appropriate and relevant for the item:
 - 1) Physical characteristics.
 - 2) In-service performance.
 - 3) Expected durability.
 - 4) Visual effect.
 - 5) Sustainable design features.
 - 6) Warranties.
 - 7) Other salient features and requirements.
 - 8) Include, as appropriate or requested, the following types of documentation for proposed substitution and for specified product in order to expedite review:
 - (a) Product Data:
 - (b) Samples.

- (c) Certificates, test, reports or similar qualification data.
 - (d) Drawings, when required to show impact on adjacent construction elements.
 - d. Impact of Substitution:
 - 1) Savings to Owner for accepting substitution.
 - 2) Change to Contract Time due to accepting substitution.
- D. Limit each request to a single proposed substitution item.
 - 1. Submit an electronic document, combining the request form with supporting data into single document.

3.02 SUBSTITUTION PROCEDURES DURING PROCUREMENT

- A. Owner will consider requests for substitutions only if submitted at least 10 days prior to the date for receipt of bids.

3.03 SUBSTITUTION PROCEDURES DURING CONSTRUCTION

- A. Under extra-ordinary conditions, submit request for Substitution for Cause immediately upon discovery of need for substitution, but not later than 14 days prior to time required for review and approval by Architect, in order to stay on approved project schedule.
- B. Substitutions will not be considered under one or more of the following circumstances:
 - 1. When they are indicated or implied on shop drawing or product data submittals, without having received prior approval.
 - 2. Without a separate written request.
 - 3. When acceptance will require revisions to Contract Documents.

3.04 RESOLUTION

- A. Architect may request additional information and documentation prior to rendering a decision. Provide this data in an expeditious manner.

END OF SECTION

**SECTION 01 30 00
ADMINISTRATIVE REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. General administrative requirements.
- B. Preconstruction meeting.
- C. Site mobilization meeting.
- D. Progress meetings.
- E. Construction progress schedule.
- F. Progress photographs.
- G. Submittals for review, information, and project closeout.
- H. Number of copies of submittals.
- I. Requests for Interpretation (RFI) procedures.
- J. Submittal procedures.

1.02 GENERAL ADMINISTRATIVE REQUIREMENTS

- A. Comply with requirements of Section 01 70 00 - Execution and Closeout Requirements for coordination of execution of administrative tasks with timing of construction activities.
- B. Make the following types of submittals to Architect:
 - 1. Requests for Interpretation (RFI).
 - 2. Requests for substitution.
 - 3. Shop drawings, product data, and samples.
 - 4. Test and inspection reports.
 - 5. Design data.
 - 6. Manufacturer's instructions and field reports.
 - 7. Applications for payment and change order requests.
 - 8. Progress schedules.
 - 9. Correction Punch List and Final Correction Punch List for Substantial Completion.
 - 10. Closeout submittals.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 PRECONSTRUCTION MEETING

- A. Owner will schedule a meeting after Notice of Award.
- B. Attendance Required:
 - 1. Owner.
 - 2. Architect.
 - 3. Contractor.
- C. Agenda:
 - 1. Execution of Owner-Contractor Agreement.
 - 2. Submission of executed bonds and insurance certificates.
 - 3. Distribution of Contract Documents.
 - 4. Submission of list of Subcontractors, list of Products, schedule of values, and progress schedule.
 - 5. Submission of initial Submittal schedule.

6. Designation of personnel representing the parties to ContractOMB and .
 7. Procedures and processing of field decisions, submittals, substitutions, applications for payments, proposal request, Change Orders, and Contract closeout procedures.
 8. Scheduling.
- D. Record minutes and distribute copies within two days after meeting to participants, with two copies to Architect, Owner, participants, and those affected by decisions made.

3.02 SITE MOBILIZATION MEETING

- A. Owner will schedule a meeting at the Project site prior to Contractor occupancy.
- B. Attendance Required:
1. Contractor.
 2. Owner.
 3. Architect.
 4. Contractor's Superintendent.
 5. Contractor's Project Manager.
 6. Major Subcontractors.
- C. Agenda:
1. Use of premises by Owner and Contractor.
 2. Owner's requirements and occupancy prior to completion.
 3. Construction facilities and controls provided by Contractor and Owner.
 4. Temporary utilities provided by Owner.
 5. Security and housekeeping procedures.
 6. Schedules.
 7. Application for payment procedures.
 8. Procedures for maintaining record documents.
 9. Requirements for start-up of equipment.
- D. Record minutes and distribute copies within two days after meeting to participants, with two copies to Architect, Owner, participants, and those affected by decisions made.

3.03 PROGRESS MEETINGS

- A. Schedule and administer meetings throughout progress of the Work at maximum weekly intervals.
- B. Attendance Required: Owner, Architect, and Contractor.
- C. Agenda:
1. Review minutes of previous meetings.
 2. Review of Work progress.
 3. Field observations, problems, and decisions.
 4. Identification of problems that impede, or will impede, planned progress.
 5. Review of submittals schedule and status of submittals.
 6. Review of RFIs log and status of responses.
 7. Review of off-site fabrication and delivery schedules.
 8. Maintenance of progress schedule.
 9. Corrective measures to regain projected schedules.
 10. Planned progress during succeeding work period.
 11. Coordination of projected progress.
 12. Maintenance of quality and work standards.
 13. Effect of proposed changes on progress schedule and coordination.
 14. Other business relating to Work.
 15. Provide three week look ahead.

3.04 CONSTRUCTION PROGRESS SCHEDULE

- A. Within 5 days after date of the Agreement, submit preliminary schedule defining planned operations for the first 60 days of Work, with a general outline for remainder of Work.
- B. If preliminary schedule requires revision after review, submit revised schedule within 10 days.
- C. Within 20 days after review of preliminary schedule, submit draft of proposed complete schedule for review.
 - 1. Include written certification that major contractors have reviewed and accepted proposed schedule.
- D. Within 5 days after joint review, submit complete schedule.
- E. Submit updated schedule with each Application for Payment.

3.05 PROGRESS PHOTOGRAPHS

- A. Submit photographs with each application for payment, taken not more than 3 days prior to submission of application for payment.
- B. Photography Type: Digital; electronic files.
- C. Provide photographs of site and construction throughout progress of work produced by an experienced photographer, acceptable to Architect.
- D. Views:
 - 1. Provide non-aerial photographs from four cardinal views at each specified time, until date of Substantial Completion.
 - 2. Consult with Architect for instructions on views required.
 - 3. Provide factual presentation.
 - 4. Provide correct exposure and focus, high resolution and sharpness, maximum depth of field, and minimum distortion.
- E. Digital Photographs: 24 bit color, minimum resolution of 1024 by 768, in JPG format; provide files unaltered by photo editing software.
 - 1. Delivery Medium: Via email.
 - 2. File Naming: Include project identification, date and time of view, and view identification.
 - 3. PDF File: Assemble all photos into printable pages in PDF format, with 2 to 3 photos per page, each photo labeled with file name; one PDF file per submittal.

3.06 REQUESTS FOR INTERPRETATION (RFI)

- A. Whenever possible, request clarifications at the next appropriate project progress meeting, with response entered into meeting minutes, rendering unnecessary the issuance of a formal RFI.
- B. Preparation: Prepare an RFI immediately upon discovery of a need for interpretation of Contract Documents. Failure to submit a RFI in a timely manner is not a legitimate cause for claiming additional costs or delays in execution of the work.
 - 1. Prepare a separate RFI for each specific item.
 - a. Review, coordinate, and comment on requests originating with subcontractors and/or materials suppliers.
 - b. Do not forward requests which solely require internal coordination between subcontractors.
 - 2. Combine RFI and its attachments into a single electronic file. PDF format is preferred.
- C. Reason for the RFI: Prior to initiation of an RFI, carefully study all Contract Documents to confirm that information sufficient for their interpretation is definitely not included.
 - 1. Include in each request Contractor's signature attesting to good faith effort to determine from Contract Documents information requiring interpretation.

2. Improper RFIs: Requests not prepared in compliance with requirements of this section, and/or missing key information required to render an actionable response. They will be returned without a response, with an explanatory notation.
- D. Content: Include identifiers necessary for tracking the status of each RFI, and information necessary to provide an actionable response.
 1. Official Project name and number, and any additional required identifiers established in Contract Documents.
 2. Owner's, Architect's, and Contractor's names.
 3. Discrete and consecutive RFI number, and descriptive subject/title.
 4. Issue date, and requested reply date.
 5. Reference to particular Contract Document(s) requiring additional information/interpretation. Identify pertinent drawing and detail number and/or specification section number, title, and paragraph(s).
 6. Annotations: Field dimensions and/or description of conditions which have engendered the request.
 7. Contractor's suggested resolution: A written and/or a graphic solution, to scale, is required in cases where clarification of coordination issues is involved, for example; routing, clearances, and/or specific locations of work shown diagrammatically in Contract Documents. If applicable, state the likely impact of the suggested resolution on Contract Time or the Contract Sum.
- E. Attachments: Include sketches, coordination drawings, descriptions, photos, submittals, and other information necessary to substantiate the reason for the request.
- F. RFI Log: Prepare and maintain a tabular log of RFIs for the duration of the project.
 1. Indicate current status of every RFI. Update log promptly and on a regular basis.
 2. Note dates of when each request is made, and when a response is received.
 3. Highlight items requiring priority or expedited response.
- G. Review Time: Architect will respond and return RFIs to Contractor within seven calendar days of receipt. For the purpose of establishing the start of the mandated response period, RFIs received after 12:00 noon will be considered as having been received on the following regular working day.
- H. Responses: Content of answered RFIs will not constitute in any manner a directive or authorization to perform extra work or delay the project. If in Contractor's belief it is likely to lead to a change to Contract Sum or Contract Time, promptly issue a notice to this effect, and follow up with an appropriate Change Order request to Owner.
 1. Response may include a request for additional information, in which case the original RFI will be deemed as having been answered, and an amended one is to be issued forthwith. Identify the amended RFI with an R suffix to the original number.
 2. Do not extend applicability of a response to specific item to encompass other similar conditions, unless specifically so noted in the response.
 3. Upon receipt of a response, promptly review and distribute it to all affected parties, and update the RFI Log.
 4. Notify Architect within seven calendar days if an additional or corrected response is required by submitting an amended version of the original RFI, identified as specified above.

3.07 SUBMITTAL SCHEDULE

- A. Submit to Architect for review a schedule for submittals in tabular format.
 1. Coordinate with Contractor's construction schedule and schedule of values.
 2. Format schedule to allow tracking of status of submittals throughout duration of construction.

3. Arrange information to include scheduled date for initial submittal, specification number and title, submittal category (for review or for information), description of item of work covered, and role and name of subcontractor.
4. Account for time required for preparation, review, manufacturing, fabrication and delivery when establishing submittal delivery and review deadline dates.
 - a. For assemblies, equipment, systems comprised of multiple components and/or requiring detailed coordination with other work, allow for additional time to make corrections or revisions to initial submittals, and time for their review.

3.08 SUBMITTALS FOR REVIEW

- A. When the following are specified in individual sections, submit them for review:
 1. Product data.
 2. Shop drawings.
 3. Samples for selection.
 4. Samples for verification.
- B. Submit to Architect for review for the limited purpose of checking for conformance with information given and the design concept expressed in the contract documents.
- C. Samples will be reviewed only for aesthetic, color, or finish selection.
- D. After review, provide copies and distribute in accordance with SUBMITTAL PROCEDURES article below and for record documents purposes described in Section 01 78 00 - Closeout Submittals.

3.09 SUBMITTALS FOR INFORMATION

- A. When the following are specified in individual sections, submit them for information:
 1. Design data.
 2. Certificates.
 3. Test reports.
 4. Inspection reports.
 5. Manufacturer's instructions.
 6. Manufacturer's field reports.
 7. Other types indicated.
- B. Submit for Architect's knowledge as contract administrator or for Owner. No action will be taken.

3.10 SUBMITTALS FOR PROJECT CLOSEOUT

- A. When the following are specified in individual sections, submit them at project closeout:
 1. Project record documents.
 2. Operation and maintenance data.
 3. Warranties.
 4. Bonds.
 5. Other types as indicated.
- B. Submit for Owner's benefit during and after project completion.

3.11 NUMBER OF COPIES OF SUBMITTALS

- A. Electronic Documents: Submit one electronic copy in PDF format; an electronically-marked up file will be returned. Create PDFs at native size and right-side up; illegible files will be rejected.
- B. Extra Copies at Project Closeout: See Section 01 78 00.
- C. Samples: Submit the number specified in individual specification sections; one of which will be retained by Architect.
 1. After review, produce duplicates.

2. Retained samples will not be returned to Contractor unless specifically so stated.

3.12 SUBMITTAL PROCEDURES

A. General Requirements:

1. Use a single transmittal for related items.
2. Submit separate packages of submittals for review and submittals for information, when included in the same specification section.
3. Preface each submittal with the appropriate specification section number and follow with a sequential identifying number. For revised submittals use original number and append a sequential numerical revision number.
4. Identify: Project; Contractor; subcontractor or supplier; pertinent drawing and detail number; and specification section number and article/paragraph, as appropriate on each copy.
5. Apply Contractor's stamp, signed or initialed certifying that review, approval, verification of products required, field dimensions, adjacent construction work, and coordination of information is in accordance with the requirements of the work and Contract Documents.
 - a. Submittals from sources other than the Contractor, or without Contractor's stamp will be returned without review.
6. Schedule submittals to expedite the Project, and coordinate submission of related items.
 - a. For each submittal for review, allow 15 days excluding delivery time to and from the Contractor.
 - b. For sequential reviews involving Architect's consultants, Owner, or another affected party, allow an additional 7 days.
7. Identify variations from Contract Documents and product or system limitations that may be detrimental to successful performance of the completed work.
8. Provide space for Contractor and Architect review stamps.
9. When revised for resubmission, identify all changes made since previous submission.
10. Distribute reviewed submittals. Instruct parties to promptly report inability to comply with requirements.
11. Incomplete submittals will not be reviewed, unless they are partial submittals for distinct portion(s) of the work, and have received prior approval for their use.
12. Submittals not requested will be recognized, and will be returned "Not Reviewed",

B. Product Data Procedures:

1. Submit only information required by individual specification sections.
2. Collect required information into a single submittal.
3. Do not submit (Material) Safety Data Sheets for materials or products.

C. Shop Drawing Procedures:

1. Prepare accurate, drawn-to-scale, original shop drawing documentation by interpreting Contract Documents and coordinating related work.
2. Generic, non-project-specific information submitted as shop drawings do not meet the requirements for shop drawings.

D. Samples Procedures:

1. Transmit related items together as single package.
2. Identify each item to allow review for applicability in relation to shop drawings showing installation locations.

END OF SECTION

SECTION 01 31 00
PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. This Section includes administrative provisions for coordinating construction operations on the project including, but not limited to, the following:
 - 1. General project coordination procedures.
 - 2. Coordination Drawings.
 - 3. Administrative and supervisory personnel.
 - 4. Project meetings.
- B. Related Sections
 - 1. Division 01 Section "Closeout Procedures" for coordinating Contract closeout.
- C. Each contractor shall participate in coordination requirements. Certain areas of responsibility will be assigned to a specific contractor.

1.03 DEFINITIONS

- A. RFI: Request from Owner, Architect, or Contractor seeking information from each other during construction.

1.04 COORDINATION

- A. Coordination: Coordinate construction operations included in various Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections that depend on each other for proper installation, connection, and operation.
 - 1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
 - 2. Coordinate installation of different components with other contractors to ensure maximum accessibility for required maintenance, service, and repair.
 - 3. Make adequate provisions to accommodate items scheduled for later installation.
- B. If necessary, prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
 - 1. Prepare similar memoranda for the Owner and separate contractors if coordination of their Work is required.
- C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:
 - 1. Preparation of the Contractor's Construction Schedule.
 - 2. Preparation of the Schedule of Values.
 - 3. Installation and removal of temporary facilities and controls.
 - 4. Delivery and processing of submittals.
 - 5. Progress meetings.

6. Pre-installation conferences.
 7. Project closeout activities.
- D. Conservation: Coordinate construction activities to ensure that operations are carried out with consideration given to conservation of energy, water, and materials.

1.05 REQUESTS FOR INFORMATION (RFIS)

- A. General: Immediately on discovery of the need for additional information or interpretation of the Contract Documents, Contractor shall prepare and submit an RFI in the form specified.
1. Architect will return RFIs submitted to Architect by other entities controlled by Contractor with no response.
 2. Coordinate and submit RFIs in a prompt manner so as to avoid delays in Contractor's work or work of subcontractors.
- B. Content of the RFI: Include a detailed, legible description of item needing information or interpretation and the following:
1. Project name
 2. Project number
 3. Date
 4. Name of Contractor
 5. Name of Architect
 6. RFI number, numbered sequentially
 7. RFI subject
 8. Specification Section number and title and related paragraphs, as appropriate
 9. Drawing number and detail references, as appropriate
 10. Field dimensions and conditions, as appropriate
 11. Contractor's suggested resolution. If Contractor's solution(s) impacts the Contract Time or the Contract Sum, Contractor shall state impact in the RFI
 12. Contractor's signature
 13. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
 - a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify Architect in writing within [10] ten days of receipt of the RFI response
- C. On receipt of Architect's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify Architect within seven days if Contractor disagrees with response.
- D. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log bi-weekly.

1.06 SUBMITTALS

- A. Coordination Drawings: Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.
1. Indicate relationship of components shown on separate Shop Drawings.
 2. Indicate required installation sequences.
 3. Refer to Division 15 Section "Basic Mechanical Materials and Methods" and Division 16 Section "Basic Electrical Materials and Methods" for specific Coordination Drawing requirements for mechanical and electrical installations.
- B. Staff Names: Within 15 days of starting construction operations, submit a list of principal staff

1. assignments, including superintendent and other personnel in attendance at the Project site. Identify individuals and their duties and responsibilities; list addresses and telephone numbers, including home and office telephone numbers. Provide names, addresses, and telephone numbers of individuals assigned as standbys in the absence of individuals assigned to the Project.
2. Post copies of list in the Project meeting room, in temporary field office, and by each temporary telephone.

1.07 PROJECT MEETINGS

- A. General: Meetings shall be schedule by the Owner and shall be conducted at the Project site, unless otherwise indicated.
 1. Attendees: Architect shall inform participants and others involved, and individuals whose presence is required, of date and time of each meeting.
 2. Agenda: Contractor shall prepare the meeting agenda and distribute the agenda to all invited attendees.
 3. Minutes: Contractor shall record significant discussions and agreements achieved and distribute the meeting minutes to everyone concerned within 5 days of the meeting.
- B. Preconstruction Conference: Owner shall schedule a preconstruction conference before starting Construction. Conference shall be held at the conference at the Project site or another convenient location. Conduct the meeting to review responsibilities and personnel assignments.
 1. Attendees: Authorized representatives of the Owner, the Architect and their consultants, the Contractor and its superintendent, and other concerned parties shall attend the conference. All participants at the conference shall be familiar with the Project and authorized to conclude matters relating to the Work.
 2. Agenda: Discuss items of significance that could affect progress, including the following:
 - a. Tentative construction schedule
 - b. Phasing
 - c. Critical work sequencing
 - d. Designation of responsible personnel
 - e. Procedures for processing field decisions and Change Orders
 - f. Procedures for processing Applications for Payment
 - g. Distribution of the Contract Documents
 - h. Submittal procedures
 - i. Preparation of Record Documents
 - j. Use of the premises
 - k. Responsibility for temporary facilities and controls
 - l. Parking availability
 - m. Office, work, and storage areas
 - n. Equipment deliveries and priorities
 - o. First aid
 - p. Security
 - q. Progress cleaning
 - r. Working hours
- C. Progress Meetings: Owner shall schedule the Progress Meetings at bi-weekly intervals.
 1. Agenda:
 - a. Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of the Project.

- b. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to the Contractor's Construction Schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
 - 1) Contractor shall issue a three-week look ahead schedule in advance of each Progress Meeting.
 - 2) Contractor shall revised the Construction Schedule after each progress meeting where revisions to the schedule have been made or recognized. At a minimum, the updated Construction Schedule shall be issued with each Application for Payment.
- c. Review present and future needs of each entity present, including the following:
 - 1) Interface requirements
 - 2) Sequence of operations
 - 3) Status of submittals
 - 4) Deliveries
 - 5) Off-site fabrication
 - 6) Access
 - 7) Site utilization
 - 8) Temporary facilities and controls
 - 9) Work hours
 - 10) Hazards and risks
 - 11) Progress cleaning
 - 12) Quality and work standard
 - 13) Change Orders
 - 14) Documentation of information for payment requests
- 2. Reporting: Contractor to record minutes and distribute copies within five days after meeting to participants, with two copies to Architect, Owner, participants, and those affected by decisions made.

PART 2 - PRODUCTS - NOT USED

PART 3 - EXECUTION - NOT USED

END OF SECTION

**SECTION 01 32 00
CONSTRUCTION PROGRESS**

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. This Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
 - 1. Preliminary Construction Schedule.
 - 2. Contractor's Construction Schedule.
 - 3. Submittals Schedule.
 - 4. Daily construction reports.
 - 5. Material location reports.
 - 6. Field condition reports.
 - 7. Preconstruction Photographs
 - 8. Construction photographs.
- B. Related Sections include the following:
 - 1. Division 01 Section "Payment Procedures" for submitting the Schedule of Values.
 - 2. Division 01 Section "Project Management and Coordination" for submitting and distributing meeting and conference minutes.
 - 3. Division 01 Section "Submittal Procedures" for submitting schedules and reports.
 - 4. Division 01 Section "Closeout Procedures" for submitting construction photographs as Project Record Documents at Project closeout.

1.03 DEFINITIONS

- A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They must start and finish on the planned early start and finish times.
 - 2. Predecessor activity is an activity that must be completed before a given activity can be started.
- B. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- C. Critical Path: The longest continuous chain of activities through the network schedule that establishes the minimum overall Project duration and contains no float.
- D. Event: The starting or ending point of an activity.
- E. Float: The measure of leeway in starting and completing an activity.
 - 1. Float time is not for the exclusive use or benefit of either Owner or Contractor, but is a jointly owned, expiring Project resource available to both parties as needed to meet schedule milestones and Contract completion date.
 - 2. Free float is the amount of time an activity can be delayed without adversely affecting the early start of the following activity.
 - 3. Total float is the measure of leeway in starting or completing an activity without adversely affecting the planned Project completion date.

- F. Major Area: A story of construction, a separate building, or a similar significant construction element.
- G. Milestone: A key or critical point in time for reference or measurement.
- H. Network Diagram: A graphic diagram of a network schedule, showing activities and activity relationships.
- I. Resource Loading: The allocation of manpower and equipment necessary for the completion of an activity as scheduled.

1.04 SUBMITTALS

- A. Qualification Data: For firms and persons specified in "Quality Assurance" Article and in-house scheduling personnel to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- B. Submittals Schedule: Submit three copies of schedule. Arrange the following information in a tubular format:
 - 1. Scheduled date for first submittal.
 - 2. Specification Section number and title.
 - 3. Submittal category (action or informational).
 - 4. Name of subcontractor.
 - 5. Description of the Work covered.
- C. Contractor's Construction Schedule: Submit three printed copies of initial schedule, one a reproducible print and one a blue- or black-line print, large enough to show entire schedule for entire construction period.
- D. CPM Reports: Concurrent with CPM schedule, submit three printed copies of each of the following computer-generated reports. Format for each activity in reports shall contain activity number, activity description, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float.
 - 1. Activity Report: List of all activities sorted by activity number and then early start date, or actual start date if known.
 - 2. Logic Report: List of preceding and succeeding activities for all activities, sorted in ascending order by activity number and then early start date, or actual start date if known.
 - 3. Total Float Report: List of all activities sorted in ascending order of total float.
- E. Photographic Documentation:
 - 1. Preconstruction Photographs: Before commencement of demolition, take color, digital photographs of Project site and surrounding properties, including existing items to remain during construction, from different vantage points, as directed by Architect.
 - 2. Periodic Construction Photographs: Take 12, color, digital photographs monthly with timing each month adjust to coincide with the cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last photographs were taken.
 - a. Photographer Qualifications: An individual who has been regularly engaged as a professional photographer of construction projects for not less than three years.
 - b. Format: 4-by-6-inch (101-by-152-mm) smooth-surface matte prints on single weight commercial grade stock.
 - c. Identification: On back of each print, provide an applied label or rubber stamped impression with the following information:
 - 1) Name of project
 - 2) Name and address of photographer
 - 3) Name of Architect

- 4) Name of Contractor
- 5) Date photograph was taken
- 6) Description of vantage point, indicating location, direction (by compass point) and elevation or story of construction.
- d. Digital Images: Submit a complete set of digital image electronic files with each submittal of prints on CD-ROM. Identify electronic media with date photographs were taken. Submit images that have same aspect ratio as the sensor, uncropped.
- e. Obtain and transfer copyright usage rights from photographer to Owner for unlimited reproduction of photographic documentation.
- f. Daily Construction Reports: Submit two copies at weekly intervals.
- g. Material Location Reports: Submit two copies at weekly intervals.
- h. Field Condition Reports: Submit two copies at weekly intervals.

1.05 QUALITY ASSURANCE

- A. Scheduling Consultant Qualifications: An experienced specialist in CPM scheduling and reporting.

1.06 COORDINATION

- A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate contractors.
- B. Coordinate Contractor's Construction Schedule with the Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.
 - 1. Secure time commitments for performing critical elements of the Work from parties involved.
 - 2. Coordinate each construction activity in the network with other activities and schedule them in proper sequence.

PART 2 - PRODUCTS

2.01 SUBMITTALS SCHEDULE

- A. Preparation: Submit a schedule of submittals, arranged in chronological order by dates required by construction schedule. Include time required for review, resubmittal, ordering, manufacturing, fabrication, and delivery when establishing dates.
 - 1. Coordinate Submittals Schedule with list of subcontracts, the Schedule of Values, and Contractor's Construction Schedule.
 - 2. Final Submittal: Submit concurrently with the first complete submittal of Contractor's Construction Schedule.

2.02 CONTRACTOR'S CONSTRUCTION SCHEDULE (CPM SCHEDULE)

- A. CPM Schedule: Prepare Contractor's Construction Schedule using a CPM network analysis diagram.
 - 1. Develop network diagram in sufficient time to submit CPM schedule so it can be accepted for use no later than 15 days after date established for the Notice to Proceed..
 - 2. Establish procedures for monitoring and updating CPM schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.
 - 3. Use "one workday" as the unit of time.
- B. CPM Schedule Preparation: Prepare a list of all activities required to complete the Work. Using the preliminary network diagram, prepare a skeleton network to identify probable critical paths. Using the preliminary network diagram, prepare a skeleton network to identify probable critical paths.

1. Activities: Indicate the estimated time duration, sequence requirements, and relationship of each activity in relation to other activities. Include estimated time frames for the following activities:
 - a. Preparation and processing of submittals
 - b. Mobilization and demobilization
 - c. Purchase of materials
 - d. Delivery
 - e. Fabrication
 - f. Utility interruptions
 - g. Installation
 - h. Work by Owner that may affect or be affected by Contractor's activities
 - i. Testing
 2. Processing: Process data to produce output data or a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the CPM schedule within the limitations of the Contract Time.
 3. Format: Mark the critical path. Locate the critical path near center of network; locate paths with most float near the edges.
 - a. Subnetworks on separate sheets are permissible for activities clearly off the critical
 - b. path.
- C. Schedule Updating: At monthly intervals, update schedule to reflect actual construction progress and activities. Issue schedule one week before each regularly scheduled progress meeting. Include the following:
1. Identification of activities that have changed.
 2. Changes in early and late start dates.
 3. Changes in early and late finish dates.
 4. Changes in activity durations in workdays.
 5. Changes in the critical path.
 6. Changes in total float or slack time.
 7. Changes in the Contract Time
 - a. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.

2.03 REPORTS

- A. Daily Construction Reports: Prepare a daily construction report recording the following information concerning events at Project site:
1. List of subcontractors at Project site.
 2. List of separate contractors at Project site.
 3. Approximate count of personnel at Project site.
 4. High and low temperatures and general weather conditions.
 5. Accidents.
 6. Meetings and significant decisions.
 7. Unusual events (refer to special reports).
 8. Stoppages, delays, shortages, and losses.
 9. Meter readings and similar recordings.
 10. Emergency procedures.
 11. Orders and requests of authorities having jurisdiction.
 12. Change Orders received and implemented.
 13. Construction Change Directives received.
 14. Services connected and disconnected.

15. Equipment or system tests and startups.
 16. Partial Completions and occupancies.
 17. Substantial Completions authorized.
- B. Material Location Reports: At monthly intervals, prepare a comprehensive list of materials delivered to and stored at Project site. List shall be cumulative, showing materials previously reported plus items recently delivered. Include with list a statement of progress on and delivery dates for materials or items of equipment fabricated or stored away from Project site.
- C. Field Condition Reports: Immediately on discovery of a difference between field conditions and the Contract Documents, prepare a detailed report. Submit with a request for information. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.

PART 3 - EXECUTION (NOT USED)

END OF SECTION

**SECTION 01 33 00
SUBMITTAL PROCEDURES**

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. Section includes requirements for the submittal schedule and administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.
 - 1. Process designated submittals for the Project electronically through designated email system.

1.03 DEFINITIONS

- A. Action Submittals: Written and graphic information and physical samples that require Architect's responsive action. Action submittals are those submittals indicated in individual Specification Sections as "action submittals."
- B. Informational Submittals: Written and graphic information and physical samples that do not require Architect's and responsive action. Submittals may be rejected for not complying with requirements. Informational submittals are those submittals indicated in individual Specification Sections as "informational submittals."
- C. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.
- D. Email System: A method to transmit certain electronic submittals between the Contractor, Architect, and Owner, via email.
 - 1. For consistency, the standard file format will be PDF. Convert paper originals and other file formats to PDF prior to submission.
 - 2. In the event of system malfunction, submittals shall be processed in accordance with the Architect's instructions, until the system malfunction has been corrected.
 - 3. For this Project, process the following submittal types through the designated email system:
 - a. Product Data.
 - b. Shop Drawings.
 - c. Product Schedules.
 - d. Qualification Data.
 - e. Certificates (Welding, Installer, Manufacturer, Product, and Material, as applicable).
 - f. Test Reports (Material, Product, Preconstruction, Compatibility, and Field, as applicable).
 - g. Research Reports.
 - h. Warranty (sample).
 - i. Design Data, including calculations.
 - j. Coordination Drawings.
 - k. Delegated-Design Services Certifications.
 - 4. For Samples, provide electronic submittal of Sample cover sheet, identifying location and actual delivery date of Samples. Deliver Samples to location (Architect's office, Project site, etc.) as directed by the Architect.
 - a. Architect will identify delivery location(s) after receipt and review of Contractor's Submittal Schedule.

1.04 SUBMITTAL SCHEDULE

- A. Submittal Schedule: Submit a schedule of submittals indicating scheduled date for each submission. Factor time required for review, ordering, manufacturing, fabrication, and delivery when establishing submission dates. Include additional time required for making corrections or revisions to submittals noted by Architect and additional time for handling and reviewing submittals required by those corrections.
 - 1. Submit concurrently with the first complete submittal of Contractor's construction schedule.
 - 2. Format: Arrange the following information in a tabular format:
 - a. Specification Section number and title.
 - b. A/E Number.
 - 1) Architect will furnish Contractor with unique "A/E Number" designation for each required submittal.
 - c. Submittal category: Action; informal.
 - d. Submittal type: Product Data, Shop Drawings, Samples, etc.
 - e. Description of the Work covered.
 - f. Scheduled date for first submittal.

1.05 COLOR SCHEDULE

- A. Color Schedule: Within 30 days after date of Notice of Award, submit a complete list of proposed manufacturers and complete product designations (i.e. model, grade, series, product line, etc.) for each item requiring color selection by Architect.

1.06 SUBMITTAL ADMINISTRATIVE REQUIREMENTS

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 - 1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Where indicated, submit all submittal items required for each Specification Section concurrently.
 - 3. Coordinate transmittal of different types of submittals for related parts of the Work so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- B. Processing Time: Allow sufficient time for submittal review, including time for resubmittals. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
- C. Electronic Submittals: Identify and incorporate information in each electronic submittal file as follows:
 - 1. Include a cover sheet on each submittal item for identification. Do not combine different submittals under same cover sheet; only one submittal is to be provided per email.
 - a. Cover Sheet: Use PDF version of sample form included in Project Manual. Complete each item on form, sign and date. Architect will furnish PDF version of sample form.
 - 2. Name submittal file as directed by Architect.
 - 3. Transmit each submittal via email using subject line as directed by Architect.
 - 4. Send submittal to designated Project-specific email address:
 - a. Use the following email address: zigmondb@studiojaed.com.
- D. Resubmittals: Make resubmittals in same form and, for non-electronic submittals, in the same number of copies as initial submittal.

1. Note date and content of revision in label or title block and clearly indicate extent of revision.
 2. Resubmit submittals until they are marked with approval notation from Architect.
 3. Refer to Supplementary Conditions for provisions allowing Owner to obtain reimbursement from the Contractor for amounts paid to the Architect for evaluation of certain resubmittals.
- E. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities.
- F. Use for Construction: Retain complete copies of submittals on Project site. Use only final action submittals that are marked with approval notation from Architect.

PART 2 - PRODUCTS

2.01 SUBMITTAL PROCEDURES, GENERAL

- A. General Submittal Procedure Requirements: Prepare and submit submittals required by individual Specification Sections. Types of submittals are indicated in individual Specification Sections.

2.02 ELECTRONIC SUBMITTAL PROCEDURES

- A. Use the designated email system for submittals in this Article.
1. Submit electronic submittals via email as PDF electronic files.
 - a. Architect will return annotated file. Annotate and retain one copy of file as an electronic Project record document file.
- B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
1. Mark submittal to show which products and options are applicable.
 2. Include the following information, as applicable:
 - a. Manufacturer's catalog cuts.
 - b. Statement of compliance with specified referenced standards.
 - c. Testing by recognized testing agency.
 3. For equipment, include the following in addition to the above, as applicable:
 - a. Printed performance curves.
 - b. Clearances required to other construction, if not indicated on accompanying Shop Drawings.
- C. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data, unless submittal based on Architect's digital data drawing files is otherwise permitted.
1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
 - a. Identification of products.
 - b. Schedules.
 - c. Compliance with specified standards.
 - d. Notation of dimensions established by field measurement.
 - e. Relationship and attachment to adjoining construction clearly indicated.
 - f. Seal and signature of professional engineer if specified.
- D. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, contact information of architects and owners, and other information specified.
- E. Certificates:

1. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification and Procedure Qualification Record on AWS forms. Include names of firms and personnel certified.
 2. Installer Certificates: Submit written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
 3. Manufacturer Certificates: Submit written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
 4. Product Certificates: Submit written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
 5. Material Certificates: Submit written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.
- F. Test Reports:
1. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
 2. Product Test Reports: Submit written reports indicating that current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
 3. Preconstruction Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
 4. Compatibility Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
 5. Field Test Reports: Submit written reports indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
- G. Research Reports: Submit written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.
- H. Warranty: Submit sample warranties as required in individual Specification Sections.
- I. Design Data: Prepare and submit written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- J. Coordination Drawing Submittals: Comply with requirements specified in Division 01 Section "Project Management and Coordination."
- K. Delegated-Design Services Certification: Submit certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.
1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

2. In addition, for a project in New Jersey, provide three paper copies of certificate, signed and sealed (with raised seal) by the responsible design professional.

2.03 NON-ELECTRONIC SUBMITTAL PROCEDURES

- A. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
 1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 2. Identification: Attach label on unexposed side of Samples that includes the following:
 - a. Generic description of Sample.
 - b. Product name and name of manufacturer.
 - c. Sample source.
 - d. Number and title of applicable Specification Section.
 3. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 4. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
Number of Samples: Submit three full sets of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Architect will return one submittal with options selected.
 5. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit three sets of Samples. Architect will return one set.
 - 1) If variation in color, pattern, texture, or other characteristic is inherent in material or product represented by a Sample, submit at least three sets of paired units that show approximate limits of variations.
- B. Subcontract List: Prepare a written summary identifying individuals or firms proposed for each portion of the Work, including those who are to furnish products or equipment fabricated to a special design. Include the following information in tabular form:
 1. Name, address, and telephone number of entity performing subcontract or supplying products
 2. Number and title of related Specification Section(s) covered by subcontract.
 3. Submit subcontract list in the following format:
 - a. Number of Copies: Four paper copies of subcontractor list, unless otherwise indicated. Architect will return one copy.
- C. Key Personnel Names: No later than 15 days after date of Notice of Award, submit a list of key personnel assignments, including superintendent and other personnel in attendance at Project site.
 1. Identify individuals and their duties and responsibilities; list addresses and telephone numbers, including emergency, office, and cellular telephone numbers and email addresses.
 - a. Number of Copies: Four paper copies of key personnel list, unless otherwise indicated.

- D. Closeout Submittals and Maintenance Material Submittals: Comply with requirements specified in Division 01 Section "Closeout Procedures."
- E. Maintenance Data: Comply with requirements specified in Division 01 Section "Operation and Maintenance Data."

2.04 DELEGATED-DESIGN SERVICES

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
- B. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to Architect.

PART 3 - EXECUTION

3.01 CONTRACTOR'S REVIEW

- A. Action and Informational Submittals: Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Identify any deviations from Contract Document requirements. Mark cover sheet with approval before submitting to Architect.
 - 1. Sign and date statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.

3.02 ARCHITECT'S ACTION

- A. General: Architect will not review submittals that do not bear Contractor's approval and will return them without action.
- B. Action Submittals: Architect will review each submittal, make marks to indicate corrections or revisions required, and return it. Architect will mark submittal appropriately to indicate action, as follows:
 - 1. Final Unrestricted Release: Where the submittal is marked "Approved," the Work covered by the submittal may proceed provided it complies with the Contract Documents. Final acceptance will depend on that compliance.
 - 2. Final-but-Restricted Release: Where the submittal is marked "Approved as Noted," the Work covered by the submittal may proceed provided it complies both with Architect's notations and corrections on the submittal and the Contract Documents. Final acceptance will depend on that compliance.
 - 3. Resubmit: Where the submittal is marked "Approved, Revise and Return Corrected Copies," the Work covered by the submittal may proceed provided it complies both with Architect's notations and corrections on the submittal and the Contract Documents. Revise submittal according to Architect's notations and corrections and return corrected copies. Final acceptance will depend on that compliance.
 - 4. Rejected: Where the submittal is marked "Rejected," do not proceed with the Work covered by the submittal. Prepare a new submittal for a product that complies with the Contract Documents.
 - 5. Incomplete - Resubmit: Where the submittal is marked "Incomplete, Submit Additional Information," do not proceed with the Work covered by the submittal. Prepare additional information requested, or required by the Contract Documents, that indicates compliance with requirements, and resubmit.
- C. Informational Submittals: Architect will review each submittal and will not return it, or will return it if it does not comply with requirements.
- D. Incomplete submittals are unacceptable, will be considered nonresponsive, and will be returned for resubmittal without review.

- E. Limit information submitted to specific products indicated. Do not submit extraneous matter. Submittals containing excessive extraneous matter will be returned for resubmittal without review.
- F. Submittals not required by the Contract Documents may be returned by the Architect without action.

END OF SECTION

**SECTION 01 35 53
SECURITY PROCEDURES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Security measures including formal security program, entry control, personnel identification, and miscellaneous restrictions.
- B. Background check must be completed with the Bureau of Criminal Identification and Investigation prior to construction start.

1.02 RELATED REQUIREMENTS

- A. Section 01 10 00 - Summary: Use of Premises and occupancy.
- B. Section 01 50 00 - Temporary Facilities and Controls: Temporary lighting and barriers and enclosures.

1.03 SECURITY PROGRAM

- A. Protect Work , existing premises and Owner's operations from theft, vandalism, and unauthorized entry.
- B. A background check must be completed with the Bureau of Criminal Identification and Investigation prior to construction start.
- C. Initiate program at project mobilization.
- D. Initiate program in coordination with Owner's existing security system at project mobilization.
- E. Maintain program throughout construction period until Owner acceptance precludes the need for Contractor security.
- F. Maintain program throughout construction period until directed by Owner.

1.04 ENTRY CONTROL

- A. Restrict entrance of persons and vehicles into Project site and existing facilities.
- B. Allow entrance only to authorized persons with proper identification.
- C. Owner will control entrance of persons and vehicles related to Owner's operations.

1.05 PERSONNEL IDENTIFICATION

- A. Provide identification badge to each person authorized to enter premises.
- B. Badge to Include: Personal photograph, name, expiration date and employer.
- C. Maintain a list of accredited persons, submit copy to Owner on request.
- D. Require return of badges at expiration of their employment on the Work.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 40 00
QUALITY REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Control of installation.
- B. Tolerances.
- C. Testing and inspection services.
- D. Manufacturers' field services.

1.02 RELATED REQUIREMENTS

- A. Section 01 42 16 - Definitions.

PART 3 EXECUTION

2.01 CONTROL OF INSTALLATION

- A. Monitor quality control over suppliers, manufacturers, products, services, site conditions, and workmanship, to produce Work of specified quality.
- B. Comply with manufacturers' instructions, including each step in sequence.
- C. Should manufacturers' instructions conflict with Contract Documents, request clarification from Architect before proceeding.
- D. Comply with specified standards as minimum quality for the Work except where more stringent tolerances, codes, or specified requirements indicate higher standards or more precise workmanship.
- E. Have Work performed by persons qualified to produce required and specified quality.
- F. Verify that field measurements are as indicated on shop drawings or as instructed by the manufacturer.
- G. Secure products in place with positive anchorage devices designed and sized to withstand stresses, vibration, physical distortion, and disfigurement.

2.02 TOLERANCES

- A. Monitor fabrication and installation tolerance control of products to produce acceptable Work. Do not permit tolerances to accumulate.
- B. Comply with manufacturers' tolerances. Should manufacturers' tolerances conflict with Contract Documents, request clarification from Architect before proceeding.
- C. Adjust products to appropriate dimensions; position before securing products in place.

2.03 MANUFACTURERS' FIELD SERVICES

- A. When specified in individual specification sections, require material or product suppliers or manufacturers to provide qualified staff personnel to observe site conditions, conditions of surfaces and installation, quality of workmanship, as applicable, and to initiate instructions when necessary.
- B. Report observations and site decisions or instructions given to applicators or installers that are supplemental or contrary to manufacturers' written instructions.

2.04 DEFECT ASSESSMENT

- A. Replace Work or portions of the Work not conforming to specified requirements.

- B. If, in the opinion of Architect, it is not practical to remove and replace the Work, Architect will direct an appropriate remedy or adjust payment.

END OF SECTION

SECTION 01 42 16
DEFINITIONS

PART 1 GENERAL

1.01 SUMMARY

- A. Other definitions are included in individual specification sections.

1.02 DEFINITIONS

- A. Furnish: To supply, deliver, unload, and inspect for damage.
- B. Install: To unpack, assemble, erect, apply, place, finish, cure, protect, clean, start up, and make ready for use.
- C. Product: Material, machinery, components, equipment, fixtures, and systems forming the work result. Not materials or equipment used for preparation, fabrication, conveying, or erection and not incorporated into the work result. Products may be new, never before used, or re-used materials or equipment.
- D. Project Manual: The book-sized volume that includes the procurement requirements (if any), the contracting requirements, and the specifications.
- E. Specifications: Same as Project Manual.
- F. Provide: To furnish and install.
- G. Supply: Same as Furnish.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 50 00
TEMPORARY FACILITIES AND CONTROLS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Temporary utilities.
- B. Temporary sanitary facilities.
- C. Temporary Controls: Barriers, enclosures, and fencing.
- D. Security requirements.
- E. Vehicular access and parking.
- F. Waste removal facilities and services.

1.02 TEMPORARY UTILITIES

- A. Owner will provide the following:
 - 1. Electrical power and metering, consisting of connection to existing facilities.
 - 2. Water supply, consisting of connection to existing facilities.
 - 3. Contractor shall be responsible to tie to existing utility services and make distribution to the construction site.

1.03 TEMPORARY SANITARY FACILITIES

- A. Provide and maintain required portable toilets and enclosures. Provide at time of project mobilization.
- B. Maintain daily in clean and sanitary condition.

1.04 BARRIERS

- A. Provide barriers to prevent unauthorized entry to construction areas, to prevent access to areas that could be hazardous to workers or the public, to allow for owner's use of site and to protect existing facilities and adjacent properties from damage from construction operations and demolition.
- B. Provide barricades required by Owner and governing authorities for public rights-of-way and for public access to existing building.
- C. Provide protection for plants designated to remain. Replace damaged plants.
- D. Protect non-owned vehicular traffic, stored materials, site, and structures from damage.
- E. Traffic Controls: Coordinate with the Owner.

1.05 FENCING

- A. Provide 6 foot high fence around on-grade storage areas; equip with vehicular and pedestrian gates with locks.

1.06 EXTERIOR ENCLOSURES

- A. Provide temporary insulated weather tight closure of exterior openings to accommodate acceptable working conditions and protection for Products, to allow for temporary heating and maintenance of required ambient temperatures identified in individual specification sections, and to prevent entry of unauthorized persons. Provide access doors with self-closing hardware and locks.

1.07 INTERIOR ENCLOSURES

- A. Provide temporary partitions as indicated and as otherwise required to separate work areas from Owner-occupied areas, to prevent penetration of dust and moisture into Owner-occupied areas, and to prevent damage to existing materials and equipment.
- B. Construction: Framing and reinforced polyethylene with plywood overlay sheet materials with closed joints and sealed edges at intersections with existing surfaces:

1.08 SECURITY

- A. Provide security and facilities to protect Work, existing facilities, and Owner's operations from unauthorized entry, vandalism, or theft.
- B. Coordinate with Owner's security program.
- C. Provide background check for any/all employees who enter building.

1.09 VEHICULAR ACCESS AND PARKING

- A. Comply with regulations relating to use of streets and sidewalks, access to emergency facilities, and access for emergency vehicles.
- B. Coordinate access and haul routes with governing authorities and Owner.
- C. Provide and maintain access to fire hydrants, free of obstructions.
- D. Existing parking areas may be used for construction parking. Owner shall designate parking areas to be used by contractor.

1.10 WASTE REMOVAL

- A. Provide waste removal facilities and services as required to maintain the site in clean and orderly condition.
- B. Provide containers with lids. Remove trash from site daily.
- C. If materials to be recycled or re-used on the project must be stored on-site, provide suitable non-combustible containers; locate containers holding flammable material outside the structure unless otherwise approved by the authorities having jurisdiction.
- D. Open free-fall chutes are not permitted. Terminate closed chutes into appropriate containers with lids.

1.11 REMOVAL OF UTILITIES, FACILITIES, AND CONTROLS

- A. Remove temporary utilities, equipment, facilities, materials, prior to Substantial Completion inspection.
- B. Clean and repair damage caused by installation or use of temporary work.
- C. Restore existing facilities used during construction to original condition.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

**SECTION 01 60 00
PRODUCT REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Transportation, handling, storage and protection.
- B. Product option requirements.
- C. Maintenance materials, including extra materials, spare parts, tools, and software.

1.02 RELATED REQUIREMENTS

- A. Section 01 25 00 - Substitution Procedures: Substitutions made during procurement and/or construction phases.
- B. Section 01 61 16 - Volatile Organic Compound (VOC) Content Restrictions: Requirements for VOC-restricted product categories.

1.03 SUBMITTALS

- A. Product Data Submittals: Submit manufacturer's standard published data. Mark each copy to identify applicable products, models, options, and other data. Supplement manufacturers' standard data to provide information specific to this Project.
- B. Shop Drawing Submittals: Prepared specifically for this Project; indicate utility and electrical characteristics, utility connection requirements, and location of utility outlets for service for functional equipment and appliances.
- C. Sample Submittals: Illustrate functional and aesthetic characteristics of the product, with integral parts and attachment devices. Coordinate sample submittals for interfacing work.
 - 1. For selection from standard finishes, submit samples of the full range of the manufacturer's standard colors, textures, and patterns.

PART 2 PRODUCTS

2.01 NEW PRODUCTS

- A. Provide new products unless specifically required or permitted by the Contract Documents.

2.02 PRODUCT OPTIONS

- A. Products Specified by Reference Standards or by Description Only: Use any product meeting those standards or description.
- B. Products Specified by Naming One or More Manufacturers: Use a product of one of the manufacturers named and meeting specifications, no options or substitutions allowed.
- C. Products Specified by Naming One or More Manufacturers with a Provision for Substitutions: Submit a request for substitution for any manufacturer not named.

2.03 MAINTENANCE MATERIALS

- A. Furnish extra materials, spare parts, tools, and software of types and in quantities specified in individual specification sections.
- B. Deliver to Project site; obtain receipt prior to final payment.

PART 3 EXECUTION

3.01 SUBSTITUTION PROCEDURES

- A. See Section 01 25 00 - Substitution Procedures.
- B. Instructions to Bidders specify time restrictions for submitting requests for substitutions during the bidding period. Comply with requirements specified in this section.

- C. Document each request with complete data substantiating compliance of proposed substitution with Contract Documents.
- D. A request for substitution constitutes a representation that the submitter:
 - 1. Has investigated proposed product and determined that it meets or exceeds the quality level of the specified product.
 - 2. Will provide the same warranty for the substitution as for the specified product.
 - 3. Will coordinate installation and make changes to other Work that may be required for the Work to be complete with no additional cost to Owner.
 - 4. Waives claims for additional costs or time extension that may subsequently become apparent.
- E. Substitution Submittal Procedure:
 - 1. Submit one electronic copy of request for substitution for consideration, including all required supporting materials. Limit each request to one proposed substitution.
 - 2. Submit shop drawings, product data, and certified test results attesting to the proposed product equivalence. Burden of proof is on proposer.
 - 3. Where color selections or physical comparison must be made, submit physical samples.
 - 4. The Architect will notify Contractor in writing of decision to accept or reject request.

3.02 TRANSPORTATION AND HANDLING

- A. Package products for shipment in manner to prevent damage; for equipment, package to avoid loss of factory calibration.
- B. If special precautions are required, attach instructions prominently and legibly on outside of packaging.
- C. Coordinate schedule of product delivery to designated prepared areas in order to minimize site storage time and potential damage to stored materials.
- D. Transport and handle products in accordance with manufacturer's instructions.
- E. Transport materials in covered trucks to prevent contamination of product and littering of surrounding areas.
- F. Promptly inspect shipments to ensure that products comply with requirements, quantities are correct, and products are undamaged.
- G. Provide equipment and personnel to handle products by methods to prevent soiling, disfigurement, or damage.
- H. Arrange for the return of packing materials, such as wood pallets, where economically feasible.

3.03 STORAGE AND PROTECTION

- A. Designate receiving/storage areas for incoming products so that they are delivered according to installation schedule and placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication.
- B. Store and protect products in accordance with manufacturers' instructions.
- C. Store with seals and labels intact and legible.
- D. Store sensitive products in weather tight, climate controlled, enclosures in an environment favorable to product.
- E. For exterior storage of fabricated products, place on sloped supports above ground.
- F. Protect products from damage or deterioration due to construction operations, weather, precipitation, humidity, temperature, sunlight and ultraviolet light, dirt, dust, and other contaminants.
- G. Comply with manufacturer's warranty conditions, if any.

- H. Cover products subject to deterioration with impervious sheet covering. Provide ventilation to prevent condensation and degradation of products.
- I. Prevent contact with material that may cause corrosion, discoloration, or staining.
- J. Provide equipment and personnel to store products by methods to prevent soiling, disfigurement, or damage.
- K. Arrange storage of products to permit access for inspection. Periodically inspect to verify products are undamaged and are maintained in acceptable condition.

END OF SECTION

SECTION 01 61 16
VOLATILE ORGANIC COMPOUND (VOC) CONTENT RESTRICTIONS

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Requirements for Indoor-Emissions-Restricted products.
- B. Requirements for VOC-Content-Restricted products.

1.02 RELATED REQUIREMENTS

- A. Section 01 30 00 - Administrative Requirements: Submittal procedures.

1.03 DEFINITIONS

- A. Indoor-Emissions-Restricted Products: All products in the following product categories, whether specified or not:
- B. VOC-Content-Restricted Products: All products in the following product categories, whether specified or not:
 - 1. Exterior and interior paints and coatings applied on site.
 - 2. Exterior and interior adhesives and sealants applied on site.
- C. Interior of Building: Anywhere inside the exterior weather barrier.
- D. Adhesives: All gunnable, trowelable, liquid-applied, and aerosol adhesives, whether specified or not; including flooring adhesives, resilient base adhesives, and pipe jointing adhesives.
- E. Sealants: All gunnable, trowelable, and liquid-applied joint sealants and sealant primers, whether specified or not; including firestopping sealants and duct joint sealers.
- F. Inherently Non-Emitting Materials: Products composed wholly of minerals or metals, unless they include organic-based surface coatings, binders, or sealants; and specifically the following:
 - 1. Concrete.
 - 2. Clay brick.
 - 3. Metals that are plated, anodized, or powder-coated.
 - 4. Glass.
 - 5. Ceramics.
 - 6. Solid wood flooring that is unfinished and untreated.

1.04 REFERENCE STANDARDS

- A. 40 CFR 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings; U.S. Environmental Protection Agency; current edition.
- B. ASTM D3960 - Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings; 2005 (Reapproved 2018).
- C. CARB (SCM) - Suggested Control Measure for Architectural Coatings; California Air Resources Board; 2019.
- D. SCAQMD 1113 - Architectural Coatings; 1977 (Amended 2016).
- E. SCAQMD 1168 - Adhesive and Sealant Applications; 1989 (Amended 2017).

1.05 SUBMITTALS

- A. See Section 01 30 00 - Administrative Requirements, for submittal procedures.
- B. Product Data: For each VOC-restricted product used in the project, submit evidence of compliance.

1.06 QUALITY ASSURANCE

- A. VOC Content Test Method: 40 CFR 59, Subpart D (EPA Method 24), or ASTM D3960, unless otherwise indicated.
 - 1. Evidence of Compliance: Acceptable types of evidence are:
 - a. Report of laboratory testing performed in accordance with requirements.
- B. Testing Agency Qualifications: Independent firm specializing in performing testing and inspections of the type specified in this section.

PART 2 PRODUCTS

2.01 MATERIALS

- A. All Products: Comply with the most stringent of federal, State, and local requirements, or these specifications.
- B. VOC-Content-Restricted Products: VOC content not greater than required by the following:
 - 1. Adhesives, Including Flooring Adhesives: SCAQMD 1168 Rule.
 - 2. Joint Sealants: SCAQMD 1168 Rule.
 - 3. Paints and Coatings: Each color; most stringent of the following:
 - a. 40 CFR 59, Subpart D.
 - b. SCAQMD 1113 Rule.
 - c. CARB (SCM).

PART 3 EXECUTION

3.01 FIELD QUALITY CONTROL

- A. Owner reserves the right to reject non-compliant products, whether installed or not, and require their removal and replacement with compliant products at no extra cost to Owner.
- B. Additional costs to restore indoor air quality due to installation of non-compliant products will be borne by Contractor.

END OF SECTION

**SECTION 01 70 00
EXECUTION AND CLOSEOUT REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Examination, preparation, and general installation procedures.
- B. Requirements for alterations work, including selective demolition.
- C. Pre-installation meetings.
- D. Cutting and patching.
- E. Surveying for laying out the work.
- F. Cleaning and protection.
- G. Starting of systems and equipment.
- H. Demonstration and instruction of Owner personnel.
- I. General requirements for maintenance service.

1.02 RELATED REQUIREMENTS

- A. Section 01 10 00 - Summary: Limitations on working in existing building; continued occupancy; work sequence; identification of salvaged and relocated materials.
- B. Section 01 30 00 - Administrative Requirements: Submittals procedures.
- C. Section 01 40 00 - Quality Requirements: Testing and inspection procedures.
- D. Section 01 50 00 - Temporary Facilities and Controls: Temporary interior partitions.
- E. Section 01 51 00 - Temporary Utilities: Temporary heating, cooling, and ventilating facilities.
- F. Section 01 74 19 - Construction Waste Management and Disposal: Additional procedures for trash/waste removal, recycling, salvage, and reuse.
- G. Section 01 78 00 - Closeout Submittals: Project record documents, operation and maintenance data, warranties and bonds.
- H. Section 01 81 13 - Sustainable Design Requirements

1.03 REFERENCE STANDARDS

- A. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2019.

1.04 SUBMITTALS

- A. See Section 01 30 00 - Administrative Requirements, for submittal procedures.
- B. Survey work: Submit name, address, and telephone number of Surveyor before starting survey work.
 - 1. On request, submit documentation verifying accuracy of survey work.
 - 2. Submit a copy of site drawing signed by the Land Surveyor, that the elevations and locations of the work are in compliance with Contract Documents.
 - 3. Submit surveys and survey logs for the project record.
- C. Cutting and Patching: Submit written request in advance of cutting or alteration that affects:
 - 1. Structural integrity of any element of Project.
 - 2. Integrity of weather exposed or moisture resistant element.
 - 3. Efficiency, maintenance, or safety of any operational element.
 - 4. Visual qualities of sight exposed elements.

5. Work of Owner or separate Contractor.

1.05 PROJECT CONDITIONS

- A. Grade site to drain. Maintain excavations free of water. Provide, operate, and maintain pumping equipment.
- B. Protect site from puddling or running water. Provide water barriers as required to protect site from soil erosion.
- C. Ventilate enclosed areas to assist cure of materials, to dissipate humidity, and to prevent accumulation of dust, fumes, vapors, or gases.
- D. Dust Control: Execute work by methods to minimize raising dust from construction operations. Provide positive means to prevent air-borne dust from dispersing into atmosphere and over adjacent property.
 - 1. Provide dust-proof enclosures to prevent entry of dust generated outdoors.
 - 2. Provide dust-proof barriers between construction areas and areas continuing to be occupied by Owner.
- E. Erosion and Sediment Control: Plan and execute work by methods to control surface drainage from cuts and fills, from borrow and waste disposal areas. Prevent erosion and sedimentation.
- F. Noise Control: Provide methods, means, and facilities to minimize noise produced by construction operations.

1.06 COORDINATION

- A. Coordinate scheduling, submittals, and work of the various sections of the Project Manual to ensure efficient and orderly sequence of installation of interdependent construction elements, with provisions for accommodating items installed later.
- B. Notify affected utility companies and comply with their requirements.
- C. Verify that utility requirements and characteristics of new operating equipment are compatible with building utilities. Coordinate work of various sections having interdependent responsibilities for installing, connecting to, and placing in service, such equipment.
- D. Coordinate space requirements, supports, and installation of mechanical and electrical work that are indicated diagrammatically on Drawings. Follow routing shown for pipes, ducts, and conduit, as closely as practicable; place runs parallel with lines of building. Utilize spaces efficiently to maximize accessibility for other installations, for maintenance, and for repairs.
- E. In finished areas except as otherwise indicated, conceal pipes, ducts, and wiring within the construction. Coordinate locations of fixtures and outlets with finish elements.
- F. Coordinate completion and clean-up of work of separate sections.
- G. After Owner occupancy of premises, coordinate access to site for correction of defective work and work not in accordance with Contract Documents, to minimize disruption of Owner's activities.

PART 2 PRODUCTS

2.01 PATCHING MATERIALS

- A. New Materials: As specified in product sections; match existing products and work for patching and extending work.
- B. Type and Quality of Existing Products: Determine by inspecting and testing products where necessary, referring to existing work as a standard.
- C. Product Substitution: For any proposed change in materials, submit request for substitution described in Section 01 60 00.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that existing site conditions and substrate surfaces are acceptable for subsequent work. Start of work means acceptance of existing conditions.
- B. Verify that existing substrate is capable of structural support or attachment of new work being applied or attached.
- C. Examine and verify specific conditions described in individual specification sections.
- D. Take field measurements before confirming product orders or beginning fabrication, to minimize waste due to over-ordering or misfabrication.
- E. Verify that utility services are available, of the correct characteristics, and in the correct locations.
- F. Prior to Cutting: Examine existing conditions prior to commencing work, including elements subject to damage or movement during cutting and patching. After uncovering existing work, assess conditions affecting performance of work. Beginning of cutting or patching means acceptance of existing conditions.

3.02 PREPARATION

- A. Clean substrate surfaces prior to applying next material or substance.
- B. Seal cracks or openings of substrate prior to applying next material or substance.
- C. Apply manufacturer required or recommended substrate primer, sealer, or conditioner prior to applying any new material or substance in contact or bond.

3.03 PREINSTALLATION MEETINGS

- A. When required in individual specification sections, convene a preinstallation meeting at the site prior to commencing work of the section.
- B. Require attendance of parties directly affecting, or affected by, work of the specific section.
- C. Notify Architect four days in advance of meeting date.
- D. Prepare agenda and preside at meeting:
 - 1. Review conditions of examination, preparation and installation procedures.
 - 2. Review coordination with related work.
- E. Record minutes and distribute copies within two days after meeting to participants, with two copies to Architect, Owner, participants, and those affected by decisions made.

3.04 LAYING OUT THE WORK

- A. Verify locations of survey control points prior to starting work.
- B. Promptly notify Architect of any discrepancies discovered.
- C. Protect survey control points prior to starting site work; preserve permanent reference points during construction.
- D. Promptly report to Architect the loss or destruction of any reference point or relocation required because of changes in grades or other reasons.
- E. Replace dislocated survey control points based on original survey control. Make no changes without prior written notice to Architect.
- F. Utilize recognized engineering survey practices.
- G. Establish elevations, lines and levels. Locate and lay out by instrumentation and similar appropriate means:

1. Site improvements including pavements; stakes for grading, fill and topsoil placement; utility locations, slopes, and invert elevations.
 2. Grid or axis for structures.
 3. Building foundation, column locations, ground floor elevations.
- H. Periodically verify layouts by same means.
- I. Maintain a complete and accurate log of control and survey work as it progresses.

3.05 GENERAL INSTALLATION REQUIREMENTS

- A. In addition to compliance with regulatory requirements, conduct construction operations in compliance with NFPA 241, including applicable recommendations in Appendix A.
- B. Install products as specified in individual sections, in accordance with manufacturer's instructions and recommendations, and so as to avoid waste due to necessity for replacement.
- C. Make vertical elements plumb and horizontal elements level, unless otherwise indicated.
- D. Install equipment and fittings plumb and level, neatly aligned with adjacent vertical and horizontal lines, unless otherwise indicated.
- E. Make consistent texture on surfaces, with seamless transitions, unless otherwise indicated.
- F. Make neat transitions between different surfaces, maintaining texture and appearance.

3.06 ALTERATIONS

- A. Drawings showing existing construction and utilities are based on casual field observation and existing record documents only.
 1. Verify that construction and utility arrangements are as shown.
 2. Report discrepancies to Architect before disturbing existing installation.
 3. Beginning of alterations work constitutes acceptance of existing conditions.
- B. Keep areas in which alterations are being conducted separated from other areas that are still occupied.
- C. Remove existing work as indicated and as required to accomplish new work.
 1. Remove items indicated on drawings.
 2. Relocate items indicated on drawings.
- D. Services (Including but not limited to HVAC, Plumbing, Fire Protection, Electrical, and Telecommunications): Remove, relocate, and extend existing systems to accommodate new construction.
 1. Maintain existing active systems that are to remain in operation; maintain access to equipment and operational components; if necessary, modify installation to allow access or provide access panel.
 2. Where existing systems or equipment are not active and Contract Documents require reactivation, put back into operational condition; repair supply, distribution, and equipment as required.
 3. Where existing active systems serve occupied facilities but are to be replaced with new services, maintain existing systems in service until new systems are complete and ready for service.
 - a. Disable existing systems only to make switchovers and connections; minimize duration of outages.
 - b. Provide temporary connections as required to maintain existing systems in service.
 4. Verify that abandoned services serve only abandoned facilities.
 5. Remove abandoned pipe, ducts, conduits, and equipment, including those above accessible ceilings; remove back to source of supply where possible, otherwise cap stub

and tag with identification; patch holes left by removal using materials specified for new construction.

- E. Protect existing work to remain.
 - 1. Prevent movement of structure; provide shoring and bracing if necessary.
 - 2. Perform cutting to accomplish removals neatly and as specified for cutting new work.
 - 3. Repair adjacent construction and finishes damaged during removal work.
- F. Adapt existing work to fit new work: Make as neat and smooth transition as possible.
 - 1. When existing finished surfaces are cut so that a smooth transition with new work is not possible, terminate existing surface along a straight line at a natural line of division and make recommendation to Architect.
 - 2. Where removal of partitions or walls results in adjacent spaces becoming one, rework floors, walls, and ceilings to a smooth plane without breaks, steps, or bulkheads.
 - 3. Where a change of plane of 1/4 inch or more occurs in existing work, submit recommendation for providing a smooth transition for Architect review and request instructions.
- G. Patching: Where the existing surface is not indicated to be refinished, patch to match the surface finish that existed prior to cutting. Where the surface is indicated to be refinished, patch so that the substrate is ready for the new finish.
- H. Refinish existing surfaces as indicated:
 - 1. Where rooms or spaces are indicated to be refinished, refinish all visible existing surfaces to remain to the specified condition for each material, with a neat transition to adjacent finishes.
 - 2. If mechanical or electrical work is exposed accidentally during the work, re-cover and refinish to match.
- I. Clean existing systems and equipment.
- J. Remove demolition debris and abandoned items from alterations areas and dispose of off-site; do not burn or bury.
- K. Do not begin new construction in alterations areas before demolition is complete.
- L. Comply with all other applicable requirements of this section.

3.07 CUTTING AND PATCHING

- A. Whenever possible, execute the work by methods that avoid cutting or patching.
- B. Perform whatever cutting and patching is necessary to:
 - 1. Complete the work.
 - 2. Fit products together to integrate with other work.
 - 3. Provide openings for penetration of mechanical, electrical, and other services.
 - 4. Match work that has been cut to adjacent work.
 - 5. Repair areas adjacent to cuts to required condition.
 - 6. Repair new work damaged by subsequent work.
 - 7. Remove samples of installed work for testing when requested.
 - 8. Remove and replace defective and non-conforming work.
- C. Execute work by methods that avoid damage to other work and that will provide appropriate surfaces to receive patching and finishing. In existing work, minimize damage and restore to original condition.
- D. Cut rigid materials using masonry saw or core drill. Pneumatic tools not allowed without prior approval.

- E. Restore work with new products in accordance with requirements of Contract Documents.
- F. Fit work air tight to pipes, sleeves, ducts, conduit, and other penetrations through surfaces.
- G. At penetrations of fire rated walls, partitions, ceiling, or floor construction, completely seal voids with fire rated material , to full thickness of the penetrated element.
- H. Patching:
 - 1. Finish patched surfaces to match finish that existed prior to patching. On continuous surfaces, refinish to nearest intersection or natural break. For an assembly, refinish entire unit.
 - 2. Match color, texture, and appearance.
 - 3. Repair patched surfaces that are damaged, lifted, discolored, or showing other imperfections due to patching work. If defects are due to condition of substrate, repair substrate prior to repairing finish.

3.08 PROGRESS CLEANING

- A. Maintain areas free of waste materials, debris, and rubbish. Maintain site in a clean and orderly condition.
- B. Remove debris and rubbish from pipe chases, plenums, attics, crawl spaces, and other closed or remote spaces, prior to enclosing the space.
- C. Broom and vacuum clean interior areas prior to start of surface finishing, and continue cleaning to eliminate dust.
- D. Collect and remove waste materials, debris, and trash/rubbish from site periodically and dispose off-site; do not burn or bury.
- E. NE CHPS - For phased, occupied renovations HEPA vacuum the carpet daily in the occupied areas.

3.09 PROTECTION OF INSTALLED WORK

- A. Protect installed work from damage by construction operations.
- B. Provide special protection where specified in individual specification sections.
- C. Provide temporary and removable protection for installed products. Control activity in immediate work area to prevent damage.
- D. Provide protective coverings at walls, projections, jambs, sills, and soffits of openings.
- E. Protect finished floors, stairs, and other surfaces from traffic, dirt, wear, damage, or movement of heavy objects, by protecting with durable sheet materials.
- F. Prohibit traffic or storage upon waterproofed or roofed surfaces. If traffic or activity is necessary, obtain recommendations for protection from waterproofing or roofing material manufacturer.
- G. Prohibit traffic from landscaped areas.
- H. Remove protective coverings when no longer needed; reuse or recycle plastic coverings if possible.

3.10 SYSTEM STARTUP

- A. Coordinate schedule for start-up of various equipment and systems.
- B. Verify that each piece of equipment or system has been checked for proper lubrication, drive rotation, belt tension, control sequence, and for conditions that may cause damage.
- C. Verify tests, meter readings, and specified electrical characteristics agree with those required by the equipment or system manufacturer.

- D. Verify that wiring and support components for equipment are complete and tested.
- E. Execute start-up under supervision of applicable Contractor personnel and manufacturer's representative in accordance with manufacturers' instructions.
- F. When specified in individual specification Sections, require manufacturer to provide authorized representative to be present at site to inspect, check, and approve equipment or system installation prior to start-up, and to supervise placing equipment or system in operation.
- G. Submit a written report that equipment or system has been properly installed and is functioning correctly.

3.11 DEMONSTRATION AND INSTRUCTION

- A. See Section 01 79 00 - Demonstration and Training.

3.12 ADJUSTING

- A. Adjust operating products and equipment to ensure smooth and unhindered operation.

3.13 FINAL CLEANING

- A. Use cleaning materials that are nonhazardous.
- B. Clean interior and exterior glass, surfaces exposed to view; remove temporary labels, stains and foreign substances, polish transparent and glossy surfaces, vacuum carpeted and soft surfaces.
- C. Remove all labels that are not permanent. Do not paint or otherwise cover fire test labels or nameplates on mechanical and electrical equipment.
- D. Clean equipment and fixtures to a sanitary condition with cleaning materials appropriate to the surface and material being cleaned.
- E. Clean filters of operating equipment.
- F. Clean debris from roofs, gutters, downspouts, and drainage systems.
- G. Clean site; sweep paved areas, rake clean landscaped surfaces.
- H. Remove waste, surplus materials, trash/rubbish, and construction facilities from the site; dispose of in legal manner; do not burn or bury.

3.14 CLOSEOUT PROCEDURES

- A. Make submittals that are required by governing or other authorities.
- B. Notify Architect when work is considered ready for Substantial Completion.
- C. Submit written certification that Contract Documents have been reviewed, work has been inspected, and that work is complete in accordance with Contract Documents and ready for Architect's review.
- D. Owner will occupy all of the building as specified in Section 01 10 00.
- E. Correct items of work listed in executed Certificates of Substantial Completion and comply with requirements for access to Owner-occupied areas.
- F. Notify Architect when work is considered finally complete.
- G. Complete items of work determined by Architect's final inspection.

3.15 MAINTENANCE

- A. Provide service and maintenance of components indicated in specification sections.

- B. Maintenance Period: As indicated in specification sections or, if not indicated, not less than one year from the Date of Substantial Completion or the length of the specified warranty, whichever is longer.
- C. Examine system components at a frequency consistent with reliable operation. Clean, adjust, and lubricate as required.
- D. Include systematic examination, adjustment, and lubrication of components. Repair or replace parts whenever required. Use parts produced by the manufacturer of the original component.
- E. Maintenance service shall not be assigned or transferred to any agent or subcontractor without prior written consent of the Owner.

END OF SECTION

SECTION 01 74 19
CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

1.01 WASTE MANAGEMENT REQUIREMENTS

- A. Owner requires that this project generate the least amount of trash and waste possible.
- B. Employ processes that ensure the generation of as little waste as possible due to error, poor planning, breakage, mishandling, contamination, or other factors.
- C. Minimize trash/waste disposal in landfills; reuse, salvage, or recycle as much waste as economically feasible.
- D. Methods of trash/waste disposal that are not acceptable are:
 - 1. Burning on the project site.
 - 2. Burying on the project site.
 - 3. Dumping or burying on other property, public or private.
 - 4. Other illegal dumping or burying.
- E. Regulatory Requirements: Contractor is responsible for knowing and complying with regulatory requirements, including but not limited to Federal, state and local requirements, pertaining to legal disposal of all construction and demolition waste materials.

1.02 RELATED REQUIREMENTS

- A. Section 01 30 00 - Administrative Requirements: Additional requirements for project meetings, reports, submittal procedures, and project documentation.
- B. Section 01 60 00 - Product Requirements: Waste prevention requirements related to delivery, storage, and handling.
- C. Section 01 70 00 - Execution and Closeout Requirements: Trash/waste prevention procedures related to demolition, cutting and patching, installation, protection, and cleaning.

1.03 DEFINITIONS

- A. Clean: Untreated and unpainted; not contaminated with oils, solvents, caulk, or the like.
- B. Construction and Demolition Waste: Solid wastes typically including building materials, packaging, trash, debris, and rubble resulting from construction, remodeling, repair and demolition operations.
- C. Hazardous: Exhibiting the characteristics of hazardous substances, i.e., ignitibility, corrosivity, toxicity or reactivity.
- D. Nonhazardous: Exhibiting none of the characteristics of hazardous substances, i.e., ignitibility, corrosivity, toxicity, or reactivity.
- E. Nontoxic: Neither immediately poisonous to humans nor poisonous after a long period of exposure.
- F. Recyclable: The ability of a product or material to be recovered at the end of its life cycle and remanufactured into a new product for reuse by others.
- G. Recycle: To remove a waste material from the project site to another site for remanufacture into a new product for reuse by others.
- H. Recycling: The process of sorting, cleansing, treating and reconstituting solid waste and other discarded materials for the purpose of using the altered form. Recycling does not include burning, incinerating, or thermally destroying waste.
- I. Return: To give back reusable items or unused products to vendors for credit.

- J. Reuse: To reuse a construction waste material in some manner on the project site.
- K. Salvage: To remove a waste material from the project site to another site for resale or reuse by others.
- L. Sediment: Soil and other debris that has been eroded and transported by storm or well production run-off water.
- M. Source Separation: The act of keeping different types of waste materials separate beginning from the first time they become waste.
- N. Toxic: Poisonous to humans either immediately or after a long period of exposure.
- O. Trash: Any product or material unable to be reused, returned, recycled, or salvaged.
- P. Waste: Extra material or material that has reached the end of its useful life in its intended use. Waste includes salvageable, returnable, recyclable, and reusable material.

1.04 SUBMITTALS

- A. See Section 01 30 00 - Administrative Requirements, for submittal procedures.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 WASTE MANAGEMENT PROCEDURES

- A. See Section 01 30 00 for additional requirements for project meetings, reports, submittal procedures, and project documentation.
- B. See Section 01 50 00 for additional requirements related to trash/waste collection and removal facilities and services.
- C. See Section 01 60 00 for waste prevention requirements related to delivery, storage, and handling.
- D. See Section 01 70 00 for trash/waste prevention procedures related to demolition, cutting and patching, installation, protection, and cleaning.

3.02 WASTE MANAGEMENT PLAN IMPLEMENTATION

- A. Manager: Designate an on-site person or persons responsible for instructing workers and overseeing and documenting results of the Waste Management Plan.
- B. Communication: Distribute copies of the Waste Management Plan to job site foreman, each subcontractor, Owner, and Architect.
- C. Instruction: Provide on-site instruction of appropriate separation, handling, and recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the project.
- D. Meetings: Discuss trash/waste management goals and issues at project meetings.
 - 1. Pre-bid meeting.
 - 2. Pre-construction meeting.
 - 3. Regular job-site meetings.
- E. Facilities: Provide specific facilities for separation and storage of materials for recycling, salvage, reuse, return, and trash disposal, for use by all contractors and installers.
 - 1. Provide containers as required.
 - 2. Provide adequate space for pick-up and delivery and convenience to subcontractors.
 - 3. Keep recycling and trash/waste bin areas neat and clean and clearly marked in order to avoid contamination of materials.

- F. Hazardous Wastes: Separate, store, and dispose of hazardous wastes according to applicable regulations.
- G. Recycling: Separate, store, protect, and handle at the site identified recyclable waste products in order to prevent contamination of materials and to maximize recyclability of identified materials. Arrange for timely pickups from the site or deliveries to recycling facility in order to prevent contamination of recyclable materials.
- H. Reuse of Materials On-Site: Set aside, sort, and protect separated products in preparation for reuse.
- I. Salvage: Set aside, sort, and protect products to be salvaged for reuse off-site.

END OF SECTION

**SECTION 01 76 10
TEMPORARY PROTECTIVE COVERINGS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Temporary protective coverings for installed floors, walls, and other surfaces.

1.02 REFERENCE STANDARDS

- A. ANSI A135.4 - Basic Hardboard; 2012 (Reaffirmed 2020).

PART 2 PRODUCTS

2.01 GENERAL

- A. Provide materials that are easily removed without damage to the surfaces covered and with the following characteristics:
 - 1. Water resistant.
 - 2. Impact resistant.
 - 3. Slip resistant.

2.02 MATERIALS

- A. Sheet Materials:
 - 1. Wood Hardboard: ANSI A135.4, tempered, 1/4 inch thick nominal.
- B. Rolled Materials:
 - 1. Self-adhering polyethylene film.
 - 2. Recycled cellulose fiberboard paper.
- C. Opening Protection Materials:
 - 1. Acrylic Plastic Sheet
- D. Tape: Type recommended by protective covering material manufacturer.

PART 3 EXECUTION

3.01 PREPARATION

- A. Remove dirt and debris from surfaces to be protected.

3.02 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Trim or overlap sheet materials to fit area to be covered.
- C. Roll out and cut rolled materials to fit area to be covered.
- D. Tape seams. Avoid taping directly to finished surfaces.
- E. Stretch self-adhering film materials to completely cover surface.
- F. Install door jamb protection to full height of opening.

3.03 REMOVAL

- A. Remove protective coverings prior to Date of Substantial Completion. Reuse or recycle materials if possible.

END OF SECTION

**SECTION 01 78 00
CLOSEOUT SUBMITTALS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Project Record Documents.
- B. Operation and Maintenance Data.
- C. Warranties and bonds.

1.02 RELATED REQUIREMENTS

- A. Section 00 65 01 - Closeout Document Checklist: List of required submittals for closeout.
- B. Section 00 72 13 - General Conditions: Performance bond and labor and material payment bonds, warranty, and correction of work.
- C. Section 01 30 00 - Administrative Requirements: Submittals procedures, shop drawings, product data, and samples.
- D. Section 01 70 00 - Execution and Closeout Requirements: Contract closeout procedures.
- E. Individual Product Sections: Specific requirements for operation and maintenance data.
- F. Individual Product Sections: Warranties required for specific products or Work.

1.03 SUBMITTALS

- A. Project Record Documents: Submit documents to Architect with claim for final Application for Payment.
- B. Operation and Maintenance Data:
 - 1. Submit two copies of preliminary draft or proposed formats and outlines of contents before start of Work. Architect will review draft and return one copy with comments.
 - 2. For equipment, or component parts of equipment put into service during construction and operated by Owner, submit completed documents within ten days after acceptance.
 - 3. Submit one copy of completed documents 15 days prior to final inspection. This copy will be reviewed and returned after final inspection, with Architect comments. Revise content of all document sets as required prior to final submission.
 - 4. Submit two sets of revised final documents in final form within 10 days after final inspection.
- C. Warranties and Bonds:
 - 1. For equipment or component parts of equipment put into service during construction with Owner's permission, submit documents within 10 days after acceptance.
 - 2. Make other submittals within 10 days after Date of Substantial Completion, prior to final Application for Payment.
 - 3. For items of Work for which acceptance is delayed beyond Date of Substantial Completion, submit within 10 days after acceptance, listing the date of acceptance as the beginning of the warranty period.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 PROJECT RECORD DOCUMENTS

- A. Maintain on site one set of the following record documents; record actual revisions to the Work:
 - 1. Drawings.
 - 2. Specifications.
 - 3. Addenda.

- 4. Change Orders and other modifications to the Contract.
- B. Ensure entries are complete and accurate, enabling future reference by Owner.
- C. Store record documents separate from documents used for construction.
- D. Record information concurrent with construction progress.
- E. Specifications: Legibly mark and record at each product section description of actual products installed, including the following:
 - 1. Changes made by Addenda and modifications.
- F. Record Drawings : Legibly mark each item to record actual construction including:
 - 1. Field changes of dimension and detail.
 - 2. Details not on original Contract drawings.

3.02 OPERATION AND MAINTENANCE DATA

- A. Source Data: For each product or system, list names, addresses and telephone numbers of Subcontractors and suppliers, including local source of supplies and replacement parts.
- B. Product Data: Mark each sheet to clearly identify specific products and component parts, and data applicable to installation. Delete inapplicable information.
- C. Drawings: Supplement product data to illustrate relations of component parts of equipment and systems, to show control and flow diagrams. Do not use Project Record Documents as maintenance drawings.
- D. Typed Text: As required to supplement product data. Provide logical sequence of instructions for each procedure, incorporating manufacturer's instructions.

3.03 OPERATION AND MAINTENANCE DATA FOR EQUIPMENT AND SYSTEMS

- A. For Each Item of Equipment and Each System:
 - 1. Description of unit or system, and component parts.
- B. Where additional instructions are required, beyond the manufacturer's standard printed instructions, have instructions prepared by personnel experienced in the operation and maintenance of the specific products.
- C. Maintenance Requirements: Include routine procedures and guide for preventative maintenance.
- D. Include manufacturer's printed operation and maintenance instructions.
- E. Additional Requirements: As specified in individual product specification sections.

3.04 ASSEMBLY OF OPERATION AND MAINTENANCE MANUALS

- A. Assemble operation and maintenance data into durable manuals for Owner's personnel use, with data arranged in the same sequence as, and identified by, the specification sections.
- B. Where systems involve more than one specification section, provide separate tabbed divider for each system.
- C. Binders: Commercial quality, 8-1/2 by 11 inch three D side ring binders with durable plastic covers; 2 inch maximum ring size. When multiple binders are used, correlate data into related consistent groupings.
- D. Cover: Identify each binder with typed or printed title OPERATION AND MAINTENANCE INSTRUCTIONS; identify title of Project; identify subject matter of contents.
- E. Project Directory: Title and address of Project; names, addresses, and telephone numbers of Architect, Consultants, Contractor and subcontractors, with names of responsible parties.

- F. Tables of Contents: List every item separated by a divider, using the same identification as on the divider tab; where multiple volumes are required, include all volumes Tables of Contents in each volume, with the current volume clearly identified.
- G. Dividers: Provide tabbed dividers for each separate product and system; identify the contents on the divider tab; immediately following the divider tab include a description of product and major component parts of equipment.
- H. Text: Manufacturer's printed data, or typewritten data on 20 pound paper.
- I. Drawings: Provide with reinforced punched binder tab. Bind in with text; fold larger drawings to size of text pages.

3.05 WARRANTIES AND BONDS

- A. Obtain warranties and bonds, executed in duplicate by responsible Subcontractors, suppliers, and manufacturers, within 10 days after completion of the applicable item of work. Except for items put into use with Owner's permission, leave date of beginning of time of warranty until the Date of Substantial completion is determined.
- B. Verify that documents are in proper form, contain full information, and are notarized.
- C. Co-execute submittals when required.
- D. Retain warranties and bonds until time specified for submittal.

END OF SECTION

**SECTION 01 79 00
DEMONSTRATION AND TRAINING**

PART 1 GENERAL

1.01 SUMMARY

- A. Demonstration of products and systems where indicated in specific specification sections.
- B. Training of Owner personnel in operation and maintenance is required for:
 - 1. HVAC systems and equipment.
 - 2. Electrical systems and equipment.

1.02 RELATED REQUIREMENTS

- A. Section 01 78 00 - Closeout Submittals: Operation and maintenance manuals.
- B. Other Specification Sections: Additional requirements for demonstration and training.

1.03 SUBMITTALS

- A. See Section 01 30 00 - Administrative Requirements, for submittal procedures.
- B. Training Plan: Owner will designate personnel to be trained; tailor training to needs and skill-level of attendees.
 - 1. Submit to Architect for transmittal to Owner.
 - 2. Submit not less than four weeks prior to start of training.
 - 3. Revise and resubmit until acceptable.
 - 4. Provide an overall schedule showing all training sessions.
 - 5. Include at least the following for each training session:
 - a. Identification, date, time, and duration.
 - b. Description of products and/or systems to be covered.
 - c. Name of firm and person conducting training; include qualifications.
 - d. Intended audience, such as job description.
 - e. Objectives of training and suggested methods of ensuring adequate training.
 - f. Methods to be used, such as classroom lecture, live demonstrations, hands-on, etc.
 - g. Media to be used, such as slides, hand-outs, etc.
 - h. Training equipment required, such as projector, projection screen, etc., to be provided by Contractor.
- C. Training Manuals: Provide training manual for each attendee; allow for minimum of two attendees per training session.
 - 1. Include applicable portion of O&M manuals.
 - 2. Include copies of all hand-outs, slides, overheads, video presentations, etc., that are not included in O&M manuals.
 - 3. Provide one extra copy of each training manual to be included with operation and maintenance data.
- D. Training Reports:
 - 1. Identification of each training session, date, time, and duration.
 - 2. Sign-in sheet showing names and job titles of attendees.
 - 3. List of attendee questions and written answers given, including copies of and references to supporting documentation required for clarification; include answers to questions that could not be answered in original training session.

1.04 QUALITY ASSURANCE

- A. Instructor Qualifications: Familiar with design, operation, maintenance and troubleshooting of the relevant products and systems.

1. Provide as instructors the most qualified trainer of those contractors and/or installers who actually supplied and installed the systems and equipment.
2. Where a single person is not familiar with all aspects, provide specialists with necessary qualifications.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 DEMONSTRATION - GENERAL

- A. Demonstrations conducted during system start-up do not qualify as demonstrations for the purposes of this section, unless approved in advance by Owner.
- B. Demonstration may be combined with Owner personnel training if applicable.
- C. Operating Equipment and Systems: Demonstrate operation in all modes, including start-up, shut-down, seasonal changeover, emergency conditions, and troubleshooting, and maintenance procedures, including scheduled and preventive maintenance.
 1. Perform demonstrations not less than two weeks prior to Substantial Completion.
 2. For equipment or systems requiring seasonal operation, perform demonstration for other season within six months.
- D. Non-Operating Products: Demonstrate cleaning, scheduled and preventive maintenance, and repair procedures.
 1. Perform demonstrations not less than two weeks prior to Substantial Completion.

3.02 TRAINING - GENERAL

- A. Conduct training on-site unless otherwise indicated.
- B. Owner will provide classroom and seating at no cost to Contractor.
- C. Provide training in minimum two hour segments.
- D. Provide minimum 8 hours of training.
- E. Training schedule will be subject to availability of Owner's personnel to be trained; re-schedule training sessions as required by Owner; once schedule has been approved by Owner failure to conduct sessions according to schedule will be cause for Owner to charge Contractor for personnel "show-up" time.
- F. Review of Facility Policy on Operation and Maintenance Data: During training discuss:
 1. The location of the O&M manuals and procedures for use and preservation; backup copies.
 2. Typical contents and organization of all manuals, including explanatory information, system narratives, and product specific information.
 3. Typical uses of the O&M manuals.
- G. Product- and System-Specific Training:
 1. Review the applicable O&M manuals.
 2. For systems, provide an overview of system operation, design parameters and constraints, and operational strategies.
 3. Review instructions for proper operation in all modes, including start-up, shut-down, seasonal changeover and emergency procedures, and for maintenance, including preventative maintenance.
 4. Provide hands-on training on all operational modes possible and preventive maintenance.
 5. Emphasize safe and proper operating requirements; discuss relevant health and safety issues and emergency procedures.
 6. Discuss common troubleshooting problems and solutions.
 7. Discuss any peculiarities of equipment installation or operation.

8. Discuss warranties and guarantees, including procedures necessary to avoid voiding coverage.
 9. Review recommended tools and spare parts inventory suggestions of manufacturers.
 10. Review spare parts and tools required to be furnished by Contractor.
 11. Review spare parts suppliers and sources and procurement procedures.
- H. Be prepared to answer questions raised by training attendees; if unable to answer during training session, provide written response within three days.

END OF SECTION

SECTION 01 81 13 SUSTAINABLE DESIGN REQUIREMENTS

PART 1 - GENERAL

1.01 GENERAL PROVISIONS

- A. Attention is directed to the CONTRACT AND GENERAL CONDITIONS and all Sections within DIVISION 01 - GENERAL REQUIREMENTS which are hereby made a part of this Section of the Specifications.

1.02 OBJECTIVES

- A. To obtain acceptable water efficiency, energy efficiency, and Indoor Air Quality (IAQ) for the completed Project and minimize the environmental impacts of construction and operation, the Contractor during the construction phase of this project shall implement the following procedures singly or in combination:
 - 1. Select products that minimize consumption of non-renewable resources, consume reduced amounts of energy and minimize amounts of pollution to produce and transport, and employ recycled and/or recyclable materials. To help purchasers incorporate environmental considerations into purchasing decisions, it is the intent of this project to conform with EPA's Five Guiding Principles on environmentally preferable purchasing. The five principles are:
 - a. Include environmental considerations as part of the normal purchasing process.
 - b. Emphasize pollution prevention early in the purchasing process.
 - c. Examine multiple environmental attributes throughout a product's or service's life cycle.
 - d. Compare relevant environmental impacts when selecting products and services.
 - e. Collect and base purchasing decisions on accurate and meaningful information about environmental performance.
 - 2. Control sources for potential IAQ pollutants by controlled selection of materials and processes used in project construction in order to attain superior IAQ.
 - 3. Products and processes that achieve the above objectives to the extent currently possible and practical have been selected and included in these Construction Documents. The Contractor is responsible to maintain and support these objectives in developing means and methods for performing the work of this Contract and in proposing product substitutions and/or changes to specified processes.

1.03 DESCRIPTION OF WORK

- A. Work Included: General requirements and procedures for compliance with Version 4.0 of the Northeast Collaborative for High Performance Schools Rhode Island (NE-CHPS RI) Guidelines, including all prerequisites and credits needed for the Project to be NE-CHPS RI verified with all RI addendums, pre-requisitions and other requirement.
 - 1. Compliance with requirements needed to obtain NE-CHPS RI prerequisites and credits may be used as one criterion to evaluate substitution requests.
 - 2. Refer to the NE-CHPS RI Scorecard attached at the end of this Section for required credits.
 - 3. Contractor shall compile the necessary calculation to show that the total percentage achieved by the project meets the required percentage for recycle content materials, rapidly renewable materials, locally produced materials, and certified wood. Contractor shall submit a proposed list within 30 days from signing of contract for review. Contractor shall submit monthly updates of these credits.
 - 4. Owner shall register project with the National CHPS organization and pay all registration and review fees.

5. Owner shall pay CHPS the costs of interpretations, certification and other administrative fees.
- B. Related Work: The following items are not included in this Section and will be performed under the designated Sections:
 1. Divisions 01 through 33 Sections for NE-CHPS requirements specific to the work of each of these Sections. Requirements may or may not include reference to NE-CHPS.

1.04 DEFINITIONS

- A. Agrifiber Products: Composite panel products derived from agricultural fiber.
- B. Composite Wood: A product consisting of wood fiber or other plant particles bonded together by a resin or binder.
- C. NE-CHPS: Northeast Collaborative for High Performance Schools Guidelines, current version.
- D. Forest-Stewardship Council (FSC) Certified Wood: Wood products bearing the FSC label come from forests that are managed in environmentally responsible, socially beneficial, and economically viable ways
- E. Chain-of-Custody Certificates: Certificates signed by manufacturers certifying that wood used to make products was obtained from forests certified by an FSC-accredited certification body to comply with FSC STD-01-001, "FSC Principles and Criteria for Forest Stewardship." Certificates shall include evidence that manufacturer is certified for chain of custody by an FSC-accredited certification body.
- F. Post-Consumer Recycled Content: Defined as materials or finished product that has served its intended consumer use and has been discarded by the consumer for recovery after use (e.g., plastic soda bottle).
- G. Post-Industrial Recycled Content: Defined as recovered industrial and manufacturing material that is recycled. Scrap raw materials that can be reused in the same manufacturing process from which they were recovered are not considered Post-Industrial Recycled Content. Fly-ash and synthetic gypsum, because they are waste products from coal burning electricity plants, are examples of Post-Industrial Recycled Content.
- H. Bio-Based Materials: Bio-based materials contain a significant amount of materials from natural fibers, plant stalks and leaves, and plant extractions. Examples are bamboo products, wheat grass cabinetry and linoleum.
- I. Salvaged or Reused Materials: Materials extracted from existing buildings in order to be reused in other buildings without be manufactured.
- J. Locally Produced Materials: Materials manufactured within five hundred miles radius of the project site. The location of manufacture refers to the final assembly of components into the building product that is furnished and installed by tradesmen.

1.05 SUBMITTALS

- A. General: Submit additional NE-CHPS RI submittals required by other Specification Sections.
- B. NE-CHPS RI submittals are in addition to other submittals. If submitted item is identical to that submitted to comply with other requirements, submit duplicate copies as a separate submittal to verify compliance with indicated NE-CHPS requirements.

PART 2 - PRODUCTS

2.01 LOW-EMITTING MATERIALS

- A. The following products and systems, where installed inside the weatherproofing system, shall meet the testing and product protocols listed below.
 1. Acceptable products in the categories listed below shall be certified in accordance with either Scientific Certification System's (SCS's) Indoor Advantage Gold program or SCS's

FloorScore Program (see www.scscertified.com/products) or GreenGuard's Children and Schools program (see <http://www.greenguard.org/en/quickSearch.aspx>). Certified products will be listed on either web site. Acceptable products may also be listed in the CHPS High Performance Products Database located at

<http://www.chps.net/dev/Drupal/node/445>. Alternatively, manufacturers may arrange for independent lab testing of materials to determine whether they meet the California Chronic Reference Exposure Levels (CRELs) as identified by the California Office of Environmental Health Hazard Assessment (OEHHA). See NE-CHPS Guidelines, page

- a. All paints and coatings that cover interior walls, floors and ceilings,
 - b. All building insulation (excluding insulation exterior to the building's vapor barrier),
 - c. Acoustical ceiling tiles and wall panels,
 - d. Gypsum wallboard.
 - e. Resilient flooring and associated adhesives.
2. Acceptable carpet systems and associated adhesives shall be certified in accordance with the Carpet and Rug Institute's Green Label Plus program (see <http://www.carpet-rug.org/commercial-customers/green-building9-and-the-environment/green-label-plus/index.cfm>). Acceptable products may also be listed in the CHPS High Performance Products Database located at <http://www.chps.net/dev/Drupal/node/445>. Alternatively, manufacturers may arrange for independent lab testing of materials to determine whether they meet the California Chronic Reference Exposure Levels (CRELs) as identified by the California Office of Environmental Health Hazard Assessment (OEHHA). See NE-CHPS Guidelines, page 83.
- B. For each low-emitting material or product, Contractor shall fill out the Materials Submittal Cover Sheet identifying the certification program that the material or product meets.
- C. For each low-emitting material or product, Contractor shall supply documentation verifying that it is certified as a low-emitting material in accordance with Scientific Certification System's FloorScore or Indoor Advantage Gold Program, or with GreenGuard's Children and Schools Program, or with the Carpet and Rug Institute's Green Label Plus Program, OR that the product is listed at www.chps.net in the CHPS Products Database, OR that the product was independently tested and has been shown to meet the Chronic Reference Exposure Levels (CRELs) for the VOCs identified by the California Office of Environmental Health Hazard Assessment (OEHHA).
- D. Do not use composite wood, agrifiber products and laminating adhesives that contain added urea-formaldehyde resins.

PART 3 - EXECUTION

3.01 NE-CHPS SCORECARD SUMMARY

- A. Comply with requirements of Scorecard Summary as developed by the Architect and attached to this Section.

3.02 MATERIALS SUBMITTAL COVER SHEET

- A. Documentation Sheet: Provide documentation for each material required for NE-CHPS Documentation as required.

END OF SECTION

**SECTION 02 41 00
DEMOLITION**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Selective demolition of building elements for alteration purposes.

1.02 RELATED REQUIREMENTS

- A. Section 01 10 00 - Summary: Limitations on Contractor's use of site and premises.
- B. Section 01 50 00 - Temporary Facilities and Controls: Site fences, security, protective barriers, and waste removal.
- C. Section 01 70 00 - Execution and Closeout Requirements: Project conditions; protection of bench marks, survey control points, and existing construction to remain; reinstallation of removed products; temporary bracing and shoring.

1.03 REFERENCE STANDARDS

- A. 29 CFR 1926 - Safety and Health Regulations for Construction; Current Edition.
- B. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2019.

PART 3 EXECUTION

2.01 SCOPE

- A. Refer to drawings for demolition scope.
- B. Break up paving within construction limits indicated on drawings to permit natural moisture drainage; leave pieces not larger than 1 square yard.
- C. Within area of new construction, remove foundation walls and footings to a minimum of 2 feet below finished grade.
- D. Remove concrete slabs on grade within construction limits indicated on drawings.
- E. Break up concrete slabs on grade within site boundaries to permit natural moisture drainage; leave pieces not larger than 1 square yard.
- F. Remove other items indicated, for salvage.

2.02 GENERAL PROCEDURES AND PROJECT CONDITIONS

- A. Comply with applicable codes and regulations for demolition operations and safety of adjacent structures and the public.
 - 1. Obtain required permits.
 - 2. Provide, erect, and maintain temporary barriers and security devices.
 - 3. Use physical barriers to prevent access to areas that could be hazardous to workers or the public.
 - 4. Conduct operations to minimize effects on and interference with adjacent structures and occupants.
 - 5. Do not close or obstruct roadways or sidewalks without permit.
 - 6. Conduct operations to minimize obstruction of public and private entrances and exits; do not obstruct required exits at any time; protect persons using entrances and exits from removal operations.
 - 7. Obtain written permission from owners of adjacent properties when demolition equipment will traverse, infringe upon or limit access to their property.
- B. Do not begin removal until receipt of notification to proceed from Owner.

- C. Do not begin removal until built elements to be salvaged or relocated have been removed.
- D. Protect existing structures and other elements that are not to be removed.
 - 1. Provide bracing and shoring.
 - 2. Prevent movement or settlement of adjacent structures.
 - 3. Stop work immediately if adjacent structures appear to be in danger.
- E. Minimize production of dust due to demolition operations; do not use water if that will result in ice, flooding, sedimentation of public waterways or storm sewers, or other pollution.
- F. If hazardous materials are discovered during removal operations, stop work and notify Architect and Owner; hazardous materials include regulated asbestos containing materials, lead, PCB's, and mercury.
- G. Perform demolition in a manner that maximizes salvage and recycling of materials.
 - 1. Dismantle existing construction and separate materials.
 - 2. Set aside reusable, recyclable, and salvageable materials; store and deliver to collection point or point of reuse.
- H. Partial Removal of Paving and Curbs: Neatly saw cut at right angle to surface.

2.03 EXISTING UTILITIES

- A. Protect existing utilities to remain from damage.
- B. Do not disrupt public utilities without permit from authority having jurisdiction.
- C. Do not close, shut off, or disrupt existing life safety systems that are in use without at least 7 days prior written notification to Owner.
- D. Do not close, shut off, or disrupt existing utility branches or take-offs that are in use without at least 7 days prior written notification to Owner.
- E. Locate and mark utilities to remain; mark using highly visible tags or flags, with identification of utility type; protect from damage due to subsequent construction, using substantial barricades if necessary.
- F. Prepare building demolition areas by disconnecting and capping utilities outside the demolition zone; identify and mark utilities to be subsequently reconnected, in same manner as other utilities to remain.

2.04 SELECTIVE DEMOLITION FOR ALTERATIONS

- A. Drawings showing existing construction and utilities are based on casual field observation and existing record documents only.
 - 1. Verify that construction and utility arrangements are as shown.
 - 2. Report discrepancies to Architect before disturbing existing installation.
 - 3. Beginning of demolition work constitutes acceptance of existing conditions that would be apparent upon examination prior to starting demolition.
- B. Maintain weatherproof exterior building enclosure except for interruptions required for replacement or modifications; take care to prevent water and humidity damage.
- C. Remove existing work as indicated and as required to accomplish new work.
 - 1. Remove rotted wood, corroded metals, and deteriorated masonry and concrete; replace with new construction specified.
 - 2. Remove items indicated on drawings.
- D. Services (Including but not limited to HVAC, Plumbing, Electrical, and Telecommunications): Remove existing systems and equipment as indicated.
 - 1. Maintain existing active systems that are to remain in operation; maintain access to equipment and operational components.

2. Where existing active systems serve occupied facilities but are to be replaced with new services, maintain existing systems in service until new systems are complete and ready for service.
 3. Verify that abandoned services serve only abandoned facilities before removal.
 4. Remove abandoned pipe, ducts, conduits, and equipment, including those above accessible ceilings; remove back to source of supply where possible, otherwise cap stub and tag with identification.
- E. Protect existing work to remain.
1. Prevent movement of structure; provide shoring and bracing if necessary.
 2. Perform cutting to accomplish removals neatly and as specified for cutting new work.
 3. Repair adjacent construction and finishes damaged during removal work.
 4. Patch as specified for patching new work.

2.05 DEBRIS AND WASTE REMOVAL

- A. Remove debris, junk, and trash from site.
- B. Remove from site all materials not to be reused on site; .
- C. Leave site in clean condition, ready for subsequent work.
- D. Clean up spillage and wind-blown debris from public and private lands.

END OF SECTION

**SECTION 26 05 05
SELECTIVE DEMOLITION FOR ELECTRICAL**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Electrical demolition.

1.02 RELATED REQUIREMENTS

- A. Section 01 70 00 - Execution and Closeout Requirements: Additional requirements for alterations work.

PART 2 PRODUCTS

2.01 MATERIALS AND EQUIPMENT

- A. Materials and equipment for patching and extending work: As specified in individual sections.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify field measurements and circuiting arrangements are as shown on Drawings.
- B. Verify that abandoned wiring and equipment serve only abandoned facilities.
- C. Demolition drawings are based on casual field observation .
- D. Report discrepancies to Owner before disturbing existing installation.
- E. Report discrepancies to Architect before disturbing existing installation.
- F. Beginning of demolition means installer accepts existing conditions.

3.02 PREPARATION

- A. Disconnect electrical systems in walls, floors, and ceilings to be removed.
- B. Coordinate utility service outages with utility company.
- C. Provide temporary wiring and connections to maintain existing systems in service during construction. When work must be performed on energized equipment or circuits, use personnel experienced in such operations.
- D. Existing Electrical Service: Maintain existing system in service until new system is complete and ready for service. Disable system only to make switchovers and connections. Minimize outage duration.
 - 1. Obtain permission from Owner at least 24 hours before partially or completely disabling system.
 - 2. Make temporary connections to maintain service in areas adjacent to work area.
- E. Existing Fire Alarm System: Maintain existing system in service until new system is accepted. Disable system only to make switchovers and connections. Minimize outage duration.
 - 1. Notify Owner before partially or completely disabling system.
 - 2. Notify local fire service.
 - 3. Make notifications at least 24 hours in advance.
 - 4. Make temporary connections to maintain service in areas adjacent to work area.

3.03 DEMOLITION AND EXTENSION OF EXISTING ELECTRICAL WORK

- A. Remove, relocate, and extend existing installations to accommodate new construction.
- B. Remove abandoned wiring to source of supply.
- C. Remove exposed abandoned conduit, including abandoned conduit above accessible ceiling finishes. Cut conduit flush with walls and floors, and patch surfaces.

- D. Disconnect abandoned outlets and remove devices. Remove abandoned outlets if conduit servicing them is abandoned and removed. Provide blank cover for abandoned outlets that are not removed.
- E. Disconnect and remove abandoned panelboards and distribution equipment.
- F. Disconnect and remove electrical devices and equipment serving utilization equipment that has been removed.
- G. Disconnect and remove abandoned luminaires. Remove brackets, stems, hangers, and other accessories.
- H. Repair adjacent construction and finishes damaged during demolition and extension work.
- I. Maintain access to existing electrical installations that remain active. Modify installation or provide access panel as appropriate.
- J. Extend existing installations using materials and methods compatible with existing electrical installations, or as specified.

3.04 CLEANING AND REPAIR

- A. See Section 01 74 19 - Construction Waste Management and Disposal for additional requirements.
- B. Clean and repair existing materials and equipment that remain or that are to be reused.
- C. Panelboards: Clean exposed surfaces and check tightness of electrical connections. Replace damaged circuit breakers and provide closure plates for vacant positions. Provide typed circuit directory showing revised circuiting arrangement.
- D. Luminaires: Remove existing luminaires for cleaning. Use mild detergent to clean all exterior and interior surfaces; rinse with clean water and wipe dry. Replace lamps , ballasts and broken electrical parts.
- E. Any lighting or ceiling-mounted devices removed during construction must be reinstalled.

END OF SECTION